ADDITIVE CHANNELS

Derivations of additive channels

Mathematical foundations and biological interpretation

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A derivation companion for readers who want to follow each mathematical result from its assumptions through its algebra to its biological meaning. The result map connects the proofs and worked calculations to the numerical reproduction code, figures, supplementary examples and interactive resources.

github.com/dortizbarrientos/additive_channels

How to use this companion

The linear share measures the fraction of variance supplied by the linear term of a stated quadratic approximation to log expected reproductive output. We call a population–fitness relationship with a large linear share an additive channel. To interpret that description, we need to know what has been averaged, which population supplies the frequency weights, and which assumptions permit a linear and a curved contribution to be separated. To predict selection, we must then change the weights to account for reproductive output. To predict a later generation, we must additionally specify inheritance.

This companion follows that sequence. It develops the calculations behind *Additive channels in curved fitness landscapes*, its mathematical appendices, mathematics guide, estimation example and visual tutorials. It expands the intermediate steps that a shorter scientific account can leave implicit: Gaussian moment contractions, population regression normal equations, matrix factorisations, normalising integrals, covariance order, fixed-point arguments and integration over uncertain breeding values. It also states the biological assumptions that make each calculation relevant.

The accompanying repository contains numerical reproductions, data generated under the stated models, and teaching resources. The proofs here are analytical. A numerical check can confirm an implementation or expose an error in a stated case; agreement in finitely many examples cannot prove a general identity. Conversely, the reported fitted estimates, simulation discrepancies and bootstrap endpoints are outputs of specified algorithms and random seeds. Their mathematical definitions and procedures can be derived, but their particular values must be reproduced computationally. Chapter 8 treats that distinction explicitly.

Three routes through the document

Read Chapters 1–5 in order to follow the theoretical argument from expected reproduction to an inheritance model. Continue with Chapters 6 and 7 to connect those quantities to observations and phenotype-space regressions. If you need one particular result, use the linked result map in Appendix A; it covers every entry in the repository's 53-record result catalogue. The equation map in Section A.2 also connects all 106 source-qualified equation labels to their detailed derivations or definitions.

Each substantial derivation identifies the mathematical tools it uses and explains the biological interpretation afterwards. The prerequisites are developed where they first become necessary. Familiarity with derivatives, finite sums, vectors and matrices will help; familiarity with Gaussian quadratic forms or matrix order is not assumed. The distinction between a selected-parent distribution and an offspring distribution is maintained throughout.

Mathematical concept	Why it is needed	Start here
Conditional expectation	Average reproductive output without confusing it with individual observations; obtain parental weights.	Chapter 1
Taylor expansion and moments	Specify the quadratic target and calculate its population variance.	Chapter 2
Least-squares projection	Identify the fitted intercept, slopes and explained variance under population frequencies.	Chapter 2

Mathematical concept	Why it is needed	Start here
Eigenvalues and coordinate changes	Separate changes in trait units from changes in biological variation.	Chapter 3
Normalisation and quadratic forms	Calculate selected-parent moments and determine when the reproductive weights are integrable.	Chapter 4
Matrix order and fixed points	Determine which covariances can persist under the specified inheritance closure.	Chapter 5
Likelihood and latent integration	Estimate the fitness relationship while accounting for uncertainty in breeding values.	Chapter 6
Conditional Gaussian moments	Convert compatible phenotype-regression moments into breeding-value-space quantities.	Chapter 7

Notation and units

Vectors are columns. Bold capital letters denote matrices unless they identify a random vector. The transpose is T. A covariance matrix is symmetric and positive semidefinite. The notation $\mathbf{G} \succ 0$ adds strict positive definiteness; inverses require it, or an explicit restriction to the subspace with positive variance.

Symbol	Meaning
$\boldsymbol{A}$, $\boldsymbol{a}$	Random breeding-value vector and a possible value. Breeding values sum population-specific average allelic effects; the inheritance model later makes stronger fixed-effect assumptions.
$\bar{\boldsymbol{A}}$, $\boldsymbol{X}$	Population mean and centred deviation $\boldsymbol{A} - \bar{\boldsymbol{A}}$.
$\mathbf{G}$	Breeding-value covariance in the population used to define the fitness description.
W, $W_A(\boldsymbol{a})$	Reproductive output and its conditional mean $\mathbb{E}[W \mid \boldsymbol{A} = \boldsymbol{a}]$.
$\ell_A(\boldsymbol{a})$	$\log W_A(\boldsymbol{a})$ where the conditional mean is positive. All logarithms are natural.
ℓ_0, $\boldsymbol{b}$, $\mathbf{H}$	Value, gradient and Hessian of ℓ_A at the reference mean. Their units are respectively log output, reciprocal trait units, and reciprocal products of trait units.
$\widetilde{\ell}_A$	The stated second-order polynomial. It equals the full log conditional mean only under an exact log-quadratic model.
L, Q_c	Linear and centred quadratic terms of that polynomial.
V_{lin}, V_{quad}, $\mathcal{A}_g$	Variances of those terms and the linear fraction $V_{\text{lin}}/(V_{\text{lin}} + V_{\text{quad}})$ when the denominator is positive. These are quantities on the log expected-output scale.
$\boldsymbol{\Gamma}$	$-\mathbf{H}$. Stabilising curvature means $\boldsymbol{\Gamma} \succeq 0$. Some selection calculations also allow indefinite curvature if the combined precision is positive definite.
$\mathbf{G}_{\text{sel}}$, $\boldsymbol{\Delta}$	Covariance and mean displacement under reproductive weighting of parents.
$\mathbf{B}$, ρ	Fixed genic covariance and the specified fraction of the selected covariance's departure from it retained after reproduction.

Symbol	Meaning
$\mathbf{P}$, $\boldsymbol{\beta}$, $\boldsymbol{\gamma}$	Phenotype covariance and linear/quadratic coefficients of a compatible raw relative-fitness regression. They are not automatically $\mathbf{G}$, $\boldsymbol{b}$ and $\mathbf{H}$.

The meaning of a subscript is stated where it is introduced. In the estimation chapter, the offspring count and prediction reliability have separate symbols. Environmental variance in a fitness assay and measurement variance in repeated trait observations also require separate biological assumptions, even when both have Gaussian distributions.

Versions and scope

The mathematical results correspond to the public code archive at commit `05c75c6`. The result catalogue, expanded checks and resource organisation used for this companion are those at commit `c860f6a`. The present companion adds explanations and proof steps; it does not change the generating models or their reference numerical results. Its source, result map and build instructions accompany this PDF in the same repository.

The assumptions strengthen at distinct points. Gaussian moments give an exact variance partition for a quadratic polynomial. Exact selected-parent formulas require an integrable exponential quadratic fitness function. The covariance recursion additionally assumes a fixed genic baseline and a specified inheritance closure. Estimation introduces its own observation model. None of these later assumptions follows merely from a large value of $\mathcal{A}_g$.

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Chapter 1

Conditional fitness, reproductive contributions and logarithms

An individual's breeding value does not determine its realised reproductive output. Individuals with the same breeding value can differ in environment, genetic background and unmeasured traits, and reproduction itself can be stochastic. We therefore begin with an average over the individuals represented at each breeding value. This chapter derives what that average supplies, why its logarithm is taken after averaging, and how environmental variation changes it. The variance calculations in Chapter 2 then describe variation in this specified quantity.

1.1 A conditional mean defines the fitness relationship

Let $\boldsymbol{A}$ denote breeding value in a specified population and let W denote nonnegative reproductive output during one selection episode. Assume

$$0 < \bar{W} = \mathbb{E}[W] < \infty.$$

The conditional expected output is

$$W_A(\boldsymbol{a}) = \mathbb{E}[W \mid \boldsymbol{A} = \boldsymbol{a}]. \tag{1.1}$$

The notation describes an expectation under the current joint distribution of breeding values and reproductive output. It does not require a deterministic map from breeding value to offspring count. Conditioning on a continuous variable is understood through a conditional distribution, rather than through the probability of the single event $\boldsymbol{A} = \boldsymbol{a}$.

The *tower property* states that an unconditional average can be obtained by first averaging conditionally and then averaging those conditional means. For example,

$$\mathbb{E}[W_A(\boldsymbol{A})] = \mathbb{E}\{\mathbb{E}[W \mid \boldsymbol{A}]\} = \mathbb{E}[W] = \bar{W}.$$

More generally, if $f(\boldsymbol{A})$ is a scalar quantity with $\mathbb{E}[|f(\boldsymbol{A})|W] < \infty$, then

$$\begin{aligned} \mathbb{E}[f(\boldsymbol{A})W] &= \mathbb{E}\{\mathbb{E}[f(\boldsymbol{A})W \mid \boldsymbol{A}]\} \\ &= \mathbb{E}\{f(\boldsymbol{A})\mathbb{E}[W \mid \boldsymbol{A}]\} \\ &= \mathbb{E}[f(\boldsymbol{A})W_A(\boldsymbol{A})]. \end{aligned} \tag{1.2}$$

The second line follows because $f(\boldsymbol{A})$ is already known when $\boldsymbol{A}$ is given. This identity is the reason conditional expected output can be used to calculate the distribution of parental breeding values represented by reproductive contributions.

Proposition 1.1 (Expected reproductive weighting)**.** *Under the assumptions above, define a probability law Q by*

$$Q(C) = \frac{\mathbb{E}[\mathbf{1}_{\{\boldsymbol{A}\in C\}}W]}{\bar{W}}$$

for each set C of breeding values. Its expectation of f is

$$\mathbb{E}_Q[f(\boldsymbol{A})] = \frac{\mathbb{E}[f(\boldsymbol{A})W_A(\boldsymbol{A})]}{\bar{W}}. \tag{1.3}$$

If the initial distribution has density p_A, the weighted density is

$$p_{\mathrm{sel}}(\boldsymbol{a}) = \frac{W_A(\boldsymbol{a})p_A(\boldsymbol{a})}{\bar{W}}. \tag{1.4}$$

Proof. Nonnegative output makes $Q(C)$ nonnegative. Taking C to be the entire space gives $Q(C) = \mathbb{E}[W]/\bar{W} = 1$, and countable additivity follows from additivity of the expectation of nonnegative indicators. Thus Q is a probability law. Equation (1.2) gives its expectation of f. When a density exists, write the numerator as

$$\mathbb{E}[f(\boldsymbol{A})W_A(\boldsymbol{A})] = \int f(\boldsymbol{a})W_A(\boldsymbol{a})p_A(\boldsymbol{a})\,d\boldsymbol{a}.$$

Dividing by $\bar{W}$ identifies the density in Equation (1.4). Its integral is one because the numerator integrates to $\mathbb{E}[W_A(\boldsymbol{A})] = \bar{W}$. □

The construction remains valid without a density. For two classes with frequencies p_1, p_2 and expected outputs w_1, w_2, their reproductive fractions are $p_j w_j/(p_1 w_1 + p_2 w_2)$. If the outputs are one and two, increasing the frequency of the second class from 1/4 to 3/4 changes mean output from

$$\tfrac{3}{4}(1) + \tfrac{1}{4}(2) = \tfrac{5}{4} \quad \text{to} \quad \tfrac{1}{4}(1) + \tfrac{3}{4}(2) = \tfrac{7}{4}.$$

Its reproductive fraction consequently changes from

$$\frac{(1/4)2}{5/4} = \frac{2}{5} \quad \text{to} \quad \frac{(3/4)2}{7/4} = \frac{6}{7}.$$

The expected output at each breeding value has not changed. Its frequency has.

Taking $f(\boldsymbol{A}) = \boldsymbol{A}$ component by component also gives a general selected-parent displacement. If the required first moments are finite,

$$\begin{aligned}\mathbb{E}_Q[\boldsymbol{A}] - \mathbb{E}[\boldsymbol{A}] &= \frac{\mathbb{E}[\boldsymbol{A}W] - \mathbb{E}[\boldsymbol{A}]\mathbb{E}[W]}{\bar{W}} \\ &= \frac{\mathrm{Cov}(\boldsymbol{A}, W)}{\bar{W}} = \frac{\mathrm{Cov}\{\boldsymbol{A}, W_A(\boldsymbol{A})\}}{\bar{W}}.\end{aligned} \tag{1.5}$$

This is the covariance part of the selection accounting developed by Price (1970). It uses no Gaussian approximation. Gaussian selected moments require the additional assumptions introduced in Chapter 4.

Biological interpretation. The weights describe how parental breeding values are represented among expected reproductive contributions. They do not specify an offspring's breeding value. That requires a transmission model. Nor should a population expectation be confused with the random ratio $\sum_i f(\boldsymbol{A}_i)W_i/\sum_i W_i$ in a finite realised sample: expectation of that ratio need not equal a ratio of expectations. The population law above is the target against which such finite-sample quantities can be compared.

1.2 What taking the logarithm changes

Where $W_A(\boldsymbol{a}) > 0$, define

$$\ell_A(\boldsymbol{a}) = \log W_A(\boldsymbol{a}). \tag{1.6}$$

For any two breeding values with positive conditional means,

$$\begin{aligned}\ell_A(\boldsymbol{a}_2) - \ell_A(\boldsymbol{a}_1) &= \log W_A(\boldsymbol{a}_2) - \log W_A(\boldsymbol{a}_1) \\ &= \log\left\{\frac{W_A(\boldsymbol{a}_2)}{W_A(\boldsymbol{a}_1)}\right\}.\end{aligned} \tag{1.7}$$

Thus equal differences in ℓ_A represent equal ratios of expected output. A difference of 0.10 corresponds to a ratio $e^{0.10}$, approximately 1.105. The gradient of ℓ_A measures a local proportional change in expected output per unit change in breeding value:

$$\frac{\partial \ell_A}{\partial a_j} = \frac{1}{W_A}\frac{\partial W_A}{\partial a_j}.$$

This is the connection to a log link for expected fitness (Morrissey and Goudie, 2022). It concerns the conditional mean, not a transformation of each observed count.

Changing the unit of reproductive output multiplies all expected outputs by one positive constant c. Then

$$\log\{cW_A(\boldsymbol{a})\} = \log c + \ell_A(\boldsymbol{a}).$$

The gradient, Hessian and variance of ℓ_A are unchanged because the added term is constant. The reproductive weights are unchanged as well:

$$\frac{cW_A(\boldsymbol{a})p_A(\boldsymbol{a})}{c\bar{W}} = \frac{W_A(\boldsymbol{a})p_A(\boldsymbol{a})}{\bar{W}}.$$

Neither result makes variance on the log scale equal to variance on the original output scale. They describe different functions.

1.3 Why averaging and taking logarithms do not commute

Proposition 1.2 (Conditional Jensen inequality)**.** *Fix a breeding value $\boldsymbol{a}$. Suppose the conditional output is positive and has finite positive mean $m = W_A(\boldsymbol{a})$. If the conditional expectation of $\log W$ is finite, then*

$$\mathbb{E}[\log W \mid \boldsymbol{A} = \boldsymbol{a}] \leq \log W_A(\boldsymbol{a}). \tag{1.8}$$

Equality holds precisely when $W = m$ with conditional probability one.

Proof. For $t > 0$, put $h(t) = t - 1 - \log t$. Its derivative is $h'(t) = 1 - 1/t$, so it decreases up to $t = 1$ and increases after $t = 1$. Its unique minimum is $h(1) = 0$. Therefore

$$\log t \leq t - 1,$$

with equality only at $t = 1$. Apply this inequality to $t = W/m$:

$$\log W \leq \log m + \frac{W - m}{m}.$$

Taking the conditional expectation gives

$$\mathbb{E}[\log W \mid \boldsymbol{A} = \boldsymbol{a}] \leq \log m + \frac{\mathbb{E}[W \mid \boldsymbol{A} = \boldsymbol{a}] - m}{m} = \log m.$$

The nonnegative difference has expectation zero only if it is zero almost surely. By the equality condition above, this requires $W/m = 1$ almost surely. □

If zero output has positive conditional probability, $\log W$ is not a finite-valued observation. It may be treated as $-\infty$ in an extended expectation, but it cannot be used as an ordinary finite response. Meanwhile $\log W_A(\boldsymbol{a})$ remains finite whenever the conditional mean is positive. For outputs zero and two with equal probability,

$$\log \mathbb{E}[W \mid \boldsymbol{A} = \boldsymbol{a}] = \log 1 = 0,$$

whereas one of the individual logarithms is not finite. If the conditional mean itself is zero, nonnegative output is zero almost surely at that breeding value. Its contribution to the selected-parent law is then zero, and ℓ_A has no finite value there. These are different boundary cases.

Biological interpretation. Variation in counts among individuals with the same breeding value is averaged before fitness differences across breeding values are described. Excluding zero counts, adding a constant to every count before logging, or averaging logged positive counts changes that target. A count likelihood with a log link can retain zero counts because it models the positive mean while leaving the observations on their count scale.

1.4 When a log observation has the required conditional mean

The distinction above does allow a useful special case. A log observation can describe the same shape as ℓ_A if the gap between its conditional mean and ℓ_A is constant across breeding values.

Proposition 1.3 (A constant conditional log gap)**.** *Assume* $0 < W_A(\boldsymbol{A}) < \infty$ *almost surely, and suppose*

$$W = W_A(\boldsymbol{A})U, \qquad U > 0, \qquad U \text{ independent of } \boldsymbol{A}, \qquad \mathbb{E}[U] = 1, \qquad \mathbb{E}[(\log U)^2] < \infty.$$

Then, with $Y = \log W$*,*

$$Y = \ell_A(\boldsymbol{A}) + \kappa + \varepsilon, \qquad \kappa = \mathbb{E}[\log U], \qquad \mathbb{E}[\varepsilon \mid \boldsymbol{A}] = 0, \tag{1.9}$$

where $\varepsilon = \log U - \kappa$ *has finite variance. Also* $\kappa \leq 0$*.*

Proof. First check that the multiplicative model has the stated conditional mean:

$$\mathbb{E}[W \mid \boldsymbol{A}] = W_A(\boldsymbol{A})\mathbb{E}[U \mid \boldsymbol{A}] = W_A(\boldsymbol{A}).$$

Taking a logarithm of this positive observation gives

$$Y = \ell_A(\boldsymbol{A}) + \log U.$$

Independence makes $\mathbb{E}[\log U \mid \boldsymbol{A}] = \mathbb{E}[\log U] = \kappa$. Subtracting κ therefore leaves a residual with zero conditional mean. The assumed second moment makes its variance finite. Finally, Jensen's inequality gives $\kappa = \mathbb{E}[\log U] \leq \log \mathbb{E}[U] = 0$. □

For an explicit example, take $Z \sim N(0, 1)$ independent of breeding value and let

$$U = \exp(\sigma Z - \sigma^2/2).$$

Completing the square in the standard normal integral gives

$$\mathbb{E}[e^{\sigma Z}] = \frac{1}{\sqrt{2\pi}} \int e^{\sigma z - z^2/2}\, dz = e^{\sigma^2/2}.$$

Hence $\mathbb{E}[U] = 1$, $\kappa = -\sigma^2/2$, and $\varepsilon = \sigma Z$ has variance σ^2. With $\sigma^2 = 4$, the residual variance is four even though the conditional mean of the log observation differs from ℓ_A by the constant -2. Under the Gaussian quadratic assumptions, the nested-R^2 result in Proposition 2.3 then applies.

Independence is a sufficient condition, not the only possibility. It is enough that $\mathbb{E}[U \mid \boldsymbol{A}] = 1$, that $\mathbb{E}[\log U \mid \boldsymbol{A}] = \kappa$ is constant, and that the resulting residual has finite variance. By contrast, if a conditionally mean-one lognormal factor has variance parameter $\sigma^2(\boldsymbol{A})$, then

$$\mathbb{E}[\log U \mid \boldsymbol{A}] = -\frac{1}{2}\sigma^2(\boldsymbol{A}).$$

When that quantity varies across breeding values, it contributes its own slope and curvature to $\mathbb{E}[Y \mid \boldsymbol{A}]$. The required constant-gap model no longer holds. None of these positive multiplicative models permits taking finite logarithms of zero offspring counts.

1.5 Deriving the environmental average

We now construct W_A from a phenotype model. This is an additional biological assumption, not a property implied by the definition of breeding value. In one trait, let

$$Z = A + E, \qquad E \sim N(0, V_E), \qquad E \text{ independent of } A.$$

Here E is an environmental deviation and $V_E > 0$ its variance. Suppose expected output conditional on the phenotype is

$$f_Z(z) = W_{\max} \exp\left\{ -\frac{(z-\theta)^2}{2\omega^2} \right\}, \qquad W_{\max} > 0, \quad \omega^2 > 0. \tag{1.10}$$

Assume that $\mathbb{E}[W \mid A, E] = f_Z(A + E)$. This makes phenotype sufficient for the conditional reproductive mean in this example. The tower property gives

$$W_A(a) = \mathbb{E}[f_Z(a + E) \mid A = a].$$

This is the environmental averaging used in the quantitative-genetic treatment of selection (Lande, 1979).

Proposition 1.4 (Gaussian environmental averaging)**.** *Under these assumptions,*

$$W_A(a) = W_{\max} \sqrt{\frac{\omega^2}{\omega^2 + V_E}} \exp\left\{ -\frac{(a-\theta)^2}{2(\omega^2 + V_E)} \right\}. \tag{1.11}$$

At a population mean a_0, its log slope and curvature are

$$b = \frac{\theta - a_0}{\omega^2 + V_E}, \qquad H = -\frac{1}{\omega^2 + V_E}. \tag{1.12}$$

Proof. Put $\delta = a - \theta$. Substituting the normal density gives

$$W_A(a) = \frac{W_{\max}}{\sqrt{2\pi V_E}} \int_{-\infty}^{\infty} \exp\left\{ -\frac{1}{2} \left(\frac{e^2}{V_E} + \frac{(e+\delta)^2}{\omega^2} \right) \right\} de. \tag{1.13}$$

To integrate, first expand the expression in parentheses:

$$\frac{e^2}{V_E} + \frac{(e+\delta)^2}{\omega^2} = \frac{\omega^2 + V_E}{V_E \omega^2} e^2 + \frac{2\delta}{\omega^2} e + \frac{\delta^2}{\omega^2}.$$

Let

$$C = \frac{\omega^2 + V_E}{V_E \omega^2}, \qquad d = \frac{V_E \delta}{\omega^2 + V_E}.$$

The first two terms are $C(e+d)^2 - Cd^2$. The remaining constant is

$$\begin{aligned} \frac{\delta^2}{\omega^2} - Cd^2 &= \frac{\delta^2}{\omega^2} - \frac{V_E \delta^2}{\omega^2(\omega^2 + V_E)} \\ &= \frac{\delta^2}{\omega^2 + V_E}. \end{aligned}$$

Therefore

$$\frac{e^2}{V_E} + \frac{(e+\delta)^2}{\omega^2} = C(e+d)^2 + \frac{\delta^2}{\omega^2 + V_E}.$$

The part independent of e can be taken outside the integral. In the remaining integral, change variable to $u = \sqrt{C}(e + d)$, so $de = du/\sqrt{C}$:

$$\int_{-\infty}^{\infty} e^{-C(e+d)^2/2}\, de = \frac{1}{\sqrt{C}} \int_{-\infty}^{\infty} e^{-u^2/2}\, du = \sqrt{\frac{2\pi}{C}}.$$

Multiplying by the original normalising factor gives

$$\frac{1}{\sqrt{2\pi V_E}} \sqrt{\frac{2\pi}{C}} = \frac{1}{\sqrt{V_E C}} = \sqrt{\frac{\omega^2}{\omega^2 + V_E}},$$

which proves Equation (1.11). Its logarithm is a constant minus $(a - \theta)^2/\{2(\omega^2 + V_E)\}$. Its first derivative is $-(a - \theta)/(\omega^2 + V_E)$; its second derivative is $-1/(\omega^2 + V_E)$. Evaluating at a_0 proves Equation (1.12). □

The expression has a continuous zero-environmental-variance limit: as V_E tends to zero, $W_A(a)$ tends to $f_Z(a)$. For the recurring example $V_E = \omega^2 = 1$, $\theta = 4$, and $a_0 = 0$,

$$W_A(a) = \frac{W_{\max}}{\sqrt{2}} e^{-(a-4)^2/4}, \qquad b = 2, \qquad H = -\frac{1}{2}.$$

The phenotype-space curvature is -1; the breeding-value-space curvature is $-1/2$. Their difference follows from averaging, before any population variance calculation is made.

1.5.1 The Jensen gap in this specific model

This example also allows the two orders of operation to be compared algebraically. Directly averaging the log phenotype-fitness function gives

$$\mathbb{E}[\log f_Z(a + E)] = \log W_{\max} - \frac{(a - \theta)^2 + V_E}{2\omega^2},$$

because $\mathbb{E}[E] = 0$ and $\mathbb{E}[E^2] = V_E$. In contrast,

$$\log \mathbb{E}[f_Z(a + E)] = \log W_{\max} - \frac{1}{2} \log\left(1 + \frac{V_E}{\omega^2}\right) - \frac{(a - \theta)^2}{2(\omega^2 + V_E)}.$$

Subtracting the first expression from the second and writing $t = V_E/\omega^2$ gives

$$\frac{1}{2}\{t - \log(1 + t)\} + \frac{(a - \theta)^2 V_E}{2\omega^2(\omega^2 + V_E)}. \tag{1.14}$$

Both terms are nonnegative. For the first, $\log(1+t) \leq t$ follows from the same logarithm inequality used above. Unless $V_E = 0$, the gap generally depends on breeding value through $(a - \theta)^2$. Averaging logged phenotype-fitness values therefore does not even preserve the breeding-value curvature in this model.

Biological interpretation. Environmental variation exposes individuals with a given breeding value to a distribution of phenotypes. Their mean output is consequently a different function from the phenotype-fitness relationship. The variance V_E in that average and the breeding-value variance G used to weight the population have different roles. Changing G on a fixed W_A is the controlled population comparison analysed later; changing V_E generally changes W_A itself. The width parameters and variances have trait-squared units, the slope b has inverse-trait units, and H has inverse-trait-squared units.

1.6 Connection to the reproducible calculations

The main and guide figure generators evaluate this environmental model, while `code/reproduce_result_checks.py` checks its integral independently in the group `conditional_mean_and_log`. That group also checks the two-class reproductive fractions. Its `noisy_regression_ratio` group illustrates the constant-gap observation model. These are numerical checks of specified examples. The identities above follow from the stated probability model and algebra; the checks do not establish them for an unrestricted biological population.

Chapter 2

Gaussian variation and the population linear projection

A fitness relationship specifies an expected output at each breeding value. A variance calculation additionally needs to know how often those breeding values occur. This chapter derives the calculation for a Gaussian population, then shows exactly which steps change when the population is not Gaussian. The regression interpretation will follow from minimising a population mean squared error. It is not an assumption that the local tangent is the fitted line.

2.1 The function and the population are separate inputs

Let the breeding-value vector be $\boldsymbol{A} \in \mathbb{R}^n$, with population mean $\boldsymbol{\mu}$ and covariance $\mathbf{G}$. Write

$$\boldsymbol{X} = \boldsymbol{A} - \boldsymbol{\mu}, \qquad \mathbb{E}[\boldsymbol{X}] = \mathbf{0}, \qquad \mathbb{E}[\boldsymbol{X}\boldsymbol{X}^{\mathsf{T}}] = \mathbf{G}. \tag{2.1}$$

Breeding values use the population-specific average effects appropriate to the trait and reference population. The following mathematics concerns the distribution of those values; it does not derive the effects from a genetic architecture.

Suppose log conditional expected reproductive output, ℓ_A, is twice continuously differentiable near $\boldsymbol{\mu}$. Its quadratic expansion is

$$\begin{aligned} \ell_A(\boldsymbol{\mu} + \boldsymbol{x}) &= q(\boldsymbol{x}) + R(\boldsymbol{x}), \\ q(\boldsymbol{x}) &= \ell_0 + \boldsymbol{b}^{\mathsf{T}}\boldsymbol{x} + \frac{1}{2}\boldsymbol{x}^{\mathsf{T}}\mathbf{H}\boldsymbol{x}, \end{aligned} \tag{2.2}$$

where $\ell_0 = \ell_A(\boldsymbol{\mu})$, $\boldsymbol{b} = \nabla\ell_A(\boldsymbol{\mu})$ and $\mathbf{H} = \nabla^2\ell_A(\boldsymbol{\mu})$. Equality of the mixed second derivatives makes $\mathbf{H}$ symmetric. The factor $1/2$ is the Taylor coefficient. Thus, in two traits, the quadratic term is

$$\frac{1}{2}H_{11}x_1^2 + H_{12}x_1x_2 + \frac{1}{2}H_{22}x_2^2.$$

The off-diagonal contribution appears twice in the matrix product before the factor $1/2$ is applied.

The function q is the same quadratic approximation denoted $\widetilde{\ell}_A$ elsewhere in the companion. Under the differentiability assumption, $R(\boldsymbol{x}) = o(\|\boldsymbol{x}\|^2)$ as $\boldsymbol{x} \to \mathbf{0}$. This is a local statement. A Gaussian population has unbounded support, so a local remainder statement alone does not establish that $R(\boldsymbol{X})$ is small in population mean square, or even that its variance is finite. The results below are exact for the random variable $q(\boldsymbol{X})$. They describe the complete $\ell_A(\boldsymbol{A})$ exactly only if the surface is quadratic on the relevant support, or approximately when an additional approximation argument is justified.

An element b_i has units of log expected output per unit of trait i; H_{ij} has units per product of trait units; and G_{ij} has the matching product of trait units. Their contractions therefore give variances in log expected output. Changing the common unit in which reproductive output is counted adds a constant to ℓ_A and does not change either variance.

Biological interpretation. The slope describes how conditional expected output changes near the population mean. Curvature describes how that slope changes. Neither is a variance until combined with the population distribution. A small curvature coefficient can contribute appreciable variation if the population occupies a sufficiently broad range, and a substantial coefficient can contribute little variation in a narrow population.

2.1.1 Why the multivariate Taylor formula has this form

Fix a displacement $\boldsymbol{x}$ whose line segment from $\boldsymbol{\mu}$ to $\boldsymbol{\mu}+\boldsymbol{x}$ lies in the differentiability neighbourhood. Define a one-variable function along that segment,

$$f(t) = \ell_A(\boldsymbol{\mu} + t\boldsymbol{x}), \qquad 0 \le t \le 1.$$

The chain rule supplies the multivariate information:

$$\begin{aligned} f'(t) &= \sum_i \partial_i \ell_A(\boldsymbol{\mu} + t\boldsymbol{x}) x_i, \\ f''(t) &= \sum_{i,j} \partial_i \partial_j \ell_A(\boldsymbol{\mu} + t\boldsymbol{x}) x_i x_j. \end{aligned} \tag{2.3}$$

Integrate f'' once to obtain $f'(s) = f'(0) + \int_0^s f''(t)\,dt$, then integrate over s from zero to one. Reversing the order of integration over the triangle $0 \le t \le s \le 1$ gives

$$f(1) = f(0) + f'(0) + \int_0^1 (1-t) f''(t)\,dt.$$

Writing $\mathbf{H}(\boldsymbol{a}) = \nabla^2 \ell_A(\boldsymbol{a})$ and subtracting the constant-Hessian part of this integral yields the exact remainder

$$R(\boldsymbol{x}) = \int_0^1 (1-t) \boldsymbol{x}^\mathsf{T} \{\mathbf{H}(\boldsymbol{\mu} + t\boldsymbol{x}) - \mathbf{H}(\boldsymbol{\mu})\} \boldsymbol{x}\,dt. \tag{2.4}$$

The integral of $1-t$ is $1/2$, which explains the coefficient in Equation (2.2). Continuity of the Hessian at $\boldsymbol{\mu}$ makes its difference tend uniformly to zero along a shrinking segment. Multiplication by $\|\boldsymbol{x}\|^2$ then gives the stated $o(\|\boldsymbol{x}\|^2)$ remainder.

If a third derivative also exists continuously, one more integration gives

$$R(\boldsymbol{x}) = \frac{1}{2} \int_0^1 (1-t)^2 D^3 \ell_A(\boldsymbol{\mu} + t\boldsymbol{x})[\boldsymbol{x}, \boldsymbol{x}, \boldsymbol{x}]\,dt. \tag{2.5}$$

Here $D^3 \ell_A[\boldsymbol{x}, \boldsymbol{x}, \boldsymbol{x}]$ abbreviates $\sum_{i,j,k} \partial_i \partial_j \partial_k \ell_A \, x_i x_j x_k$. If the absolute value of this trilinear form is at most $M \|\boldsymbol{x}\|^3$ along the segment, then

$$|R(\boldsymbol{x})| \le \frac{M}{2} \|\boldsymbol{x}\|^3 \int_0^1 (1-t)^2\,dt = \frac{M}{6} \|\boldsymbol{x}\|^3. \tag{2.6}$$

The derivative bound must hold where the segment runs. A bound established only in a ball of radius a_0 controls $\mathbb{E}[R(\boldsymbol{X})^2\,\mathbf{1}\{\|\boldsymbol{X}\| \le a_0\}]$, but leaves $\mathbb{E}[R(\boldsymbol{X})^2\,\mathbf{1}\{\|\boldsymbol{X}\| > a_0\}]$ uncontrolled. Because a nondegenerate Gaussian places positive probability outside every finite ball, a population approximation needs an additional tail argument or a global derivative bound. We do not replace that requirement with the observation that a local quadratic fit is close near the mean.

2.2 Deriving the Gaussian moments

Assume first that $\boldsymbol{X} \sim \mathcal{N}(\mathbf{0}, \mathbf{G})$ with $\mathbf{G} \succ 0$. Section 2.6 removes the positive-definiteness restriction by working only in directions that vary. We need moments through order four because squaring the quadratic term produces products of four breeding-value deviations.

The moment-generating function is

$$M(\boldsymbol{t}) = \mathbb{E}[\exp(\boldsymbol{t}^\mathsf{T} \boldsymbol{X})].$$

It is useful because differentiating the exponential with respect to t_i brings down a factor X_i. Completing the square in the normal density gives

$$-\frac{1}{2}\boldsymbol{x}^{\mathsf{T}}\mathbf{G}^{-1}\boldsymbol{x}+\boldsymbol{t}^{\mathsf{T}}\boldsymbol{x}=-\frac{1}{2}(\boldsymbol{x}-\mathbf{G}\boldsymbol{t})^{\mathsf{T}}\mathbf{G}^{-1}(\boldsymbol{x}-\mathbf{G}\boldsymbol{t})+\frac{1}{2}\boldsymbol{t}^{\mathsf{T}}\mathbf{G}\boldsymbol{t}. \tag{2.7}$$

To check the equality, expand the first quadratic on the right: its two cross terms each contribute $\boldsymbol{t}^{\mathsf{T}}\boldsymbol{x}/2$, and its constant term cancels half of $\boldsymbol{t}^{\mathsf{T}}\mathbf{G}\boldsymbol{t}$. The integral of the shifted normal density is one. Consequently,

$$M(\boldsymbol{t})=\exp\!\left(\frac{1}{2}\boldsymbol{t}^{\mathsf{T}}\mathbf{G}\boldsymbol{t}\right). \tag{2.8}$$

Differentiation under this Gaussian integral is legitimate: on a bounded neighbourhood of $\boldsymbol{t}$, a polynomial times $\exp(\boldsymbol{t}^{\mathsf{T}}\boldsymbol{x})$ is dominated by an integrable Gaussian with a slightly smaller positive precision. In particular, all moments used here are finite.

For the derivatives, write $u_i=(\mathbf{G}\boldsymbol{t})_i$. Then $\partial u_i/\partial t_j=G_{ij}$, and

$$\begin{aligned}\partial_i M &= u_i M,\\ \partial_j\partial_i M &= (G_{ij}+u_iu_j)M,\\ \partial_k\partial_j\partial_i M &= (G_{ij}u_k+G_{ik}u_j+G_{jk}u_i+u_iu_ju_k)M.\end{aligned} \tag{2.9}$$

At $\boldsymbol{t}=\mathbf{0}$, every u_i is zero and $M=1$. Therefore

$$\mathbb{E}[X_i]=0,\qquad \mathbb{E}[X_iX_j]=G_{ij},\qquad \mathbb{E}[X_iX_jX_k]=0. \tag{2.10}$$

Differentiate the last line of Equation (2.9) once more. At zero, the derivative of $G_{ij}u_k$ is $G_{ij}G_{kl}$; the next two terms give $G_{ik}G_{jl}$ and $G_{jk}G_{il}$. Terms retaining any u_i vanish. We have proved the fourth-moment identity:

$$\mathbb{E}[X_iX_jX_kX_l]=G_{ij}G_{kl}+G_{ik}G_{jl}+G_{il}G_{jk}. \tag{2.11}$$

The three terms correspond to the three ways to pair four positions. This is the order-four case of the Gaussian product-moment result associated with Isserlis (1918); the calculation above establishes the particular identity needed here rather than treating it as an unexplained rule. Putting all four indices equal gives $\mathbb{E}[X_i^4]=3G_{ii}^2$.

Subtracting the product of the two second moments yields

$$\begin{aligned}\operatorname{Cov}(X_iX_j,X_kX_l)&=\mathbb{E}[X_iX_jX_kX_l]-\mathbb{E}[X_iX_j]\mathbb{E}[X_kX_l]\\ &=G_{ik}G_{jl}+G_{il}G_{jk}.\end{aligned} \tag{2.12}$$

This covariance of products is the direct input to the quadratic variance. It includes correlations among traits; independence of the X_i is not required.

2.3 From moments to the variance partition

Define the linear term, the uncentred quadratic term, and its centred version by

$$L=\boldsymbol{b}^{\mathsf{T}}\boldsymbol{X},\qquad Q=\frac{1}{2}\boldsymbol{X}^{\mathsf{T}}\mathbf{H}\boldsymbol{X},\qquad Q_c=Q-\mathbb{E}[Q]. \tag{2.13}$$

Since $\mathbb{E}[\boldsymbol{X}]=\mathbf{0}$, $\mathbb{E}[L]=0$. For the quadratic mean,

$$\begin{aligned}\mathbb{E}[Q]&=\frac{1}{2}\sum_{i,j}H_{ij}\mathbb{E}[X_iX_j]=\frac{1}{2}\sum_{i,j}H_{ij}G_{ij}\\ &=\frac{1}{2}\sum_i(\mathbf{HG})_{ii}=\frac{1}{2}\operatorname{tr}(\mathbf{HG}).\end{aligned} \tag{2.14}$$

The penultimate equality uses $G_{ij} = G_{ji}$. Thus

$$q(\boldsymbol{X}) = \mathbb{E}[q(\boldsymbol{X})] + L + Q_c, \qquad \mathbb{E}[q(\boldsymbol{X})] = \ell_0 + \tfrac{1}{2}\operatorname{tr}(\mathbf{HG}).$$

Centring Q changes its mean but not its variance; it makes the comparison of the two varying components explicit.

For the linear term,

$$\begin{aligned}\operatorname{Var}(L) &= \mathbb{E}\left[\left(\sum_i b_i X_i\right)\left(\sum_j b_j X_j\right)\right] \\ &= \sum_{i,j} b_i b_j G_{ij} = \boldsymbol{b}^{\mathsf{T}}\mathbf{G}\boldsymbol{b}. \end{aligned} \tag{2.15}$$

For the quadratic term, covariance is bilinear, so

$$\begin{aligned}\operatorname{Var}(Q_c) &= \frac{1}{4}\sum_{i,j,k,l} H_{ij}H_{kl}\operatorname{Cov}(X_iX_j, X_kX_l) \\ &= \frac{1}{4}\sum_{i,j,k,l} H_{ij}H_{kl}(G_{ik}G_{jl} + G_{il}G_{jk}). \end{aligned} \tag{2.16}$$

There are two sums in the final line. Interchanging the dummy indices k and l in the first makes it equal to the second because $H_{kl} = H_{lk}$. For the second sum, regroup the scalar factors:

$$\begin{aligned}\sum_{i,j,k,l} H_{ij}H_{kl}G_{il}G_{jk} &= \sum_{i,k}\left(\sum_j H_{ij}G_{jk}\right)\left(\sum_l H_{kl}G_{li}\right) \\ &= \sum_{i,k}(\mathbf{HG})_{ik}(\mathbf{HG})_{ki} \\ &= \sum_i(\mathbf{HGHG})_{ii} = \operatorname{tr}(\mathbf{HGHG}). \end{aligned} \tag{2.17}$$

Again $G_{il} = G_{li}$ was used. No step interchanges the order of two matrices. The two equal sums and the factor $1/4$ therefore give

$$\operatorname{Var}(Q_c) = \frac{1}{2}\operatorname{tr}[(\mathbf{HG})^2]. \tag{2.18}$$

The square is a matrix product, not an elementwise square.

Finally,

$$\begin{aligned}\operatorname{Cov}(L, Q_c) &= \mathbb{E}[LQ_c] = \frac{1}{2}\sum_{i,j,k} b_i H_{jk}\mathbb{E}[X_iX_jX_k] \\ &= 0. \end{aligned} \tag{2.19}$$

The constant subtracted in Q_c contributes nothing because $\mathbb{E}[L] = 0$. Equivalently, the linear term is odd under $\boldsymbol{X} \mapsto -\boldsymbol{X}$ and the quadratic term is even, so their product integrates to zero under a centrally symmetric Gaussian density.

Proposition 2.1 (Gaussian variance partition)**.** *For a centred Gaussian vector with covariance $\mathbf{G} \succeq 0$, a fixed vector $\boldsymbol{b}$ and a fixed symmetric matrix $\mathbf{H}$, define q, L and Q_c by Equations (2.2) and (2.13). Then*

$$\begin{aligned} V_{\text{lin}} &= \operatorname{Var}(L) = \boldsymbol{b}^{\mathsf{T}}\mathbf{G}\boldsymbol{b}, \\ V_{\text{quad}} &= \operatorname{Var}(Q_c) = \frac{1}{2}\operatorname{tr}[(\mathbf{HG})^2], \\ \operatorname{Var}(q) &= V_{\text{lin}} + V_{\text{quad}}. \end{aligned} \tag{2.20}$$

If the total is positive, the first-order variance share is

$$\mathcal{A}_g = \frac{V_{\text{lin}}}{V_{\text{lin}} + V_{\text{quad}}}. \tag{2.21}$$

Proof. For $\mathbf{G} \succ 0$, the preceding moment calculations establish each identity; the variance of a sum adds twice the covariance in Equation (2.19), which is zero. For singular $\mathbf{G}$, apply the same calculation to its positive-variance coordinates, as justified in Section 2.6. □

Uncorrelated terms need not be independent. For example, if X is standard normal, X and $X^2 - 1$ have zero covariance, yet the second is determined once the first is known. The decomposition above is an orthogonal decomposition in mean square; it does not describe two independent biological mechanisms.

Biological interpretation. Both components are variation among breeding values in a conditional expected-fitness function. They are not the additive and epistatic genetic variance components of the underlying trait, and they are not a partition of the variance of realised reproductive counts. The environmental and other variation averaged at each breeding value has already entered the definition of that conditional mean.

2.4 Why the population-fitted line has this share

For a square-integrable response Y and centred predictors $\boldsymbol{X}$, a population affine fit minimises

$$J(k, \boldsymbol{d}) = \mathbb{E}[(Y - k - \boldsymbol{d}^{\mathsf{T}}\boldsymbol{X})^2].$$

An affine fit includes both an intercept k and a slope vector $\boldsymbol{d}$. Expanding this quadratic objective, or differentiating with respect to its coefficients, gives

$$\begin{aligned} \frac{\partial J}{\partial k} &= -2\mathbb{E}[Y - k - \boldsymbol{d}^{\mathsf{T}}\boldsymbol{X}], \\ \nabla_{\boldsymbol{d}} J &= -2\mathbb{E}[\boldsymbol{X}(Y - k - \boldsymbol{d}^{\mathsf{T}}\boldsymbol{X})]. \end{aligned} \tag{2.22}$$

Setting these derivatives to zero yields the normal equations

$$k = \mathbb{E}[Y], \qquad \mathbf{G}\boldsymbol{d} = \operatorname{Cov}(\boldsymbol{X}, Y). \tag{2.23}$$

The objective is a convex quadratic in the coefficients. When $\mathbf{G} \succ 0$, these equations give its unique minimum. More directly, if $\widehat{Y} = k + \boldsymbol{d}^{\mathsf{T}}\boldsymbol{X}$ satisfies them, its residual is orthogonal to constants and to every component of $\boldsymbol{X}$. For any other affine fit $\widehat{Y} + h$,

$$\begin{aligned} \mathbb{E}[(Y - \widehat{Y} - h)^2] &= \mathbb{E}[(Y - \widehat{Y})^2] + \mathbb{E}[h^2] - 2\mathbb{E}[(Y - \widehat{Y})h] \\ &= \mathbb{E}[(Y - \widehat{Y})^2] + \mathbb{E}[h^2]. \end{aligned} \tag{2.24}$$

This verifies optimality without relying on a derivative test and shows why the normal equations are an orthogonal-projection condition.

Apply the equations to $Y = q(\boldsymbol{X})$. The mean was calculated above, and

$$\begin{aligned} \operatorname{Cov}(\boldsymbol{X}, q) &= \operatorname{Cov}(\boldsymbol{X}, \boldsymbol{b}^{\mathsf{T}}\boldsymbol{X}) + \operatorname{Cov}(\boldsymbol{X}, Q_c) \\ &= \mathbf{G}\boldsymbol{b} + \mathbf{0}. \end{aligned} \tag{2.25}$$

The last vector is zero component by component by the same third-moment calculation used in Equation (2.19). We obtain

Proposition 2.2 (Population affine projection)**.** *Under the assumptions of Proposition 2.1, the best population affine prediction of $q(\boldsymbol{X})$ is*

$$\widehat{q}_1(\boldsymbol{X}) = \ell_0 + \frac{1}{2}\operatorname{tr}(\mathbf{HG}) + \boldsymbol{b}^\top \boldsymbol{X}. \tag{2.26}$$

Its residual is Q_c. If $\operatorname{Var}(q) > 0$, its population coefficient of determination is

$$R_1^2 = 1 - \frac{\mathbb{E}[(q - \widehat{q}_1)^2]}{\operatorname{Var}(q)} = \frac{V_{\text{lin}}}{V_{\text{lin}} + V_{\text{quad}}} = \mathcal{A}_g. \tag{2.27}$$

Proof. The normal equations give the displayed intercept and $\boldsymbol{d} = \boldsymbol{b}$ on the varying support. Subtracting the fit from q leaves Q_c. The residual has zero mean and variance V_{quad}, whereas $\operatorname{Var}(q) = V_{\text{lin}} + V_{\text{quad}}$. Substitution gives the stated R^2. □

The tangent plane at the population mean is $\ell_0 + \boldsymbol{b}^\top \boldsymbol{X}$. Its slope agrees with the population fit under these assumptions, but its intercept omits the mean curvature contribution. For one trait with $H < 0$, the fitted intercept is lower by $|H|G/2$. Comparing the curve to its tangent without this intercept adjustment does not give the fitted regression residual.

If $V_{\text{lin}} > 0$, the correlation expression follows separately:

$$\begin{aligned} \operatorname{Corr}(L, L + Q_c)^2 &= \frac{\operatorname{Cov}(L, L + Q_c)^2}{\operatorname{Var}(L)\operatorname{Var}(L + Q_c)} \\ &= \frac{V_{\text{lin}}^2}{V_{\text{lin}}(V_{\text{lin}} + V_{\text{quad}})} = \mathcal{A}_g. \end{aligned} \tag{2.28}$$

This form requires a nonconstant L. The regression form still assigns $R_1^2 = 0$ to a constant fit when the target varies.

Biological interpretation. Every individual has equal weight in the population mean squared error. Common breeding values collectively receive more weight because more individuals have them. A large share therefore means that a linear description accounts for much of the quadratic target's variation under those frequencies. It neither identifies gene action nor establishes an accurate prediction after reproductive reweighting.

2.5 What happens when the response includes noise?

Suppose the observed response has the model

$$Y = q(\boldsymbol{X}) + \kappa + \varepsilon, \qquad \mathbb{E}[\varepsilon \mid \boldsymbol{X}] = 0, \qquad \mathbb{E}[\varepsilon^2] = \sigma_\varepsilon^2 < \infty, \tag{2.29}$$

where κ is constant. Conditional noise variance may depend on $\boldsymbol{X}$; constant variance and independence are stronger assumptions than this argument needs. If $f(\boldsymbol{X})$ is any square-integrable function for which the product is integrable, conditioning gives

$$\mathbb{E}[f(\boldsymbol{X})\varepsilon] = \mathbb{E}\{f(\boldsymbol{X})\mathbb{E}[\varepsilon \mid \boldsymbol{X}]\} = 0. \tag{2.30}$$

In particular, the residual is orthogonal to the intercept, all linear predictors, and the complete quadratic conditional mean.

The best quadratic prediction is $q + \kappa$, with residual ε. The best affine prediction is $\widehat{q}_1 + \kappa$, with residual $Q_c + \varepsilon$. Hence

$$\begin{aligned} \operatorname{Var}(Y) &= V_{\text{lin}} + V_{\text{quad}} + \sigma_\varepsilon^2, \\ \mathbb{E}[(Y - \widehat{q}_1 - \kappa)^2] &= V_{\text{quad}} + \sigma_\varepsilon^2. \end{aligned} \tag{2.31}$$

We can now derive both R^2 values rather than assuming they have the same interpretation as the noiseless share.

Proposition 2.3 (The nested-regression ratio)**.** *Under Gaussian predictors, the quadratic target, and Equation (2.29), let R_1^2 and R_{1+2}^2 denote the population coefficients of determination of the affine and complete quadratic fits. If $V_{\mathrm{lin}} + V_{\mathrm{quad}} > 0$, then*

$$\begin{aligned} R_1^2 &= \frac{V_{\mathrm{lin}}}{V_{\mathrm{lin}} + V_{\mathrm{quad}} + \sigma_\varepsilon^2}, \\ R_{1+2}^2 &= \frac{V_{\mathrm{lin}} + V_{\mathrm{quad}}}{V_{\mathrm{lin}} + V_{\mathrm{quad}} + \sigma_\varepsilon^2}, \\ \frac{R_1^2}{R_{1+2}^2} &= \mathcal{A}_g. \end{aligned} \tag{2.32}$$

Proof. Subtract the appropriate mean squared residual from $\mathrm{Var}(Y)$ to obtain each explained variance. Both use the same total response variance. Dividing cancels that denominator, and the positive quadratic-target variance ensures that R_{1+2}^2 is not zero. □

For $b = 2$, $H = -0.5$, $G = 1$ and $\sigma_\varepsilon^2 = 4$, the components are 4 and 0.125. Thus

$$R_1^2 = \frac{4}{8.125} = \frac{32}{65}, \qquad R_{1+2}^2 = \frac{4.125}{8.125} = \frac{33}{65}, \qquad \frac{R_1^2}{R_{1+2}^2} = \frac{32}{33}.$$

Neither individual R^2 equals 32/33. If the quadratic target is constant but the noise varies, both R^2 values are zero and their ratio is undefined. An infinite residual variance also prevents this variance calculation.

Chapter 1, especially Proposition 1.3, specifies when a logged individual response can have the conditional mean in Equation (2.29). The required gap between $\mathbb{E}[\log W \mid \boldsymbol{A}]$ and $\log \mathbb{E}[W \mid \boldsymbol{A}]$ must be constant in breeding value, and the log error must have a finite second moment. Those are additional observation-model conditions. A general fitness response, particularly a count with zeros, does not satisfy them merely because its expected value is positive.

Biological interpretation. A ratio of two fit statistics can recover the share only if both fits target the same conditional relationship with the stated residual model. The algebra does not turn variation in realised reproduction into breeding-value variance, and it does not justify logging individual counts. Directly modelling expected output remains a separate route.

2.6 Singular covariance and the boundary cases

A covariance matrix can be singular when some recorded trait combinations do not vary. An inverse in all recorded coordinates would then be meaningless, but the variance calculation remains well defined. Let $r = \mathrm{rank}(\mathbf{G})$ and choose orthonormal columns of an $n \times r$ matrix $\mathbf{U}$ spanning its positive-eigenvalue subspace. Then

$$\mathbf{G} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\mathsf{T}, \qquad \mathbf{U}^\mathsf{T}\mathbf{U} = \mathbf{I}_r, \qquad \boldsymbol{\Lambda} \succ 0. \tag{2.33}$$

If $\boldsymbol{v}$ lies in the null space of $\mathbf{G}$, then $\mathrm{Var}(\boldsymbol{v}^\mathsf{T}\boldsymbol{X}) = 0$ and its mean is zero. Therefore $\boldsymbol{v}^\mathsf{T}\boldsymbol{X} = 0$ almost surely. It follows that $\boldsymbol{X} = \mathbf{U}\boldsymbol{Z}$ almost surely, where $\boldsymbol{Z} = \mathbf{U}^\mathsf{T}\boldsymbol{X} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Lambda})$. In these occupied coordinates,

$$\boldsymbol{b}_r = \mathbf{U}^\mathsf{T}\boldsymbol{b}, \qquad \mathbf{H}_r = \mathbf{U}^\mathsf{T}\mathbf{H}\mathbf{U}, \qquad q = \ell_0 + \boldsymbol{b}_r^\mathsf{T}\boldsymbol{Z} + \tfrac{1}{2}\boldsymbol{Z}^\mathsf{T}\mathbf{H}_r\boldsymbol{Z}. \tag{2.34}$$

All positive-definite derivations apply to $\boldsymbol{\Lambda}$. Substitution and cyclicity of trace recover the formulas in the original coordinates. The normal equation only requires $\mathbf{G}(\boldsymbol{d} - \boldsymbol{b}) = \mathbf{0}$: slope coefficients outside the occupied subspace are not uniquely identified, but fitted values are unique because those extra directions contribute zero almost surely.

The boundary distinctions are consequently precise:

- If $V_{\mathrm{lin}} = 0 < V_{\mathrm{quad}}$, the fitted response is constant and its R^2 is zero. The correlation in Equation (2.28) is undefined because its first argument has zero variance.
- If $V_{\mathrm{quad}} = 0 < V_{\mathrm{lin}}$, the quadratic target is affine on the occupied support and $\mathcal{A}_g = 1$.
- If both components are zero, the target is constant almost surely; the share and target-regression R^2 are undefined, not zero.

In particular, $V_{\mathrm{lin}} = 0$ means that the gradient has no component in a varying direction. It need not mean that every component of the gradient in an enlarged, partly unoccupied trait space is zero.

2.7 What changes outside a Gaussian population?

The conditioning and reproductive-weighting definitions do not require Gaussian breeding values. The moment formulas do. It is useful to derive the corrections explicitly so that a departure from normality has a mathematical meaning rather than serving as a general caveat.

Let $\boldsymbol{X}$ now be any centred vector with finite fourth moments. Retain $\mathbf{G} = \mathbb{E}[\boldsymbol{X}\boldsymbol{X}^{\mathsf{T}}]$, and define the third central moment tensor and fourth cumulant tensor by

$$\begin{aligned} M_{ijk} &= \mathbb{E}[X_i X_j X_k], \\ K_{ijkl} &= \mathbb{E}[X_i X_j X_k X_l] - G_{ij}G_{kl} - G_{ik}G_{jl} - G_{il}G_{jk}. \end{aligned} \tag{2.35}$$

The tensor K records the part of the fourth moment that the Gaussian pairings do not explain. The linear variance and quadratic mean still use only second moments, so their previous formulas remain valid. Repeating the covariance calculation, however, gives

$$\begin{aligned} \mathrm{Cov}(L, Q_c) &= \frac{1}{2}\sum_{i,j,k} b_i H_{jk} M_{ijk}, \\ \mathrm{Var}(Q_c) &= \frac{1}{2}\,\mathrm{tr}[(\mathbf{HG})^2] + \frac{1}{4}\sum_{i,j,k,l} H_{ij}H_{kl}K_{ijkl}. \end{aligned} \tag{2.36}$$

These are exact identities under the stated finite-moment assumption. The total variance includes twice the first line in addition to the linear and actual quadratic variances. Central symmetry makes $M_{ijk} = 0$ but need not make $K_{ijkl} = 0$.

The slope of the best affine fit can also change. Set

$$u_i = \mathrm{Cov}(X_i, Q_c) = \frac{1}{2}\sum_{j,k} H_{jk} M_{ijk}. \tag{2.37}$$

On a positive-definite occupied support,

$$\begin{aligned} \mathrm{Cov}(\boldsymbol{X}, q) &= \mathbf{G}\boldsymbol{b} + \boldsymbol{u}, \\ \boldsymbol{d} &= \boldsymbol{b} + \mathbf{G}^{-1}\boldsymbol{u}, \\ \mathrm{Var}(\boldsymbol{d}^{\mathsf{T}}\boldsymbol{X}) &= \boldsymbol{b}^{\mathsf{T}}\mathbf{G}\boldsymbol{b} + 2\boldsymbol{b}^{\mathsf{T}}\boldsymbol{u} + \boldsymbol{u}^{\mathsf{T}}\mathbf{G}^{-1}\boldsymbol{u}. \end{aligned} \tag{2.38}$$

Thus the variance of the local linear term and the variance explained by the best affine fit are different quantities in general. One can still calculate the actual population regression using the observed distribution; one cannot infer that its R^2 equals the Gaussian expression simply by substituting an observed covariance.

Example 2.4 (Symmetry preserves orthogonality but not the fourth moment)**.** Let $X = -1$ or 1 with equal probabilities, $b = 1$ and $H = -1$. Then $G = 1$, $\mathbb{E}[X^3] = 0$ and $\mathbb{E}[X^4] = 1$, so the fourth

cumulant is $1 - 3 = -2$. Since $X^2 = 1$ always, $Q_c = -(X^2 - 1)/2 = 0$. The quadratic variance is exactly zero. Equation (2.36) gives the same answer, $1/2 + (-2)/4 = 0$. The response is perfectly affine on these two points, with $R^2 = 1$, whereas substituting $G = 1$ into the Gaussian formula would give $1/(1 + 1/2) = 2/3$.

Example 2.5 (Asymmetry changes the fitted slope)**.** Let $X = -1$ with probability $2/3$ and $X = 2$ with probability $1/3$. Then $\mathbb{E}[X] = 0$, $G = 2$, $\mathbb{E}[X^3] = 2$, and $\mathbb{E}[X^4] = 6$. Choose $b = 1$ and $H = -1/2$, so $q = X - X^2/4$. The centred quadratic term is $Q_c = -(X^2 - 2)/4$. Direct averaging gives

$$\mathrm{Var}(L) = 2, \qquad \mathrm{Var}(Q_c) = \frac{1}{8}, \qquad \mathrm{Cov}(L, Q_c) = -\frac{1}{2}.$$

Consequently $\mathrm{Var}(q) = 2 + 1/8 - 1 = 9/8$. Here $u = H\mathbb{E}[X^3]/2 = -1/2$, and the fitted slope is $d = 1 + (-1/2)/2 = 3/4$. The intercept is $\mathbb{E}[q] = -1/2$. The fitted line $-1/2 + 3X/4$ equals q at both support points, so its R^2 is one. The Gaussian expression instead uses quadratic variance $H^2G^2/2 = 1/2$ and would report $2/(2 + 1/2) = 0.8$.

Biological interpretation. Changes in breeding-value frequencies can affect third and fourth moments as well as covariance. The Gaussian formulas isolate a tractable case, not a property of all populations with the same covariance. A finite multilocus simulation can assess the approximation in its specified design; it cannot prove Gaussianity or make higher moments irrelevant.

2.8 Connection to executable checks

The groups `gaussian_projection`, `noisy_regression_ratio`, and `boundaries_and_non_gaussian_cases` in `code/reproduce_result_checks.py` independently integrate or fit small specified populations. They check the displayed values and implementations. The general results above follow from their proofs, not from passing a finite set of examples. Sample correlations from a finite Gaussian sample can differ from the population identity because the sample's third and fourth moments are not exactly their population values.

Chapter 3

Covariance, coordinates and the geometry of the linear share

There are two distinct ways to change a calculation involving several traits. We can change the population while keeping the trait coordinates fixed, or we can describe the same population in different coordinates. The first can change the variance share. The second must leave it unchanged if every quantity is transformed consistently. This chapter proves both statements and explains the orientation comparisons used in the visual examples.

Throughout, the population is Gaussian and the target is the specified quadratic function q from Chapter 2. A comparison on a fixed surface also holds the population mean, gradient and Hessian fixed unless a change is explicitly stated. The results are exact for that quadratic function; applying them to a local expansion of a wider surface retains the remainder qualifications in Section 2.1.1.

3.1 Proportional change in covariance

Let $\mathbf{G}_0 \succeq 0$ be a reference covariance and define a family of Gaussian populations with the same mean and covariance $\mathbf{G}_s = s\mathbf{G}_0$, where $s > 0$. This changes every variance and covariance by the same factor. A standard deviation changes by $\sqrt{s}$; the parameter s is a variance multiplier, not a standard-deviation multiplier. Put

$$L_0 = \boldsymbol{b}^{\mathsf{T}}\mathbf{G}_0\boldsymbol{b}, \qquad Q_0 = \frac{1}{2}\operatorname{tr}[(\mathbf{H}\mathbf{G}_0)^2].$$

Proposition 3.1 (Proportional-covariance scaling)**.** *With the mean, $\boldsymbol{b}$ and $\mathbf{H}$ fixed, the two components under $\mathbf{G}_s = s\mathbf{G}_0$ are*

$$V_{\mathrm{lin}}(s) = sL_0, \qquad V_{\mathrm{quad}}(s) = s^2Q_0. \tag{3.1}$$

If $L_0 + Q_0 > 0$, then

$$\mathcal{A}_g(s) = \frac{L_0}{L_0 + sQ_0}. \tag{3.2}$$

If both L_0 and Q_0 are positive, the share decreases strictly with s, so proportional contraction increases it.

Proof. The linear expression contains one copy of the covariance: $\boldsymbol{b}^{\mathsf{T}}(s\mathbf{G}_0)\boldsymbol{b} = sL_0$. The quadratic expression contains two:

$$\frac{1}{2}\operatorname{tr}\{\mathbf{H}(s\mathbf{G}_0)\mathbf{H}(s\mathbf{G}_0)\} = s^2Q_0.$$

Cancelling the common positive factor s in the component ratio gives Equation (3.2). Differentiating yields

$$\frac{d\mathcal{A}_g}{ds} = -\frac{L_0Q_0}{(L_0 + sQ_0)^2}, \tag{3.3}$$

which is strictly negative precisely when both components are positive. □

The absolute quantities and their ratio therefore change at different rates. For example,

$$\begin{aligned} \frac{dV_{\text{lin}}}{ds} &= L_0, & \frac{dV_{\text{quad}}}{ds} &= 2sQ_0, \\ 1-\mathcal{A}_g(s) &= \frac{sQ_0}{L_0+sQ_0}, & \frac{d(1-\mathcal{A}_g)}{ds} &= \frac{L_0Q_0}{(L_0+sQ_0)^2}. \end{aligned} \tag{3.4}$$

The omitted fraction does not scale as s^2, although the absolute quadratic variance does. For $L_0 > 0$, its small-s behaviour is $1-\mathcal{A}_g(s) = sQ_0/L_0 + O(s^2)$.

If $L_0, Q_0 > 0$, the mathematical limits are

$$\lim_{s\downarrow 0} \mathcal{A}_g(s) = 1, \qquad \lim_{s\to\infty} \mathcal{A}_g(s) = 0.$$

At $s = 0$ itself, however, both variances vanish and the share is undefined. The first limit is not the share of a population with no variation. If $L_0 = 0 < Q_0$, the share remains zero for every positive s; if $Q_0 = 0 < L_0$, it remains one. Thus a narrow population at a stationary point does not acquire a linear contribution merely by becoming narrower. Large-s limits describe the specified quadratic family, not an assurance that a local expansion remains accurate over arbitrarily broad populations.

Biological interpretation. For a fixed local relationship and mean, narrowing the breeding-value distribution can improve the fraction represented by a line while reducing the amount of linearly described variation. There is no contradiction: the numerator becomes smaller, but the curved contribution becomes smaller faster. An actual comparison among populations must also consider changes in the mean, conditional expected-fitness surface and population-specific average effects. Holding these inputs fixed is how this calculation isolates the frequency-distribution effect.

3.1.1 Why an arbitrary contraction is different

The condition $\mathbf{G}_1 \preceq \mathbf{G}_0$ says that variance has not increased in any linear trait combination. It does not require all directions to contract by the same fraction. Consider

$$\boldsymbol{b} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \mathbf{H} = -\mathbf{I}_2, \qquad \mathbf{G}_0 = \mathbf{I}_2, \qquad \mathbf{G}_1 = \begin{pmatrix} 0.1 & 0 \\ 0 & 0.9 \end{pmatrix}.$$

The difference $\mathbf{G}_0 - \mathbf{G}_1 = \operatorname{diag}(0.9, 0.1)$ is positive definite. The initial components are

$$V_{\text{lin},0} = 1, \qquad V_{\text{quad},0} = \tfrac{1}{2}(1^2 + 1^2) = 1, \qquad \mathcal{A}_{g,0} = \tfrac{1}{2}.$$

After the contraction,

$$V_{\text{lin},1} = 0.1, \qquad V_{\text{quad},1} = \tfrac{1}{2}(0.1^2 + 0.9^2) = 0.41, \qquad \mathcal{A}_{g,1} = \frac{0.1}{0.51} \simeq 0.1961.$$

Both absolute components decreased, yet the share decreased too. Most of the contraction was in the direction supplying the linear variation; substantial variance remained in a direction supplying only curvature. This example prevents extending Equation (3.3) to every covariance change described informally as a reduction in variation.

3.2 An invertible change of trait coordinates

Let $\boldsymbol{X}' = \mathbf{D}\boldsymbol{X}$, where $\mathbf{D}$ is invertible. This is a new description of the same individuals, not a new population. The covariance becomes

$$\mathbf{G}' = \mathbb{E}[\boldsymbol{X}'\boldsymbol{X}'^{\mathsf{T}}] = \mathbf{D}\mathbf{G}\mathbf{D}^{\mathsf{T}}. \tag{3.5}$$

The scalar target cannot change when its argument is relabelled. Since $\boldsymbol{X} = \mathbf{D}^{-1}\boldsymbol{X}'$, its expression is

$$q = \ell_0 + \boldsymbol{b}^{\mathsf{T}}\mathbf{D}^{-1}\boldsymbol{X}' + \frac{1}{2}(\boldsymbol{X}')^{\mathsf{T}}\mathbf{D}^{-\mathsf{T}}\mathbf{H}\mathbf{D}^{-1}\boldsymbol{X}'. \tag{3.6}$$

Reading off its coefficients gives

$$\boldsymbol{b}' = \mathbf{D}^{-\mathsf{T}}\boldsymbol{b}, \qquad \mathbf{H}' = \mathbf{D}^{-\mathsf{T}}\mathbf{H}\mathbf{D}^{-1}, \tag{3.7}$$

where $\mathbf{D}^{-\mathsf{T}} = (\mathbf{D}^{-1})^{\mathsf{T}}$. Transforming a covariance while leaving the derivative coefficients in their old units would describe a different calculation.

Proposition 3.2 (Coordinate invariance)**.** *Under the simultaneous transformations above, the linear variance, quadratic variance, quadratic mean contribution and defined share are unchanged.*

Proof. For the linear variance, adjacent inverse factors cancel:

$$\begin{aligned}(\boldsymbol{b}')^{\mathsf{T}}\mathbf{G}'\boldsymbol{b}' &= \boldsymbol{b}^{\mathsf{T}}\mathbf{D}^{-1}(\mathbf{D}\mathbf{G}\mathbf{D}^{\mathsf{T}})\mathbf{D}^{-\mathsf{T}}\boldsymbol{b} \\ &= \boldsymbol{b}^{\mathsf{T}}\mathbf{G}\boldsymbol{b}.\end{aligned} \tag{3.8}$$

For the quadratic term,

$$\begin{aligned}\mathbf{H}'\mathbf{G}' &= \mathbf{D}^{-\mathsf{T}}\mathbf{H}\mathbf{G}\mathbf{D}^{\mathsf{T}}, \\ (\mathbf{H}'\mathbf{G}')^2 &= \mathbf{D}^{-\mathsf{T}}\mathbf{H}\mathbf{G}(\mathbf{D}^{\mathsf{T}}\mathbf{D}^{-\mathsf{T}})\mathbf{H}\mathbf{G}\mathbf{D}^{\mathsf{T}} \\ &= \mathbf{D}^{-\mathsf{T}}(\mathbf{H}\mathbf{G})^2\mathbf{D}^{\mathsf{T}}.\end{aligned} \tag{3.9}$$

For compatible matrices, the trace identity

$$\operatorname{tr}(\mathbf{A}\mathbf{B}) = \sum_{i,j} A_{ij}B_{ji} = \operatorname{tr}(\mathbf{B}\mathbf{A})$$

allows a cyclic rotation of factors inside a trace. Applying it gives

$$\begin{aligned}\operatorname{tr}[(\mathbf{H}'\mathbf{G}')^2] &= \operatorname{tr}\{(\mathbf{H}\mathbf{G})^2\mathbf{D}^{\mathsf{T}}\mathbf{D}^{-\mathsf{T}}\} \\ &= \operatorname{tr}[(\mathbf{H}\mathbf{G})^2].\end{aligned} \tag{3.10}$$

The same argument with one copy of $\mathbf{H}'\mathbf{G}'$ proves invariance of $\operatorname{tr}(\mathbf{H}\mathbf{G})$. A cyclic rotation is not permission to rearrange factors in an arbitrary order: matrices need not commute. The unchanged components give an unchanged ratio whenever its denominator is positive. □

For a one-trait unit conversion $X' = 10X$, the transformed quantities are $G' = 100G$, $b' = b/10$ and $H' = H/100$. Taking $G = 1$, $b = 2$ and $H = -0.5$ gives $G' = 100$, $b' = 0.2$ and $H' = -0.005$, and hence

$$(0.2)^2(100) = 4, \qquad \tfrac{1}{2}(-0.005)^2(100)^2 = 0.125.$$

The numerical coefficients change, while the variances in log expected output do not. The proof also applies to singular covariance matrices; the coordinate transformation preserves their rank and maps the occupied support to the same population expressed differently.

3.3 Whitening and the curvature spectrum

For $\mathbf{G} \succ 0$, its spectral decomposition can be written $\mathbf{G} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\mathsf{T}$ with orthogonal $\mathbf{U}$ and positive diagonal $\mathbf{\Lambda}$. Define the symmetric square root and its inverse by

$$\mathbf{G}^{1/2} = \mathbf{U}\mathbf{\Lambda}^{1/2}\mathbf{U}^\mathsf{T}, \qquad \mathbf{G}^{-1/2} = \mathbf{U}\mathbf{\Lambda}^{-1/2}\mathbf{U}^\mathsf{T}.$$

The rescaled variable $\boldsymbol{Z} = \mathbf{G}^{-1/2}\boldsymbol{X}$ has covariance $\mathbf{I}$. For Gaussian $\boldsymbol{X}$, its density is proportional to $\exp(-\sum_i z_i^2/2)$, a product of standard normal densities. Thus the coordinates of $\boldsymbol{Z}$ are independent standard normals. This covariance-removing transformation is called whitening.

Substitution $\boldsymbol{X} = \mathbf{G}^{1/2}\boldsymbol{Z}$ gives

$$q = \ell_0 + \boldsymbol{c}^\mathsf{T}\boldsymbol{Z} + \frac{1}{2}\boldsymbol{Z}^\mathsf{T}\mathbf{S}\boldsymbol{Z}, \qquad \boldsymbol{c} = \mathbf{G}^{1/2}\boldsymbol{b}, \qquad \mathbf{S} = \mathbf{G}^{1/2}\mathbf{H}\mathbf{G}^{1/2}. \tag{3.11}$$

The symmetric matrix $\mathbf{S}$ has real eigenvalues μ_i. Write $\mathbf{S} = \mathbf{O}\,\mathrm{diag}(\mu_i)\mathbf{O}^\mathsf{T}$ and $\boldsymbol{Y} = \mathbf{O}^\mathsf{T}\boldsymbol{Z}$. Orthogonal rotation preserves a standard normal density, so the Y_i are again independent standard normals. In these coordinates,

$$L = (\mathbf{O}^\mathsf{T}\boldsymbol{c})^\mathsf{T}\boldsymbol{Y}, \qquad Q_c = \frac{1}{2}\sum_i \mu_i(Y_i^2 - 1). \tag{3.12}$$

Since $\mathrm{Var}(Y_i) = 1$, $\mathrm{Var}(Y_i^2) = 3 - 1 = 2$, and distinct squared coordinates are independent, these forms give

$$V_{\mathrm{lin}} = \|\boldsymbol{c}\|^2, \qquad V_{\mathrm{quad}} = \frac{1}{2}\sum_i \mu_i^2. \tag{3.13}$$

This also supplies a direct explanation of nonnegativity. Although $\mathbf{HG}$ need not be symmetric,

$$\begin{aligned} \mathrm{tr}[(\mathbf{HG})^2] &= \mathrm{tr}[(\mathbf{G}^{1/2}\mathbf{H}\mathbf{G}^{1/2})^2] \\ &= \mathrm{tr}(\mathbf{S}^2) = \sum_i \mu_i^2 \geq 0. \end{aligned} \tag{3.14}$$

For positive-definite $\mathbf{G}$, $\mathbf{HG} = \mathbf{G}^{-1/2}\mathbf{S}\mathbf{G}^{1/2}$, so it is similar to $\mathbf{S}$. For singular $\mathbf{G}$, the whitening and similarity statements are made on the occupied support; the trace identity itself still holds in the full coordinates.

If $\|\boldsymbol{c}\|$ and the eigenvalues μ_i are fixed, rotating $\boldsymbol{c}$ relative to the eigenvectors of $\mathbf{S}$ cannot change the share: neither scalar on the right of Equation (3.13) changes. The relative orientation can affect selected-parent displacement, however, because that calculation uses an inverse matrix acting on the gradient; see Section 4.5.1.

Biological interpretation. The number of named traits does not by itself determine how linear the population description is. What matters here is the gradient and curvature after accounting for the amount and association of variation in the population. Adding a trait coordinate that never varies adds no variance contribution. Relabelling the same varying coordinates cannot change the share either.

3.4 Physical orientation and the Rayleigh quotient

Now keep the trait axes fixed and change a biological input. For a unit direction $\boldsymbol{u}$, the variance of the breeding-value combination $\boldsymbol{u}^\mathsf{T}\boldsymbol{X}$ is $\boldsymbol{u}^\mathsf{T}\mathbf{G}\boldsymbol{u}$. It is a directional evolvability measure in the specified trait units (Hansen and Houle, 2008). Write a gradient of fixed Euclidean length as

$\boldsymbol{b} = a\boldsymbol{u}$, with $a > 0$. If $\mathbf{G}$ has eigenvectors $\boldsymbol{v}_i$ and eigenvalues λ_i, expand $\boldsymbol{u} = \sum_i z_i \boldsymbol{v}_i$, where $\sum_i z_i^2 = 1$. Then

$$V_{\text{lin}} = a^2 \boldsymbol{u}^{\mathsf{T}} \mathbf{G} \boldsymbol{u} = a^2 \sum_i \lambda_i z_i^2. \tag{3.15}$$

The weighted mean of the λ_i lies between their minimum and maximum. Hence

$$a^2 \lambda_{\min}(\mathbf{G}) \le V_{\text{lin}} \le a^2 \lambda_{\max}(\mathbf{G}). \tag{3.16}$$

The upper equality holds when the gradient lies entirely in the largest-eigenvalue eigenspace; if that eigenvalue is simple, this means the corresponding eigenvector, with either sign. The lower equality is analogous. This is the Rayleigh-quotient argument, including its equality conditions. It fixes the gradient's ordinary length, not the length of the whitened gradient.

Curvature's orientation relative to covariance can also matter. In a two-trait coordinate system where $\mathbf{G} = \operatorname{diag}(g_1, g_2)$, matrix multiplication gives

$$V_{\text{quad}} = \frac{1}{2}\{H_{11}^2 g_1^2 + 2H_{12}^2 g_1 g_2 + H_{22}^2 g_2^2\}. \tag{3.17}$$

Rotating a fixed set of curvature eigenvalues relative to these axes changes the entries H_{ij} and can change this variance. If instead $\mathbf{H} = h\mathbf{I}$, then $V_{\text{quad}} = h^2 \operatorname{tr}(\mathbf{G}^2)/2$, which depends only on the covariance eigenvalues. A rotation of $\mathbf{G}$ with those eigenvalues fixed leaves the quadratic variance unchanged.

This last case explains the two-trait rotation figure. Choose axes along and perpendicular to the gradient, with gradient length a. Let the population's larger-variance eigenvector make angle ϕ with the gradient, and let its principal variances be g_1 and g_2. Resolving the unit gradient direction along the two covariance eigenvectors gives

$$V_{\text{lin}} = a^2(g_1 \cos^2\phi + g_2 \sin^2\phi), \qquad V_{\text{quad}} = \frac{h^2}{2}(g_1^2 + g_2^2) \quad \text{when } \mathbf{H} = h\mathbf{I}. \tag{3.18}$$

At $\phi = 90°$, the linear variance is $a^2 g_2$, not zero unless the smaller principal variance is zero. A section along the long axis can therefore have zero slope while the complete fitted plane still varies in the perpendicular direction.

These physical comparisons do not contradict the fixed-spectrum statement in Section 3.3. Changing the relative position of $\mathbf{G}$, $\boldsymbol{b}$ and $\mathbf{H}$ in fixed trait coordinates need not preserve $\left\|\mathbf{G}^{1/2}\boldsymbol{b}\right\|$ or the eigenvalues of $\mathbf{G}^{1/2}\mathbf{H}\mathbf{G}^{1/2}$. Each comparison must state what stays fixed before drawing an orientation conclusion.

3.5 The regression triangle is in a space of functions

The geometry of a population regression is different from the physical geometry of trait axes. For centred square-integrable functions U and V of breeding value, define an inner product and length by

$$\langle U, V \rangle = \mathbb{E}[UV], \qquad \|U\|_{L^2} = \sqrt{\mathbb{E}[U^2]}.$$

Functions that agree almost surely are the same element for this calculation. The population distribution enters the inner product: changing its frequencies changes which differences matter to the fit. For the Gaussian quadratic target,

$$q - \mathbb{E}[q] = L + Q_c, \qquad \langle L, Q_c \rangle = 0.$$

Expanding the square proves the right-triangle identity

$$\|q - \mathbb{E}[q]\|_{L^2}^2 = \|L\|_{L^2}^2 + \|Q_c\|_{L^2}^2 = V_{\text{lin}} + V_{\text{quad}}. \tag{3.19}$$

When $V_{\mathrm{lin}} > 0$, define the angle ψ between the centred target and its linear projection through their normalised inner product. Then

$$\begin{aligned}\cos\psi &= \frac{\langle L, L + Q_c\rangle}{\|L\|_{L^2}\,\|L + Q_c\|_{L^2}} \\ &= \frac{V_{\mathrm{lin}}}{\sqrt{V_{\mathrm{lin}}}\sqrt{V_{\mathrm{lin}} + V_{\mathrm{quad}}}}, \qquad \cos^2\psi = \mathcal{A}_g. \end{aligned} \tag{3.20}$$

The triangle's side lengths are standard deviations, not the component variances themselves. It represents the relation among functions over a population, not an angle between two trait vectors or a direction of evolution. If $V_{\mathrm{lin}} = 0 < V_{\mathrm{quad}}$, the ordinary angle to a zero projection is undefined; a plotted 90° angle is only a limiting convention. If both variances vanish, no such triangle is defined.

Biological interpretation. A large share states that the linear fit's mean squared residual is small relative to the quadratic target's population variance. Neither a small function-space angle nor a thin breeding-value ellipse specifies the descendants of the population. Selection changes the weighting of the parental distribution, and inheritance requires information beyond either geometric picture.

3.6 Connection to executable checks

`code/verify_mathematics.py` checks coordinate transformations, proportional scaling and the explicit anisotropic counterexample. The `orientation_and_absolute_scale` group in `code/reproduce_result_checks.py` checks fixed whitened norms separately from a physical-coordinate Rayleigh comparison. `tutorials/geometry/code/geometry.py` integrates the specified Gaussian populations to check the visual examples. These finite computations illustrate the distinctions established in the derivations; they do not substitute for the invariance or scaling proofs.

Chapter 4

Reproductive weighting and the selected-parent distribution

A population fit asks how well a function describes individuals under their existing frequencies. Selection asks how those frequencies change when weighted by expected reproductive output. This chapter performs that second calculation exactly for a Gaussian population and a log-quadratic conditional mean. It then derives a separate error bound for ignoring curvature. No inheritance rule is used: the distribution obtained here describes parents weighted by their reproductive contributions.

4.1 The exact model and its integrability condition

Let $\boldsymbol{X} = \boldsymbol{A} - \boldsymbol{\mu} \sim \mathcal{N}(\mathbf{0}, \mathbf{G})$ with $\mathbf{G} \succ 0$. The singular case will be treated on the occupied support. Assume in this chapter that conditional expected output is exactly

$$W_A(\boldsymbol{\mu} + \boldsymbol{x}) = \exp\left(\ell_0 + \boldsymbol{b}^\mathsf{T}\boldsymbol{x} - \frac{1}{2}\boldsymbol{x}^\mathsf{T}\boldsymbol{\Gamma}\boldsymbol{x}\right), \qquad \boldsymbol{\Gamma} = -\mathbf{H} = \boldsymbol{\Gamma}^\mathsf{T}. \tag{4.1}$$

This assumption is stronger than having a quadratic Taylor approximation near the mean. It allows the selection integrals to use this function over the whole occupied support. The matrix $\boldsymbol{\Gamma}$ need not initially be positive semidefinite. Positive semidefiniteness will be imposed when deriving covariance contraction under stabilising curvature.

As derived in Proposition 1.1, the reproductively weighted density is

$$p_{\mathrm{sel}}(\boldsymbol{x}) = \frac{p(\boldsymbol{x})W_A(\boldsymbol{\mu} + \boldsymbol{x})}{\overline{W}}, \qquad \overline{W} = \int_{\mathbb{R}^n} p(\boldsymbol{x})W_A(\boldsymbol{\mu} + \boldsymbol{x})\, d\boldsymbol{x}. \tag{4.2}$$

A probability density requires $0 < \overline{W} < \infty$. Positivity of the fitness function does not by itself ensure finiteness.

Multiplying the normal density by Equation (4.1) gives

$$p(\boldsymbol{x})W_A(\boldsymbol{\mu} + \boldsymbol{x}) = \frac{e^{\ell_0}}{(2\pi)^{n/2}\det(\mathbf{G})^{1/2}} \exp\left(-\frac{1}{2}\boldsymbol{x}^\mathsf{T}\mathbf{J}\boldsymbol{x} + \boldsymbol{b}^\mathsf{T}\boldsymbol{x}\right), \qquad \mathbf{J} = \mathbf{G}^{-1} + \boldsymbol{\Gamma}. \tag{4.3}$$

The matrix $\mathbf{J}$ is the combined precision: it collects the quadratic coefficient from population frequency and the quadratic coefficient from expected output. "Precision" here means inverse covariance in a normal density. Its sign determines whether the product can be normalised.

Proposition 4.1 (Integrability of the Gaussian quadratic weight)**.** *For $\mathbf{G} \succ 0$, finite ℓ_0 and $\boldsymbol{b}$, and symmetric $\boldsymbol{\Gamma}$, the normalising integral in Equation (4.2) is finite if and only if $\mathbf{J} = \mathbf{G}^{-1} + \boldsymbol{\Gamma} \succ 0$.*

Proof. Diagonalise the symmetric matrix as $\mathbf{J} = \mathbf{O}\,\mathrm{diag}(\lambda_i)\mathbf{O}^\mathsf{T}$, where $\mathbf{O}$ is orthogonal. Set $\boldsymbol{y} = \mathbf{O}^\mathsf{T}\boldsymbol{x}$ and $\boldsymbol{d} = \mathbf{O}^\mathsf{T}\boldsymbol{b}$. The Jacobian has absolute determinant one, and the nonconstant exponential becomes

$$\prod_{i=1}^{n} \exp(-\lambda_i y_i^2/2 + d_i y_i).$$

Because it is nonnegative, integration can be performed one coordinate at a time. If every $\lambda_i > 0$, completing each scalar square gives a finite Gaussian integral. If some $\lambda_i < 0$, that coordinate has

a positive quadratic exponent and its integral diverges. If $\lambda_i = 0$, the integral is $\int_{\mathbb{R}} e^{d_i y_i}\, dy_i$; it diverges whether $d_i = 0$ or not. Thus strict positivity in every occupied direction is necessary as well as sufficient. □

In particular, a zero eigenvalue at the boundary cannot be rescued by putting a zero gradient in that direction. The integrand would then be constant along an unbounded direction, which still has infinite integral.

Biological interpretation. Population frequencies can make a locally increasing expected-output function integrable even if the fitness curvature is not stabilising in every direction. Conversely, an assumed output function that increases too fast relative to the Gaussian frequency decline gives infinite population mean output. Such a model cannot define the selected group by the proposed normalisation.

4.2 Complete the square, including the normalising constant

Assume henceforth that $\mathbf{J} \succ 0$. Define

$$\mathbf{V} = \mathbf{J}^{-1}, \qquad \boldsymbol{m} = \mathbf{V}\boldsymbol{b}. \tag{4.4}$$

Since $\mathbf{J}\boldsymbol{m} = \boldsymbol{b}$, expansion gives

$$\begin{aligned}(\boldsymbol{x}-\boldsymbol{m})^{\mathsf{T}}\mathbf{J}(\boldsymbol{x}-\boldsymbol{m}) &= \boldsymbol{x}^{\mathsf{T}}\mathbf{J}\boldsymbol{x} - \boldsymbol{x}^{\mathsf{T}}\mathbf{J}\boldsymbol{m} - \boldsymbol{m}^{\mathsf{T}}\mathbf{J}\boldsymbol{x} + \boldsymbol{m}^{\mathsf{T}}\mathbf{J}\boldsymbol{m} \\ &= \boldsymbol{x}^{\mathsf{T}}\mathbf{J}\boldsymbol{x} - 2\boldsymbol{b}^{\mathsf{T}}\boldsymbol{x} + \boldsymbol{b}^{\mathsf{T}}\mathbf{V}\boldsymbol{b}. \end{aligned} \tag{4.5}$$

Therefore

$$-\frac{1}{2}\boldsymbol{x}^{\mathsf{T}}\mathbf{J}\boldsymbol{x} + \boldsymbol{b}^{\mathsf{T}}\boldsymbol{x} = -\frac{1}{2}(\boldsymbol{x}-\boldsymbol{m})^{\mathsf{T}}\mathbf{J}(\boldsymbol{x}-\boldsymbol{m}) + \frac{1}{2}\boldsymbol{b}^{\mathsf{T}}\mathbf{V}\boldsymbol{b}. \tag{4.6}$$

The two cross terms combine because $\mathbf{J}$ is symmetric; no commutation of $\mathbf{G}$ and $\boldsymbol{\Gamma}$ has been assumed.

To integrate the remaining centred quadratic, diagonalise $\mathbf{J}$ as in the preceding proof. Translation by $\boldsymbol{m}$ and orthogonal rotation leave the integration element unchanged. The product of one-dimensional Gaussian integrals is

$$\begin{aligned}\int_{\mathbb{R}^n} e^{-(\boldsymbol{x}-\boldsymbol{m})^{\mathsf{T}}\mathbf{J}(\boldsymbol{x}-\boldsymbol{m})/2}\, d\boldsymbol{x} &= \prod_i \sqrt{\frac{2\pi}{\lambda_i}} \\ &= (2\pi)^{n/2}\det(\mathbf{J})^{-1/2}. \end{aligned} \tag{4.7}$$

Substituting this into Equation (4.3) gives the exact population mean output:

$$\overline{W} = e^{\ell_0}\det(\mathbf{G})^{-1/2}\det(\mathbf{J})^{-1/2}\exp\left(\frac{1}{2}\boldsymbol{b}^{\mathsf{T}}\mathbf{J}^{-1}\boldsymbol{b}\right). \tag{4.8}$$

Dividing by this constant cancels every term independent of $\boldsymbol{x}$. The selected density is

$$p_{\mathrm{sel}}(\boldsymbol{x}) = \frac{\det(\mathbf{J})^{1/2}}{(2\pi)^{n/2}}\exp\left[-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{m})^{\mathsf{T}}\mathbf{J}(\boldsymbol{x}-\boldsymbol{m})\right]. \tag{4.9}$$

This establishes the distribution, not only its first two moments.

Proposition 4.2 (Selected-parent Gaussian moments)**.** *Under the exact Gaussian/log-quadratic model and positive combined precision,*

$$\begin{aligned}\mathbf{G}_{\mathrm{sel}} &= (\mathbf{G}^{-1} + \boldsymbol{\Gamma})^{-1}, \\ \boldsymbol{\Delta}_{\mathrm{full}} &= \mathbb{E}[\boldsymbol{A} \mid \mathrm{sel}] - \boldsymbol{\mu} = \mathbf{G}_{\mathrm{sel}}\boldsymbol{b}. \end{aligned} \tag{4.10}$$

The selected breeding-value distribution is $\mathcal{N}(\boldsymbol{\mu} + \boldsymbol{\Delta}_{\mathrm{full}}, \mathbf{G}_{\mathrm{sel}})$.

Proof. Equation (4.9) is the normal density with mean $\boldsymbol{m}$ and covariance $\mathbf{J}^{-1}$. Returning from centred $\boldsymbol{X}$ to $\boldsymbol{A} = \boldsymbol{\mu} + \boldsymbol{X}$ adds $\boldsymbol{\mu}$ to the mean and leaves the covariance unchanged. □

The covariance is independent of $\boldsymbol{b}$ within this exact family, while the selected mean is not. This does not establish slope independence of selected covariance for an arbitrary fitness surface or a non-Gaussian population. Those cases need their own weighted-moment calculation.

4.2.1 A second route through derivatives of the normaliser

With $\mathbf{G}$, $\boldsymbol{\Gamma}$ and ℓ_0 fixed, regard $\overline{W}$ as a function of the coefficient vector $\boldsymbol{b}$. Differentiation under the integral brings down a factor X_i, just as in the moment-generating function calculation. Positive combined precision keeps these polynomially weighted integrals finite. Hence

$$\begin{aligned}\frac{\partial \log \overline{W}}{\partial b_i} &= \frac{\mathbb{E}[X_i W_A]}{\mathbb{E}[W_A]} = \mathbb{E}[X_i \mid \mathrm{sel}],\\ \frac{\partial^2 \log \overline{W}}{\partial b_i \partial b_j} &= \mathbb{E}[X_i X_j \mid \mathrm{sel}] - \mathbb{E}[X_i \mid \mathrm{sel}]\mathbb{E}[X_j \mid \mathrm{sel}].\end{aligned} \tag{4.11}$$

These are derivatives with respect to a model coefficient, not spatial derivatives with respect to breeding value. From Equation (4.8), the only $\boldsymbol{b}$-dependent term in $\log \overline{W}$ is $\boldsymbol{b}^{\mathsf{T}}\mathbf{J}^{-1}\boldsymbol{b}/2$. Its gradient is $\mathbf{J}^{-1}\boldsymbol{b}$ and its Hessian is $\mathbf{J}^{-1}$, recovering both selected moments independently from the normalising expression.

4.3 Stabilising curvature, saddle curvature and occupied support

Define the symmetric covariance-weighted curvature

$$\mathbf{T} = \mathbf{G}^{1/2}\boldsymbol{\Gamma}\mathbf{G}^{1/2}. \tag{4.12}$$

Multiplication shows

$$\begin{aligned}\mathbf{J} &= \mathbf{G}^{-1/2}(\mathbf{I} + \mathbf{T})\mathbf{G}^{-1/2},\\ \mathbf{G}_{\mathrm{sel}} &= \mathbf{G}^{1/2}(\mathbf{I} + \mathbf{T})^{-1}\mathbf{G}^{1/2}.\end{aligned} \tag{4.13}$$

The inverse in the second line reverses the order of the factors in the first. This factorisation works even when $\mathbf{G}$ and $\boldsymbol{\Gamma}$ do not commute. Congruence by an invertible matrix preserves positive definiteness, so integrability is equivalent to $\mathbf{I} + \mathbf{T} \succ 0$. If τ_i are the eigenvalues of $\mathbf{T}$, the condition is $\tau_i > -1$ for every occupied direction.

The normaliser can consequently also be written as

$$\overline{W} = e^{\ell_0} \det(\mathbf{I} + \mathbf{T})^{-1/2} \exp\left(\frac{1}{2}\boldsymbol{b}^{\mathsf{T}}\mathbf{G}_{\mathrm{sel}}\boldsymbol{b}\right). \tag{4.14}$$

To see this determinant identity explicitly, $\det(\mathbf{J}) = \det(\mathbf{G})^{-1}\det(\mathbf{I} + \mathbf{T})$ in Equation (4.13); its covariance determinant then cancels the one in Equation (4.8).

Corollary 4.3 (Covariance contraction under stabilising curvature)**.** *If* $\boldsymbol{\Gamma} \succeq 0$*, then* $\mathbf{0} \prec \mathbf{G}_{\mathrm{sel}} \preceq \mathbf{G}$ *for* $\mathbf{G} \succ 0$*.*

Proof. Positive semidefiniteness of $\boldsymbol{\Gamma}$ implies $\mathbf{T} \succeq 0$. The eigenvalues of $(\mathbf{I} + \mathbf{T})^{-1}$ are then $1/(1 + \tau_i)$, all positive and no greater than one. Therefore

$$\mathbf{0} \prec (\mathbf{I} + \mathbf{T})^{-1} \preceq \mathbf{I}.$$

For any vector $\boldsymbol{v}$, premultiplication and postmultiplication by $\mathbf{G}^{1/2}$ preserve the corresponding nonnegative quadratic forms. Equation (4.13) gives the stated covariance order. □

This order says that variance in every fixed linear combination of traits does not increase. It does not require every individual covariance entry to decrease numerically.

An indefinite $\mathbf{\Gamma}$ can still give a valid selected Gaussian. For example, take $\mathbf{G} = \mathbf{I}_2$ and $\mathbf{\Gamma} = \mathrm{diag}(0.4, -0.2)$. The combined precision is $\mathrm{diag}(1.4, 0.8) \succ 0$, giving selected variances $5/7$ and $5/4$. One direction contracts and the other expands. With $\boldsymbol{b} = (0.6, -0.4)^\mathsf{T}$, the mean shift is $(3/7, -1/2)^\mathsf{T}$. Replacing the second curvature by -1 gives a zero precision direction and a divergent normaliser, even if that direction's gradient is also zero.

For singular $\mathbf{G}$, use the support construction in Section 2.6: $\boldsymbol{X} = \mathbf{U}\boldsymbol{Z}$, with positive-definite covariance $\mathbf{\Lambda}$ for $\boldsymbol{Z}$. Replace $\boldsymbol{b}$ by $\mathbf{U}^\mathsf{T}\boldsymbol{b}$ and $\mathbf{\Gamma}$ by $\mathbf{U}^\mathsf{T}\mathbf{\Gamma}\mathbf{U}$ in every integral and inverse. The normalising density is then r-dimensional, where $r = \mathrm{rank}(\mathbf{G})$, and selected moments are mapped back by $\mathbf{U}$. A gradient or Hessian component outside the occupied support cannot affect this weighted distribution. Reweighting parents also cannot create variation outside their existing support.

Biological interpretation. The selected covariance describes variation among contributing parents. The stabilising-curvature result is a within-episode statement. It does not specify covariance after recombination, segregation, mutation or other processes of inheritance, and it is not a general claim that population covariance must decline across generations.

4.4 Ignoring curvature changes a different calculation

If curvature is omitted, the reproductive weight is proportional to $\exp(\boldsymbol{b}^\mathsf{T}\boldsymbol{x})$, and the same square-completion calculation with $\mathbf{\Gamma} = \mathbf{0}$ gives

$$\mathbf{\Delta}_{\mathrm{lin}} = \mathbf{G}\boldsymbol{b}, \qquad \mathbf{G}_{\mathrm{sel,lin}} = \mathbf{G}. \tag{4.15}$$

Using the population-fitted affine log function instead of the tangent adds the intercept $\mathrm{tr}(\mathbf{HG})/2$. Its exponential is a common positive factor and cancels from the selection normalisation. Thus both choices give the same first-order parental mean shift here, despite their different quality as fits on the log scale.

For one trait with initial variance $g > 0$, curvature $H = -\gamma$, and $1 + \gamma g > 0$, the exact formulas reduce to

$$g_{\mathrm{sel}} = \frac{g}{1+\gamma g}, \qquad \Delta_{\mathrm{full}} = \frac{gb}{1+\gamma g}, \qquad \Delta_{\mathrm{lin}} = gb. \tag{4.16}$$

If $b \neq 0$, subtraction gives

$$\Delta_{\mathrm{lin}} - \Delta_{\mathrm{full}} = \gamma g\, \Delta_{\mathrm{full}}, \qquad \frac{|\Delta_{\mathrm{lin}} - \Delta_{\mathrm{full}}|}{|\Delta_{\mathrm{full}}|} = |\gamma g|. \tag{4.17}$$

For positive stabilising curvature, ignoring curvature overestimates the magnitude of this scalar shift. For an integrable negative γ, it underestimates it. In several traits the two shift vectors need not be parallel, so a single coordinatewise percentage is not the general result.

Example 4.4 (A share of 32/33 and a displacement error of 1/2)**.** Take $b = 2$, $H = -0.5$ and $g = 1$. The variance components and their share are

$$V_{\mathrm{lin}} = 4, \qquad V_{\mathrm{quad}} = \tfrac{1}{2}(0.5)^2 = 0.125, \qquad \mathcal{A}_g = \frac{4}{4.125} = \frac{32}{33} \simeq 0.9697.$$

The exact selected variance is $1/(1+0.5) = 2/3$, and the exact shift is $2/(1+0.5) = 4/3$. Ignoring curvature gives a shift of 2, so its error relative to the exact shift is

$$\frac{2 - 4/3}{4/3} = \frac{2/3}{4/3} = \frac{1}{2}.$$

The linear fit describes about 97% of variation in the quadratic log target, yet the resulting reproductive weighting overestimates the selected-parent displacement by 50%.

For the same $g = 1$ and $\gamma = 0.5$, raising b from 2 to 10 raises V_{lin} from 4 to 100 while leaving $V_{\text{quad}} = 0.125$. The share becomes $100/100.125 \simeq 0.9988$. The shifts are now 10 and $10/1.5$, but their relative discrepancy remains $1/2$. Increasing the slope's magnitude is therefore not an accuracy guarantee.

Nor does high population R^2 guarantee agreement in every ranking. In the first example, omit the irrelevant constant and write $q(x) = 2x - x^2/4$. At two upper-tail breeding values,

$$q(4) = 4 > q(5) = 3.75,$$

whereas the fitted line $2x - 1/4$ orders 5 above 4. These are tail values of the specified standard normal population; the example does not estimate the frequency of ranking reversals. It shows that a variance-explained criterion is not an all-pairs ordering condition.

Biological interpretation. The frequency-weighted residual variance assigns little aggregate weight to a rare part of the population. Reproductive weighting additionally depends on its expected output, so the same part can contribute differently to the selected group. The displayed example computes both operations exactly. It makes no claim about offspring breeding values: their distribution still requires a model of what parents transmit.

4.5 Deriving the displacement-error bound

We now compare the two shifts in a norm that accounts for the population's covariance. Define

$$\|\boldsymbol{v}\|_{\mathbf{G}^{-1}} = \sqrt{\boldsymbol{v}^{\mathsf{T}}\mathbf{G}^{-1}\boldsymbol{v}} = \left\|\mathbf{G}^{-1/2}\boldsymbol{v}\right\|, \qquad \boldsymbol{c} = \mathbf{G}^{1/2}\boldsymbol{b}. \tag{4.18}$$

This measures displacement after removing the covariance scale. A displacement along a low-variance direction is large in this norm relative to one of the same trait-coordinate length along a high-variance direction. Unlike ordinary Euclidean length in arbitrarily chosen trait units, the quadratic form is invariant when displacement and covariance undergo the same invertible coordinate transformation:

$$(\mathbf{D}\boldsymbol{v})^{\mathsf{T}}(\mathbf{D}\mathbf{G}\mathbf{D}^{\mathsf{T}})^{-1}(\mathbf{D}\boldsymbol{v}) = \boldsymbol{v}^{\mathsf{T}}\mathbf{G}^{-1}\boldsymbol{v}.$$

Let $\mathbf{M} = (\mathbf{I} + \mathbf{T})^{-1}$, with $\mathbf{T}$ defined in Equation (4.12). The factorisation already proved gives

$$\begin{aligned} \boldsymbol{\Delta}_{\text{full}} &= \mathbf{G}^{1/2}\mathbf{M}\mathbf{G}^{1/2}\boldsymbol{b} = \mathbf{G}^{1/2}\mathbf{M}\boldsymbol{c}, \\ \boldsymbol{\Delta}_{\text{lin}} &= \mathbf{G}^{1/2}\boldsymbol{c}. \end{aligned} \tag{4.19}$$

Multiplying by $\mathbf{G}^{-1/2}$ therefore gives

$$\begin{aligned} \mathbf{G}^{-1/2}\boldsymbol{\Delta}_{\text{full}} &= \mathbf{M}\boldsymbol{c}, \\ \mathbf{G}^{-1/2}(\boldsymbol{\Delta}_{\text{lin}} - \boldsymbol{\Delta}_{\text{full}}) &= (\mathbf{I} - \mathbf{M})\boldsymbol{c}. \end{aligned} \tag{4.20}$$

From $(\mathbf{I} + \mathbf{T})\mathbf{M} = \mathbf{I}$, subtract $\mathbf{M}$ to obtain $\mathbf{I} - \mathbf{M} = \mathbf{T}\mathbf{M}$. Thus the two whitened vectors are

$$\mathbf{G}^{-1/2}\boldsymbol{\Delta}_{\text{full}} = \mathbf{M}\boldsymbol{c}, \qquad \mathbf{G}^{-1/2}(\boldsymbol{\Delta}_{\text{lin}} - \boldsymbol{\Delta}_{\text{full}}) = \mathbf{T}\mathbf{M}\boldsymbol{c}. \tag{4.21}$$

This step uses the inverse identity, not a presumed commutation of the original covariance and curvature matrices.

Theorem 4.5 (Relative selected-parent displacement bound)**.** *Assume the exact Gaussian/log-quadratic model, positive combined precision, and* $\boldsymbol{c} = \mathbf{G}^{1/2}\boldsymbol{b} \neq \mathbf{0}$*. Then*

$$\frac{\|\boldsymbol{\Delta}_{\text{lin}} - \boldsymbol{\Delta}_{\text{full}}\|_{\mathbf{G}^{-1}}}{\|\boldsymbol{\Delta}_{\text{full}}\|_{\mathbf{G}^{-1}}} \leq \|\mathbf{T}\|_2 \leq \|\mathbf{T}\|_{\text{F}} = \sqrt{2V_{\text{quad}}}. \tag{4.22}$$

For singular covariance the theorem applies on its occupied support, with a nonzero gradient component there. Stabilising curvature is sufficient but not necessary: the bound also holds for an integrable indefinite curvature matrix.

Proof. Since $\mathbf{M}$ is invertible and $\boldsymbol{c} \neq \mathbf{0}$, the vector $\boldsymbol{y} = \mathbf{M}\boldsymbol{c}$ is nonzero. Equation (4.21) turns the left-hand side into

$$\frac{\|\mathbf{T}\boldsymbol{y}\|}{\|\boldsymbol{y}\|}.$$

The spectral norm is defined as the supremum of this ratio over all nonzero vectors, giving the first inequality.

For a symmetric matrix, diagonalise $\mathbf{T} = \mathbf{O}\,\text{diag}(\tau_i)\mathbf{O}^{\mathsf{T}}$. Its spectral norm is $\max_i |\tau_i|$, while its squared Frobenius norm is $\text{tr}(\mathbf{T}^{\mathsf{T}}\mathbf{T}) = \sum_i \tau_i^2$. A largest squared term cannot exceed the sum of all the squared terms, proving the second inequality. Finally, trace cyclicity gives

$$\begin{aligned} \|\mathbf{T}\|_{\text{F}}^2 &= \text{tr}(\mathbf{T}^2) \\ &= \text{tr}(\mathbf{G}^{1/2}\boldsymbol{\Gamma}\mathbf{G}\boldsymbol{\Gamma}\mathbf{G}^{1/2}) \\ &= \text{tr}(\boldsymbol{\Gamma}\mathbf{G}\boldsymbol{\Gamma}\mathbf{G}) = \text{tr}(\mathbf{H}\mathbf{G}\mathbf{H}\mathbf{G}) = 2V_{\text{quad}}, \end{aligned} \tag{4.23}$$

where $\mathbf{H} = -\boldsymbol{\Gamma}$ changes both curvature signs. Taking square roots proves the final equality. Restricting every step to the positive-variance support proves the singular version. □

If the gradient has zero component on the support, both shifts are zero. Their absolute discrepancy is zero, but the displayed relative error is $0/0$ and is undefined. It should not be assigned a finite relative accuracy solely by continuity from a chosen gradient direction.

4.5.1 Eigenbasis formula and equality conditions

The bound can be read more closely by resolving the whitened gradient into curvature eigenvectors. Let c_i be its coefficients in the orthonormal eigenbasis of $\mathbf{T}$. The exact selected displacement has coefficients $y_i = c_i/(1+\tau_i)$, because integrability ensures $1 + \tau_i > 0$. Therefore the squared relative error is exactly

$$\mathcal{E}^2 = \frac{\displaystyle\sum_i \frac{\tau_i^2 c_i^2}{(1+\tau_i)^2}}{\displaystyle\sum_i \frac{c_i^2}{(1+\tau_i)^2}}. \tag{4.24}$$

It is a weighted mean of the squared curvature eigenvalues, with weights proportional to the squared components of the exact displacement. If $\boldsymbol{c}$ lies along a single curvature eigenvector k, then $\mathcal{E} = |\tau_k|$. Equality with the spectral-norm bound requires the displacement, and equivalently the gradient, to lie entirely in eigenspaces whose eigenvalues have maximum absolute value. Repeated eigenvalues need not select a unique direction. Equality between the spectral and Frobenius norms requires at most one nonzero eigenvalue. In one trait both norm bounds are therefore exact, recovering Equation (4.17).

This makes precise why fixed share need not mean fixed displacement accuracy. If $\|\boldsymbol{c}\| = 1$ and $\mathbf{T} = \text{diag}(0.1, 0.8)$, then every gradient orientation has

$$V_{\text{lin}} = 1, \qquad V_{\text{quad}} = \tfrac{1}{2}(0.1^2 + 0.8^2) = 0.325, \qquad \mathcal{A}_g = \frac{1}{1.325}.$$

Gradients along the first and second eigenvectors nevertheless give relative errors 0.1 and 0.8. The variance-share statement fixes a gradient length and curvature spectrum. The selected-parent calculation also depends on their relative orientation through the inverse factors in Equation (4.24).

4.5.2 Why absolute curvature matters

The upper bound involves $\sqrt{2V_{\text{quad}}}$, not the fraction $V_{\text{quad}}/(V_{\text{lin}} + V_{\text{quad}})$. A small fraction may result from a large linear component rather than small absolute curvature. More strongly, consider multiplying the nonconstant natural-log expected-output function by $t > 0$ while holding the population fixed. This replaces $\boldsymbol{b}$ and $\mathbf{H}$ by $t\boldsymbol{b}$ and $t\mathbf{H}$. Both variance components are multiplied by t^2, so their share is unchanged. In the one-trait stabilising case, however, the relative displacement error becomes $t\gamma g$. It can increase while the share stays exactly the same.

This is a change in the expected-fitness relationship: up to a common factor, expected output is raised to the power t. It is not a change of trait units or merely a change of logarithm notation. Multiplying all expected outputs by a common constant instead adds a log intercept, leaving both the share and the normalised reproductive weights unchanged.

Biological interpretation. The share answers how much of a specified quadratic description is represented linearly across the population. The bound answers how much selected-parent displacement can change when curvature is omitted under the exact distributional model. Reporting the two absolute components alongside the share preserves the information needed for this separate check. Extending the calculation to descendants still requires the inheritance assumptions developed in the next chapter.

4.6 Connection to executable checks

The `selected_parent_moments` group in `code/reproduce_result_checks.py` starts with quadrature nodes from the original population and reweights them by expected output. It compares the resulting moments with the formulas above for stabilising and integrable saddle examples. The same file's `orientation_and_absolute_scale` group checks equal-share examples with different displacement errors. The scalar 32/33 and 50% calculation is also integrated independently in `tutorials/introduction/check_examples.py`. These numerical checks assess implementations for specified cases; the normalisation and bound for the whole stated model follow from the derivations in this chapter.

Chapter 5

How selection and reproduction change covariance

Selection changes which parents contribute to reproduction. To calculate the genetic variation among their offspring, we also need to specify what they transmit. These are separate biological steps. A distribution of parental breeding values weighted by expected reproductive output describes the first step; segregation, recombination and mating determine the second.

This chapter follows covariance through both steps under a stated inheritance model. The central question is whether variation can persist when stabilising selection repeatedly reduces covariance among contributing parents. We first identify the genetic variance that the model holds fixed, then derive the update. Matrix order gives a positive covariance equilibrium without requiring the trait axes to be independent. For one trait, a stronger argument gives a unique equilibrium and convergence from every nonnegative starting variance. The boundary with no reduction of associations after selection shows why these conclusions depend on reproduction.

5.1 What is held fixed when allelic effects are fixed?

Let the breeding value be an n-component column vector. In the additive genetic architecture considered here, a copy of a designated allele at locus j contributes the fixed vector $\boldsymbol{\alpha}_j$. If $N_j \in \{0, 1, 2\}$ is its diploid allele count, then, up to a constant vector,

$$\boldsymbol{A} = \sum_{j=1}^{m} \boldsymbol{\alpha}_j N_j, \qquad \boldsymbol{X} = \boldsymbol{A} - \mathbb{E}[\boldsymbol{A}] = \sum_{j=1}^{m} \boldsymbol{\alpha}_j (N_j - 2p_j), \tag{5.1}$$

where both homologous allele copies have marginal frequency p_j. The fixed-effect assumption makes this the same linear map in each generation. It is an extra assumption about genetic architecture: a population-specific regression definition of breeding value does not, by itself, guarantee fixed allelic coefficients across generations.

Write $N_j = Z_{j1} + Z_{j2}$, where Z_{jh} is a zero–one indicator for the allele on homologue h. Under random association of the two homologues,

$$\mathbb{E}[N_j] = 2p_j, \tag{5.2}$$

$$\mathrm{Var}(N_j) = \mathrm{Var}(Z_{j1}) + \mathrm{Var}(Z_{j2}) + 2\,\mathrm{Cov}(Z_{j1}, Z_{j2}) \tag{5.3}$$

$$= 2p_j(1 - p_j). \tag{5.4}$$

If allele counts at different loci are also uncorrelated, the covariance of the sum in Equation (5.1) is

$$\mathbf{B} = 2\sum_{j=1}^{m} p_j(1 - p_j)\boldsymbol{\alpha}_j \boldsymbol{\alpha}_j^{\mathsf{T}}. \tag{5.5}$$

This is the genic covariance baseline used in the model. Each outer product $\boldsymbol{\alpha}_j \boldsymbol{\alpha}_j^{\mathsf{T}}$ describes how one allele's effects contribute to variances and covariances of traits; the factor $2p_j(1-p_j)$ describes variation in its copy number. An allele with effects on several traits can therefore contribute off-diagonal genic covariance even when different loci are independent.

Actual allele counts need not have these random-association properties. Expanding the covariance without assuming independence gives

$$\mathbf{G} = \sum_{j} \mathrm{Var}(N_j)\boldsymbol{\alpha}_j \boldsymbol{\alpha}_j^{\mathsf{T}} + \sum_{j<k} \mathrm{Cov}(N_j, N_k)(\boldsymbol{\alpha}_j \boldsymbol{\alpha}_k^{\mathsf{T}} + \boldsymbol{\alpha}_k \boldsymbol{\alpha}_j^{\mathsf{T}}), \tag{5.6}$$

$$\mathbf{G} = \mathbf{B} + \mathbf{D}. \tag{5.7}$$

Here $\mathbf{D}$ is the total departure from the genic baseline. It includes the cross-locus terms and, if homologues are associated within loci, the terms $2\,\mathrm{Cov}(Z_{j1}, Z_{j2})\boldsymbol{\alpha}_j\boldsymbol{\alpha}_j^{\mathsf{T}}$. Unlike a covariance matrix, $\mathbf{D}$ need not be positive semidefinite: associations can reduce variation in a trait combination. This distinction underlies the effect of selection-induced disequilibrium on quantitative genetic variance (Bulmer, 1971, 1974).

Example 5.1 (Fixed effects with a changing genic variance)**.** Consider one trait and one locus with allelic effect $\alpha = 1$. At $p = 0.5$, the genic variance is $2(0.5)(0.5) = 0.50$. At $p = 0.8$, it is $2(0.8)(0.2) = 0.32$. The effect is unchanged, but its variance contribution is smaller.

Biological interpretation. Holding the allelic effects fixed keeps breeding-value coordinates comparable across generations. Holding $\mathbf{B}$ fixed additionally removes changes in genic covariance caused by changing allele frequencies. The latter can represent an approximation over a short interval, or a formal infinitesimal model in which genic covariance remains constant. It is not a consequence of additive gene action alone. Mutation, drift, migration and changes in allelic effects would require further terms or a different model (Barton and Turelli, 1991; Walsh and Lynch, 2018).

5.2 From selected parents to the next generation

At generation t, assume that breeding values before selection are Gaussian, with mean $\boldsymbol{\mu}_t$ and positive-definite covariance $\mathbf{G}_t$. Write $\boldsymbol{x} = \boldsymbol{a} - \boldsymbol{\mu}_t$. In this chapter the log conditional mean of reproductive output is exactly quadratic on the population's support:

$$\ell_A(\boldsymbol{\mu}_t + \boldsymbol{x}) = \ell_{0,t} + \boldsymbol{b}_t^{\mathsf{T}}\boldsymbol{x} - \frac{1}{2}\boldsymbol{x}^{\mathsf{T}}\boldsymbol{\Gamma}\boldsymbol{x}, \qquad \boldsymbol{\Gamma} \succeq \mathbf{0}. \tag{5.8}$$

Thus $\boldsymbol{\Gamma} = -\mathbf{H}$ is the sign-reversed Hessian and describes stabilising curvature. The exact assumption matters: substituting a Taylor expansion does not make the selected distribution exactly Gaussian.

Weighting the Gaussian density by $W_A(\boldsymbol{a}) = \exp\{\ell_A(\boldsymbol{a})\}$ adds the quadratic precision term $\boldsymbol{\Gamma}$. The resulting covariance among parents, under these reproductive weights, is

$$\mathbf{G}_t^{\mathrm{sel}} = (\mathbf{G}_t^{-1} + \boldsymbol{\Gamma})^{-1}. \tag{5.9}$$

The precision is positive definite because $\mathbf{G}_t^{-1} \succ \mathbf{0}$ and $\boldsymbol{\Gamma} \succeq \mathbf{0}$. Consequently the expected-output weights have a finite normalising integral for every finite $\boldsymbol{b}_t$. Completing the square changes the mean, but the coefficient of the centred quadratic determines covariance and contains no $\boldsymbol{b}_t$.

Equation (5.9) has not yet described offspring. To do so, hold the genic covariance at $\mathbf{B} \succ \mathbf{0}$ and make the following assumption about the departure from this baseline:

$$\mathbf{D}_{t+1} = \rho\mathbf{D}_t^{\mathrm{sel}}, \qquad \mathbf{D}_t^{\mathrm{sel}} = \mathbf{G}_t^{\mathrm{sel}} - \mathbf{B}, \qquad 0 \le \rho < 1. \tag{5.10}$$

The selected departure includes associations already present as well as their change during the current episode of selection. The assumption retains the same fraction ρ of this whole departure after reproduction. Substitution into $\mathbf{G}_{t+1} = \mathbf{B} + \mathbf{D}_{t+1}$ gives

$$\mathbf{G}_{t+1} = \mathbf{B} + \rho(\mathbf{G}_t^{\mathrm{sel}} - \mathbf{B}) \tag{5.11}$$

$$= (1 - \rho)\mathbf{B} + \rho(\mathbf{G}_t^{-1} + \boldsymbol{\Gamma})^{-1}. \tag{5.12}$$

This derivation identifies what the inheritance model contributes. The Gaussian selection calculation supplies $\mathbf{G}_t^{\mathrm{sel}}$; Equation (5.10) specifies how reproduction uses it. Neither a variance share nor a fitness regression supplies that transmission assumption.

5.2.1 Why one half is a useful benchmark

A moment calculation explains the familiar freely recombining, random-mating infinitesimal benchmark (Bulmer, 1971). Suppose two contributing parents are independent draws from the same selected distribution. In a model for their offspring, write

$$\boldsymbol{A}_{\mathrm{o}} = \frac{\boldsymbol{A}_1 + \boldsymbol{A}_2}{2} + \boldsymbol{M}, \qquad \mathbb{E}[\boldsymbol{M} \mid \text{parents}] = \mathbf{0}, \tag{5.13}$$

where $\boldsymbol{M}$ is the Mendelian segregation deviation. The conditional zero mean makes its covariance with the midparent value zero. If its mean conditional covariance is $\mathbf{B}/2$, the law of total covariance gives

$$\begin{aligned} \mathrm{Cov}(\boldsymbol{A}_{\mathrm{o}}) &= \mathrm{Cov}\left(\frac{\boldsymbol{A}_1 + \boldsymbol{A}_2}{2}\right) + \mathbb{E}[\mathrm{Cov}(\boldsymbol{M} \mid \text{parents})] && (5.14)\\ &= \frac{1}{4}(\mathbf{G}_t^{\mathrm{sel}} + \mathbf{G}_t^{\mathrm{sel}}) + \frac{1}{2}\mathbf{B} && (5.15)\\ &= \frac{1}{2}\mathbf{G}_t^{\mathrm{sel}} + \frac{1}{2}\mathbf{B}. && (5.16) \end{aligned}$$

This is Equation (5.12) with $\rho = 1/2$. The two ingredients—independent parents and a specified segregation covariance—are part of the model. For example, at an independently segregating locus a heterozygous parent transmits the allele with variance $1/4$ in copy number. Averaging over two parents with heterozygote frequency $2p_j(1-p_j)$ gives $p_j(1-p_j)$, half the locus's genic copy-number variance. Summing the corresponding effect outer products gives $\mathbf{B}/2$ under independent segregation across loci and the fixed single-locus heterozygosities assumed by this infinitesimal closure.

The more general scalar ρ in Equation (5.12) summarises retention; it is not derived from a single recombination fraction. Different locus pairs have different recombination rates and contributions to covariance. Nonrandom mating can also change the covariance between parents and the segregation term. These mechanisms do not, in general, rescale every covariance departure by the same number (Bulmer, 1974; Barton and Turelli, 1991). In particular, this model supplies no universal conversion from a selfing rate or an inbreeding coefficient to ρ.

Biological interpretation. When selected associations reduce variance, reproduction can restore some variance by reducing those associations. The term $(1-\rho)\mathbf{B}$ does not represent new mutations or new alleles entering the population. It expresses the assumed return towards the covariance that the same allele frequencies and effects would produce under random association. Maintaining Gaussian breeding-value distributions in successive generations is also part of this closure; a rule for covariance alone does not prove that distributional assumption.

5.3 How to compare covariance matrices

For one trait, a smaller variance has an immediate meaning. With several traits, comparing individual matrix entries is insufficient: a covariance matrix may lose variance in one trait combination and gain it in another. The useful ordering compares every linear combination at once; it is the positive-semidefinite order used in matrix and convex analysis (Boyd and Vandenberghe, 2004).

Definition 5.2 (Loewner order)**.** For symmetric matrices $\mathbf{C}$ and $\mathbf{D}$, write $\mathbf{C} \preceq \mathbf{D}$ when $\mathbf{D} - \mathbf{C}$ is positive semidefinite, that is,

$$\boldsymbol{v}^{\mathsf{T}}\mathbf{C}\boldsymbol{v} \le \boldsymbol{v}^{\mathsf{T}}\mathbf{D}\boldsymbol{v} \quad \text{for every } \boldsymbol{v} \in \mathbb{R}^n. \tag{5.17}$$

Strict positive definiteness is written $\mathbf{C} \prec \mathbf{D}$.

Because $\mathrm{Var}(\boldsymbol{v}^{\mathsf{T}}\boldsymbol{A}) = \boldsymbol{v}^{\mathsf{T}}\mathbf{G}\boldsymbol{v}$, the order says that every specified linear combination of traits has at least as much variance under $\mathbf{D}$ as under $\mathbf{C}$. It is a partial order: two covariances need not be comparable. The same comparison is preserved by a change of coordinates. Indeed, if $\mathbf{D} - \mathbf{C} \succeq \mathbf{0}$, then

$$\boldsymbol{v}^{\mathsf{T}}\mathbf{R}(\mathbf{D} - \mathbf{C})\mathbf{R}^{\mathsf{T}}\boldsymbol{v} = (\mathbf{R}^{\mathsf{T}}\boldsymbol{v})^{\mathsf{T}}(\mathbf{D} - \mathbf{C})(\mathbf{R}^{\mathsf{T}}\boldsymbol{v}) \geq 0. \tag{5.18}$$

Multiplication on the two sides in this form is called a congruence. It does not require the matrices to commute.

Lemma 5.3 (Stabilising selection reduces covariance)**.** *If* $\mathbf{G} \succ \mathbf{0}$ *and* $\mathbf{\Gamma} \succeq \mathbf{0}$*, then*

$$\mathbf{0} \prec (\mathbf{G}^{-1} + \mathbf{\Gamma})^{-1} \preceq \mathbf{G}. \tag{5.19}$$

If $\mathbf{\Gamma} \succ \mathbf{0}$*, the upper inequality is strict. Equality of the two complete covariance matrices occurs exactly when* $\mathbf{\Gamma} = \mathbf{0}$*.*

Proof. A positive-definite symmetric matrix has an orthogonal eigenvector basis with positive eigenvalues. Taking their positive square roots defines $\mathbf{G}^{1/2}$ and its inverse $\mathbf{G}^{-1/2}$. Define

$$\mathbf{K} = \mathbf{G}^{1/2}\mathbf{\Gamma}\mathbf{G}^{1/2} \succeq \mathbf{0}. \tag{5.20}$$

Factor the precision in the order shown:

$$\mathbf{G}^{-1} + \mathbf{\Gamma} = \mathbf{G}^{-1/2}(\mathbf{I} + \mathbf{K})\mathbf{G}^{-1/2}, \tag{5.21}$$

$$(\mathbf{G}^{-1} + \mathbf{\Gamma})^{-1} = \mathbf{G}^{1/2}(\mathbf{I} + \mathbf{K})^{-1}\mathbf{G}^{1/2}. \tag{5.22}$$

The eigenvalues κ_i of $\mathbf{K}$ are nonnegative. Therefore the eigenvalues of $(\mathbf{I} + \mathbf{K})^{-1}$ are $1/(1 + \kappa_i)$, which lie in $(0, 1]$. It follows that

$$\mathbf{0} \prec (\mathbf{I} + \mathbf{K})^{-1} \preceq \mathbf{I}. \tag{5.23}$$

Congruence by $\mathbf{G}^{1/2}$ gives Equation (5.19). If $\mathbf{\Gamma} \succ \mathbf{0}$, every $\kappa_i > 0$, so every eigenvalue $1/(1+\kappa_i) < 1$ and the reduction is positive definite. Equality of the full matrices requires every $\kappa_i = 0$. Since $\mathbf{G}^{1/2}$ is invertible, this is equivalent to $\mathbf{\Gamma} = \mathbf{0}$. □

This proof is an instance of the reversal of Loewner order by inversion: if $\mathbf{P} \succeq \mathbf{Q} \succ \mathbf{0}$, then $\mathbf{P}^{-1} \preceq \mathbf{Q}^{-1}$. To see the general statement, apply the same argument to $\mathbf{Q}^{-1/2}\mathbf{P}\mathbf{Q}^{-1/2} \succeq \mathbf{I}$ and undo the congruence. The proof diagonalises one symmetric matrix; it never assumes that $\mathbf{G}$ and $\mathbf{\Gamma}$ share eigenvectors.

Biological interpretation. Under the Gaussian quadratic model, stabilising selection cannot increase variance in any trait combination among contributing parents. A reduction in one episode of selection does not imply a reduction from one generation to the next: reproduction subsequently changes the associations. Nor does zero curvature along one coordinate guarantee an unchanged marginal variance there, because that trait can covary with a trait under selection.

5.4 A positive matrix equilibrium exists

Define the covariance update on positive-definite matrices by

$$F(\mathbf{G}) = (1 - \rho)\mathbf{B} + \rho(\mathbf{G}^{-1} + \mathbf{\Gamma})^{-1}. \tag{5.24}$$

Assume throughout this section that $\mathbf{B} \succ \mathbf{0}$, $\mathbf{\Gamma} \succeq \mathbf{0}$ and $0 \leq \rho < 1$ are fixed. The lower matrix

$$\mathbf{L} = (1 - \rho)\mathbf{B} \tag{5.25}$$

is positive definite. Consider the set

$$\mathcal{K} = \{\mathbf{G} = \mathbf{G}^{\mathsf{T}} : \mathbf{L} \preceq \mathbf{G} \preceq \mathbf{B}\}. \tag{5.26}$$

The two bounds concern every trait combination, not entrywise intervals.

Proposition 5.4 (Existence and bounds)**.** *The map F has at least one positive-definite fixed point in $\mathcal{K}$. Every positive-definite fixed point of F, whether or not considered initially within $\mathcal{K}$, satisfies*

$$(1-\rho)\mathbf{B} \preceq \mathbf{G}^* \preceq \mathbf{B}. \tag{5.27}$$

Proof. We first show that F leaves the set $\mathcal{K}$ invariant. For $\mathbf{G} \in \mathcal{K}$, the selected covariance is positive definite and no larger than $\mathbf{G}$ by Lemma 5.3. Hence

$$F(\mathbf{G}) \succeq (1-\rho)\mathbf{B} = \mathbf{L}, \tag{5.28}$$
$$F(\mathbf{G}) \preceq (1-\rho)\mathbf{B} + \rho\mathbf{G} \tag{5.29}$$
$$\preceq (1-\rho)\mathbf{B} + \rho\mathbf{B} = \mathbf{B}. \tag{5.30}$$

Thus $F(\mathcal{K}) \subseteq \mathcal{K}$.

Next, regard real symmetric $n \times n$ matrices as a Euclidean space with $n(n+1)/2$ coordinates, one for each independent matrix entry. The set $\mathcal{K}$ is nonempty because it contains $\mathbf{B}$. It is convex: for $\mathbf{G}_1, \mathbf{G}_2 \in \mathcal{K}$ and $0 \le \theta \le 1$, each of

$$\theta\mathbf{G}_1 + (1-\theta)\mathbf{G}_2 - \mathbf{L} = \theta(\mathbf{G}_1 - \mathbf{L}) + (1-\theta)(\mathbf{G}_2 - \mathbf{L}), \tag{5.31}$$
$$\mathbf{B} - \theta\mathbf{G}_1 - (1-\theta)\mathbf{G}_2 = \theta(\mathbf{B} - \mathbf{G}_1) + (1-\theta)(\mathbf{B} - \mathbf{G}_2) \tag{5.32}$$

is a sum of positive-semidefinite matrices.

It is also closed: a limit of matrices satisfying each quadratic-form inequality in Equation (5.26) still satisfies it. It is bounded because $\mathbf{0} \preceq \mathbf{G} \preceq \mathbf{B}$ implies $\lambda_{\max}(\mathbf{G}) \le \lambda_{\max}(\mathbf{B})$ and therefore

$$\|\mathbf{G}\|_{\mathrm{F}}^2 = \sum_{i=1}^{n} \lambda_i(\mathbf{G})^2 \le n\lambda_{\max}(\mathbf{B})^2. \tag{5.33}$$

Closed and bounded subsets of a finite-dimensional Euclidean space are compact. In this setting, compactness means that any sequence in the set has a convergent subsequence whose limit stays in the set.

Finally, matrix inversion is continuous wherever the determinant is nonzero. Throughout $\mathcal{K}$,

$$\lambda_{\min}(\mathbf{G}) \ge (1-\rho)\lambda_{\min}(\mathbf{B}) > 0, \tag{5.34}$$

so the first inversion in F is well-defined. The second is also well-defined because $\mathbf{G}^{-1} + \mathbf{\Gamma} \succ \mathbf{0}$. Matrix sums and scalar multiplication are continuous, so F is continuous on $\mathcal{K}$.

Brouwer's fixed-point theorem states that a continuous map from a nonempty compact convex subset of a finite-dimensional Euclidean space into itself has a fixed point (Granas and Dugundji, 2003). All its hypotheses have now been checked. The resulting $\mathbf{G}^* \in \mathcal{K}$ is positive definite because it is no smaller than $\mathbf{L} \succ \mathbf{0}$.

To bound any other positive-definite fixed point, no assumption about its starting position is needed. The equation $\mathbf{G}^* = F(\mathbf{G}^*)$ immediately gives $\mathbf{G}^* \succeq (1-\rho)\mathbf{B}$. Applying Equation (5.19) at $\mathbf{G}^*$ gives

$$\mathbf{G}^* \preceq (1-\rho)\mathbf{B} + \rho\mathbf{G}^*, \tag{5.35}$$
$$(1-\rho)\mathbf{G}^* \preceq (1-\rho)\mathbf{B}, \tag{5.36}$$
$$\mathbf{G}^* \preceq \mathbf{B}. \tag{5.37}$$

Division by $1-\rho$ preserves the inequality because this scalar is strictly positive. This proves both bounds for every positive fixed point. □

Biological interpretation. For any trait combination $\boldsymbol{v}^{\mathsf{T}}\boldsymbol{A}$, an equilibrium variance lies between $(1-\rho)\boldsymbol{v}^{\mathsf{T}}\mathbf{B}\boldsymbol{v}$ and $\boldsymbol{v}^{\mathsf{T}}\mathbf{B}\boldsymbol{v}$. Thus fixed positive genic covariance and incomplete retention prevent a covariance equilibrium from losing all variation in a genetic direction. If $\rho > 0$, its lower bound is strict because the selected covariance is positive definite. If $\rho = 0$, the next covariance is exactly $\mathbf{B}$. With $\rho > 0$, the full matrix $\mathbf{G}^* = \mathbf{B}$ is possible exactly when $\boldsymbol{\Gamma} = \mathbf{0}$.

The argument proves existence and bounds. Brouwer's theorem does not say that the fixed point is unique or that repeated updates converge to it. Invariance says that a trajectory starting in $\mathcal{K}$ stays there; it does not say that every other starting covariance enters that interval after one update. We make no general matrix uniqueness or convergence claim here. The scalar case permits a separate, stronger proof.

5.5 One trait: the fixed point and every trajectory

Let g be a breeding-value variance, $B > 0$ the genic variance, and $\gamma \geq 0$ the stabilising curvature. The scalar map is

$$F(g) = (1-\rho)B + \rho\frac{g}{1+\gamma g}, \qquad g \geq 0, \quad 0 \leq \rho < 1. \tag{5.38}$$

It also describes an isotropic model with $\mathbf{G} = g\mathbf{I}$, $\mathbf{B} = B\mathbf{I}$ and $\boldsymbol{\Gamma} = \gamma\mathbf{I}$ in fixed trait coordinates. The product γg is dimensionless: g and B have units of trait squared, while γ has inverse squared-trait units. Although the precision calculation used $g > 0$, the fraction in Equation (5.38) has the continuous value zero at $g = 0$. The scalar recursion therefore includes the zero starting variance, followed by $F(0) = (1-\rho)B > 0$.

5.5.1 Solving the fixed-point equation

Set $g^* = F(g^*)$ and first take $\gamma > 0$. Since $1 + \gamma g^* > 0$, multiplying by the denominator preserves the equation:

$$g^* = (1-\rho)B + \rho\frac{g^*}{1+\gamma g^*}, \tag{5.39}$$

$$g^* + \gamma(g^*)^2 = (1-\rho)B + (1-\rho)\gamma B g^* + \rho g^*, \tag{5.40}$$

$$\gamma(g^*)^2 + (1-\rho)(1-\gamma B)g^* - (1-\rho)B = 0. \tag{5.41}$$

The leading coefficient is positive and the constant term is negative. Thus the product of the two roots is negative, so exactly one root is positive. Write $c = (1-\rho)(1-\gamma B)$ and $d = (1-\rho)B > 0$. The positive root is

$$g^* = \frac{-c + \sqrt{c^2 + 4\gamma d}}{2\gamma}. \tag{5.42}$$

When $c \geq 0$, subtracting two nearly equal positive numbers in this formula can lose floating-point precision. Multiplying numerator and denominator by $\sqrt{c^2 + 4\gamma d} + c$ gives the equivalent form

$$g^* = \frac{2d}{\sqrt{c^2 + 4\gamma d} + c}. \tag{5.43}$$

Use the rationalised form when $c \geq 0$ and the first form when $c < 0$ to avoid that subtraction. This is an arithmetic choice, not a different equilibrium. As γ tends to zero with $\rho < 1$ fixed, the rationalised expression tends to B, agreeing with the direct zero-curvature calculation below.

5.5.2 Why the scalar iteration converges

An equilibrium equation alone does not tell us where a trajectory goes. To answer that question, compare two possible current variances $g, h \geq 0$:

$$F(g) - F(h) = \rho\left(\frac{g}{1+\gamma g} - \frac{h}{1+\gamma h}\right) \tag{5.44}$$

$$= \rho\frac{g(1+\gamma h) - h(1+\gamma g)}{(1+\gamma g)(1+\gamma h)} \tag{5.45}$$

$$= \frac{\rho(g-h)}{(1+\gamma g)(1+\gamma h)}. \tag{5.46}$$

The two mixed terms cancel. Since each denominator factor is at least one,

$$|F(g) - F(h)| \leq \rho|g-h|. \tag{5.47}$$

A map that reduces every distance by at most a constant strictly below one is a contraction.

Proposition 5.5 (Global scalar convergence)**.** *For $B > 0$, $\gamma \geq 0$ and $0 \leq \rho < 1$, the map in Equation (5.38) has a unique fixed point on $[0, \infty)$. Every nonnegative starting variance converges to it, with error bound*

$$|g_t - g^*| \leq \rho^t |g_0 - g^*|, \qquad t \geq 1. \tag{5.48}$$

Proof. The map takes $[0, \infty)$ into itself. This set is complete in the usual distance $|g-h|$: a Cauchy sequence of nonnegative real numbers has a nonnegative real limit. These properties and Equation (5.47) are the hypotheses of the Banach contraction theorem (Banach, 1922; Granas and Dugundji, 2003). Its argument can be seen directly in the iterates.

For $0 < \rho < 1$, repeated use of the bound gives

$$|g_{k+1} - g_k| \leq \rho^k |g_1 - g_0|. \tag{5.49}$$

For integers $p \geq 1$, the triangle inequality then gives

$$|g_{t+p} - g_t| \leq \sum_{k=t}^{t+p-1} |g_{k+1} - g_k| \tag{5.50}$$

$$\leq |g_1 - g_0| \sum_{k=t}^{t+p-1} \rho^k \tag{5.51}$$

$$\leq \frac{\rho^t}{1-\rho}|g_1 - g_0|. \tag{5.52}$$

The final quantity tends to zero as $t \to \infty$, independently of p. The iterates are therefore Cauchy and converge to a value $g^* \geq 0$. Continuity gives $F(g^*) = \lim_t F(g_t) = \lim_t g_{t+1} = g^*$.

If two fixed points g^* and h^* existed, then

$$|g^* - h^*| = |F(g^*) - F(h^*)| \leq \rho|g^* - h^*|. \tag{5.53}$$

As $1 - \rho > 0$, their distance must be zero. Finally, compare each iterate with this fixed point in Equation (5.47) and apply the inequality t times to obtain Equation (5.48). If $\rho = 0$, $F(g) = B$ for every g, so all conclusions follow after one update. □

The convergence proof needs neither a positive starting variance nor a starting point inside $[(1-\rho)B, B]$. It also explains the direction of change. For $0 < \rho < 1$, the exact difference formula gives

$$F(g) - g^* = \frac{\rho(g - g^*)}{(1+\gamma g)(1+\gamma g^*)}. \tag{5.54}$$

The sign of the deviation from equilibrium is preserved and its magnitude strictly decreases unless it is already zero. Thus a start below g^* increases monotonically towards it; a start above g^* decreases monotonically towards it. Neither trajectory crosses the fixed point. At $\rho = 0$, it reaches the fixed point immediately.

Biological interpretation. Stabilising selection never increases the selected-parent variance in this model, but reproduction need not leave the next generation below its preselection value. With $B = 1$, $\gamma = 0.3$, $\rho = 0.9$ and $g_t = 0.1$, selection gives $0.1/1.03$, and reproduction gives

$$g_{t+1} = 0.1 + 0.9\frac{0.1}{1.03} = 0.1873786408. \tag{5.55}$$

The starting variance is below the equilibrium. Convergence describes both increasing and decreasing trajectories; it does not imply that covariance must fall in every population.

5.5.3 Boundary cases and the worked iteration

With no retention, $\rho = 0$, reproduction returns the variance to B after one step. With no curvature, $\gamma = 0$ and $\rho < 1$,

$$g_{t+1} = (1 - \rho)B + \rho g_t, \tag{5.56}$$

$$g_{t+1} - B = \rho(g_t - B), \tag{5.57}$$

$$g_t = B + \rho^t(g_0 - B). \tag{5.58}$$

Therefore B is the unique fixed point, even if the original population contains covariance departures that reproduction gradually removes. The matrix version of this particular boundary is equally direct: $\mathbf{G}_t = \mathbf{B} + \rho^t(\mathbf{G}_0 - \mathbf{B})$ when $\mathbf{\Gamma} = \mathbf{0}$. This explicit boundary calculation is separate from any general matrix convergence claim.

For the worked example in the mathematics guide, $B = 1$, $\gamma = 0.5$, $\rho = 0.5$ and $g_0 = 1$. The first selected variance is $2/3$, so the first offspring variance is $g_1 = 1/2 + (1/2)(2/3) = 5/6$. Further substitutions give

Generation	Exact variance	Decimal variance
0	1	1.000000
1	5/6	0.833333
2	27/34	0.794118
3	149/190	0.784211
4	827/1058	0.781664

The fixed-point polynomial reduces to $2(g^*)^2 + g^* - 2 = 0$, giving

$$g^* = \frac{\sqrt{17} - 1}{4} = 0.7807764064. \tag{5.59}$$

These iterates remain between 0.5 and 1 because that interval is invariant and contains the starting value. The numerical sequence illustrates the proven convergence; it is not its proof.

5.6 Retaining every covariance departure

The boundary $\rho = 1$ removes the baseline term entirely. It must be analysed separately: the positive lower bound $(1 - \rho)\mathbf{B}$ and the strict scalar contraction constant used above are then lost. Assume $\mathbf{G}_0 \succ \mathbf{0}$ and fixed $\mathbf{\Gamma} \succeq \mathbf{0}$. The update becomes

$$\mathbf{G}_{t+1} = (\mathbf{G}_t^{-1} + \mathbf{\Gamma})^{-1}. \tag{5.60}$$

Inverting both sides gives the simple precision recursion

$$\mathbf{G}_{t+1}^{-1} = \mathbf{G}_t^{-1} + \mathbf{\Gamma}. \tag{5.61}$$

The initial case is $\mathbf{G}_0^{-1} = \mathbf{G}_0^{-1} + 0\mathbf{\Gamma}$. If $\mathbf{G}_t^{-1} = \mathbf{G}_0^{-1} + t\mathbf{\Gamma}$, one further update gives $\mathbf{G}_{t+1}^{-1} = \mathbf{G}_0^{-1} + (t+1)\mathbf{\Gamma}$. Induction therefore proves

$$\mathbf{G}_t = (\mathbf{G}_0^{-1} + t\mathbf{\Gamma})^{-1}. \tag{5.62}$$

No simultaneous diagonalisation of $\mathbf{G}_0$ and $\mathbf{\Gamma}$ was used.

If $\mathbf{\Gamma} \succ \mathbf{0}$, then for $t > 0$

$$\mathbf{0} \prec \mathbf{G}_t \preceq (t\mathbf{\Gamma})^{-1} = \frac{1}{t}\mathbf{\Gamma}^{-1}. \tag{5.63}$$

The right-hand side tends to zero, so covariance disappears in every direction. If $\mathbf{\Gamma} = \mathbf{0}$, every covariance stays at its starting value. In one trait with $\gamma > 0$, $g_t = g_0/(1 + t\gamma g_0)$ for a positive start. The example $g_0 = 1$, $\gamma = 0.5$ gives $g_4 = 1/3$. At this boundary the continuous scalar map also leaves $g_0 = 0$ at zero.

5.6.1 Singular curvature: what variation actually remains?

Suppose $\mathbf{\Gamma}$ has rank r with $0 < r < n$. Let the columns of $\mathbf{U}$ be orthonormal eigenvectors with positive curvature, and let the columns of $\mathbf{V}$ be an orthonormal basis for its null space. Then $\mathbf{Q} = [\mathbf{U}\ \mathbf{V}]$ is orthogonal and

$$\mathbf{\Gamma} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^{\mathsf{T}}, \qquad \mathbf{\Lambda} \succ \mathbf{0}, \qquad \mathbf{Q}^{\mathsf{T}}\mathbf{G}_0\mathbf{Q} = \begin{pmatrix} \mathbf{C} & \mathbf{J} \\ \mathbf{J}^{\mathsf{T}} & \mathbf{D} \end{pmatrix}. \tag{5.64}$$

Here $\mathbf{C}$ is the initial covariance of the coordinates with positive curvature, $\mathbf{D}$ that of the zero-curvature coordinates, and $\mathbf{J}$ their cross-covariance. The block letters in this subsection are local matrix names; $\mathbf{D}$ here is not the association departure used earlier.

Proposition 5.6 (The exact retained covariance)**.** *Under these assumptions,*

$$\lim_{t\to\infty} \mathbf{G}_t = \mathbf{V}(\mathbf{D} - \mathbf{J}^{\mathsf{T}}\mathbf{C}^{-1}\mathbf{J})\mathbf{V}^{\mathsf{T}}. \tag{5.65}$$

The matrix in parentheses is positive definite on the null-space coordinates. It equals their initial Gaussian conditional covariance given the positive-curvature coordinates, rather than their marginal covariance $\mathbf{D}$ *in general.*

Proof. For $t > 0$ define

$$\mathbf{M}_t = t^{-1}\mathbf{\Lambda}^{-1} + \mathbf{U}^{\mathsf{T}}\mathbf{G}_0\mathbf{U} = t^{-1}\mathbf{\Lambda}^{-1} + \mathbf{C}. \tag{5.66}$$

Both terms are positive definite, so this matrix can be inverted. We will use the identity

$$(\mathbf{G}_0^{-1} + t\mathbf{U}\mathbf{\Lambda}\mathbf{U}^{\mathsf{T}})^{-1} = \mathbf{G}_0 - \mathbf{G}_0\mathbf{U}\mathbf{M}_t^{-1}\mathbf{U}^{\mathsf{T}}\mathbf{G}_0. \tag{5.67}$$

This is a form of the matrix inverse update identity, often called the Woodbury identity. Here it can be verified without invoking a general formula. Multiplying its right-hand side on the left by $\mathbf{G}_0^{-1} + t\mathbf{U}\mathbf{\Lambda}\mathbf{U}^{\mathsf{T}}$ gives

$$\mathbf{I} + \mathbf{U}\big[t\mathbf{\Lambda} - \mathbf{M}_t^{-1} - t\mathbf{\Lambda}\mathbf{C}\mathbf{M}_t^{-1}\big]\mathbf{U}^{\mathsf{T}}\mathbf{G}_0. \tag{5.68}$$

The bracket vanishes because

$$t\mathbf{\Lambda}\mathbf{M}_t = \mathbf{I} + t\mathbf{\Lambda}\mathbf{C}, \qquad (\mathbf{I} + t\mathbf{\Lambda}\mathbf{C})\mathbf{M}_t^{-1} = t\mathbf{\Lambda}. \tag{5.69}$$

This establishes Equation (5.67), with matrix products in their stated order.

As $t \to \infty$, $\mathbf{M}_t \to \mathbf{C} \succ \mathbf{0}$, so $\mathbf{M}_t^{-1} \to \mathbf{C}^{-1}$. Expressing the limiting identity in the $\mathbf{Q}$ coordinates gives

$$\lim_{t\to\infty} \mathbf{Q}^\mathsf{T}\mathbf{G}_t\mathbf{Q} = \begin{pmatrix} \mathbf{C} & \mathbf{J} \\ \mathbf{J}^\mathsf{T} & \mathbf{D} \end{pmatrix} - \begin{pmatrix} \mathbf{C} \\ \mathbf{J}^\mathsf{T} \end{pmatrix} \mathbf{C}^{-1} \begin{pmatrix} \mathbf{C} & \mathbf{J} \end{pmatrix} \tag{5.70}$$

$$= \begin{pmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{D} - \mathbf{J}^\mathsf{T}\mathbf{C}^{-1}\mathbf{J} \end{pmatrix}. \tag{5.71}$$

Transforming back gives Equation (5.65).

The matrix $\mathbf{S} = \mathbf{D} - \mathbf{J}^\mathsf{T}\mathbf{C}^{-1}\mathbf{J}$ is the Schur complement of $\mathbf{C}$. For a nonzero null-coordinate vector $\boldsymbol{z}$, use the positive definiteness of the full initial covariance to obtain

$$\boldsymbol{z}^\mathsf{T}\mathbf{S}\boldsymbol{z} = \begin{pmatrix} -\mathbf{C}^{-1}\mathbf{J}\boldsymbol{z} \\ \boldsymbol{z} \end{pmatrix}^\mathsf{T} \begin{pmatrix} \mathbf{C} & \mathbf{J} \\ \mathbf{J}^\mathsf{T} & \mathbf{D} \end{pmatrix} \begin{pmatrix} -\mathbf{C}^{-1}\mathbf{J}\boldsymbol{z} \\ \boldsymbol{z} \end{pmatrix} > 0. \tag{5.72}$$

Finally, write $\boldsymbol{X}_{\text{curved}} = \mathbf{U}^\mathsf{T}\boldsymbol{X}_0$ and $\boldsymbol{X}_{\text{null}} = \mathbf{V}^\mathsf{T}\boldsymbol{X}_0$ for the centred initial coordinates. In a joint Gaussian distribution, subtracting the conditional linear mean of the null coordinates leaves residual $\boldsymbol{X}_{\text{null}} - \mathbf{J}^\mathsf{T}\mathbf{C}^{-1}\boldsymbol{X}_{\text{curved}}$. Its covariance is $\mathbf{S}$, and its covariance with $\boldsymbol{X}_{\text{curved}}$ is zero. Joint Gaussianity then makes the residual independent of those coordinates, proving the conditional covariance interpretation. □

Example 5.7 (A trait with no curvature can still lose variance)**.** Let

$$\mathbf{G}_0 = \begin{pmatrix} 1 & 0.4 \\ 0.4 & 2 \end{pmatrix}, \qquad \mathbf{\Gamma} = \begin{pmatrix} 0.5 & 0 \\ 0 & 0 \end{pmatrix}. \tag{5.73}$$

The second trait has zero curvature but initially covaries with the first. Its retained variance is $2 - 0.4^2/1 = 1.84$, so

$$\lim_{t\to\infty} \mathbf{G}_t = \begin{pmatrix} 0 & 0 \\ 0 & 1.84 \end{pmatrix}. \tag{5.74}$$

Keeping the original marginal variance 2 would ignore the initial association with the trait whose variation is removed. When $\mathbf{J} = \mathbf{0}$, marginal and conditional covariance do coincide.

Biological interpretation. With no restoration, persistent covariance is confined to trait combinations with zero curvature. Its amount depends on the original associations with combinations under stabilising selection. This is a covariance limit under the specified model, not a statement that allele frequencies, mean traits or reproductive output reach a joint equilibrium. If the initial covariance is singular, first restrict the calculation to its positive-variance support. At $\rho = 1$ no baseline term adds variation outside that support; use the curvature restricted to the same coordinates before taking inverses.

5.7 A covariance trajectory does not determine the fitness gradient

The recursion contains $\mathbf{G}_t$, $\mathbf{B}$, $\mathbf{\Gamma}$ and ρ, but no directional gradient $\boldsymbol{b}_t$. This independence follows from the exact Gaussian and quadratic selection calculation in Equation (5.9). It is not a general property of selection on non-Gaussian populations or nonquadratic fitness functions.

The first-order variance share still depends on the gradient. In the isotropic case with n traits,

$$V_{\text{lin}}(t) = g_t \left\|\boldsymbol{b}_t\right\|^2, \tag{5.75}$$

$$V_{\text{quad}}(t) = \frac{1}{2} n\gamma^2 g_t^2, \tag{5.76}$$

$$\mathcal{A}_g(t) = \frac{\left\|\boldsymbol{b}_t\right\|^2}{\left\|\boldsymbol{b}_t\right\|^2 + \frac{1}{2}n\gamma^2 g_t}, \qquad g_t > 0, \tag{5.77}$$

where the denominator must be positive. If $\boldsymbol{b}_t = \mathbf{0}$ but $\gamma > 0$, the share is zero. If both terms vanish, the variance share is undefined. Under fixed nonzero gradient and positive curvature, a decrease in g_t increases the share even though both component variances decrease. A start below the scalar equilibrium instead produces increasing g_t and a decreasing share under that same fixed gradient.

Figure 4 of the paper chooses $n = 4$, $g_0 = 4$, $B = 1$, $\gamma = 0.30$ and $\rho = 0.90$. The scalar equilibrium is $g^* = 0.4723532349$. To calculate the share, it separately specifies

$$\|\boldsymbol{b}_t\| = \begin{cases} 1, & t \le 40, \\ \exp[-0.06(t-40)], & t > 40. \end{cases} \tag{5.78}$$

While the gradient is maintained, the share rises from 0.581395 to 0.921639 by generation 40. It subsequently falls as the imposed gradient decreases, reaching approximately 0.00870449 at generation 100. The figure combines a derived covariance trajectory with a supplied gradient history. That history is not an output of the covariance recursion.

A changing local gradient need not mean that the fitness surface changes. For a fixed quadratic function $\ell_A(\boldsymbol{a}) = c + \boldsymbol{d}^\mathsf{T}\boldsymbol{a} - \frac{1}{2}\boldsymbol{a}^\mathsf{T}\boldsymbol{\Gamma}\boldsymbol{a}$, differentiation gives

$$\boldsymbol{b}_t = \boldsymbol{d} - \boldsymbol{\Gamma}\boldsymbol{\mu}_t. \tag{5.79}$$

Movement of the population mean changes the local gradient on the same surface. Predicting that movement would require its own transmission assumptions and mean recursion. The covariance equilibrium proved here does not provide them.

5.8 Checks and the scope of the conclusions

The repository script `code/reproduce_result_checks.py` records the group `inheritance_matrix_and_limits`. It checks the single-locus variances 0.50 and 0.32, twelve matrix interval cases with noncommuting baseline and curvature, one numerical fixed point and its bounds, scalar contractions and several starting values, and the correlated singular-curvature limit 1.84. The chosen matrices are

$$\mathbf{B} = \begin{pmatrix} 1 & 0.25 \\ 0.25 & 0.6 \end{pmatrix}, \qquad \boldsymbol{\Gamma} = \begin{pmatrix} 0.6 & -0.15 \\ -0.15 & 0.3 \end{pmatrix}, \qquad \rho = 0.65. \tag{5.80}$$

They are positive definite and do not commute. Thus their numerical check does not rely on independent trait directions. It complements the proof for all matrices satisfying the assumptions; a finite list of checked cases could not establish that general result.

The source `code/figures.py` generates the main trajectory, while `code/guide_figures.py` generates the scalar worked iteration. The main reproduction script also checks archived forward-time population simulations. Those simulations compare a current-population Gaussian variance formula with empirical moments. Their agreement does not independently validate the fixed-$\mathbf{B}$ inheritance closure derived here.

The covariance results therefore answer a conditional biological question: what happens when a fixed genic baseline, stabilising curvature and a specified retention rule act on Gaussian breeding-value distributions? Within that model, incomplete retention allows a positive covariance equilibrium, and the scalar trajectory converges to it from either side. To ask about changing allele frequencies, linkage architecture or mating systems, those mechanisms must enter the model explicitly. The derivations identify the assumptions that connect selection among parents to variation in the next generation.

Chapter 6

Estimating the share from repeated measurements and counts

The theoretical share requires a breeding-value distribution and a conditional fitness relationship. Neither is directly observed in the worked example. Instead, each genotype has repeated trait measurements and one offspring count. This chapter derives the estimator from that observation model. It then derives predictions on the observed-data scale, the specified shrinkage comparison, and the quartic calculation that separates a local approximation from variation in a complete curve.

The probability calculations are exact under the stated model. Parameter estimates are obtained by numerical optimisation, and bootstrap intervals describe a fitted sampling model. Deriving a likelihood does not prove that its assumptions hold in a biological dataset, that a numerical optimiser finds a global optimum, or that a particular interval has its nominal coverage.

6.1 Specify the observations and the latent quantity

Let A_i be the breeding value of genotype i, let μ be the population mean, and write $X_i = A_i - \mu$. There are n unrelated genotypes and m independent trait measurements per genotype. The teaching model is

$$\begin{aligned} &X_i \sim N(0, G), \\ &Y_{ij} = \mu + X_i + e_{ij}, \qquad\qquad e_{ij} \sim N(0, V_E), \qquad j = 1, \ldots, m. \end{aligned} \tag{6.1}$$

The X_i are independent across genotypes, and all measurement errors are independent of them and of one another. Here V_E is measurement-environment variance. In this example it is not an additional environmental variance to be integrated into an underlying phenotype-fitness function.

Let R_i denote the offspring count, with observed value r_i. It is the instance of reproductive output W in this observation model. A separate assay gives

$$\begin{aligned} R_i \mid X_i = x &\sim \text{Poisson}\{\lambda(x)\}, \\ \lambda(x) = W_A(\mu + x) &= \exp\{\alpha + bx + \tfrac{1}{2}Hx^2\}. \end{aligned} \tag{6.2}$$

Conditional on X_i, reproduction is independent of the environments in the trait measurements. The parameter $\alpha = \ell_A(\mu)$ is log expected output at the population mean. Thus e^α is not generally the population mean output $\mathbb{E}[\lambda(X)]$.

The generating example has $n = 800$, $m = 3$, $G = V_E = 1$, $\mu = 0$, $\alpha = \log 4$, $b = 0.7$, and $H = -0.7$. The trait has centimetre units, G, V_E have cm^2 units, b has cm^{-1} units, and H has cm^{-2} units. The true breeding values are saved separately and are not supplied to the estimator.

Biological interpretation. Repeated measurements distinguish variation within genotypes from variation between them. Calling the latter additive genetic variance requires the additional assumption that the genetic contribution to this trait is wholly additive. Permanent environmental differences, dominance, epistasis or a shared environment with the reproductive assay would change the model. Clonal replication alone does not eliminate those possibilities. The example isolates the estimation problem under explicit assumptions; it is not a general identification argument for clonal data.

6.2 Calibrate the measurement model

Define the genotype mean and the mean across genotypes by

$$\bar{Y}_i = \frac{1}{m}\sum_{j=1}^{m} Y_{ij}, \qquad \bar{Y} = \frac{1}{n}\sum_{i=1}^{n} \bar{Y}_i.$$

Because $\bar{Y}_i = \mu + X_i + \bar{e}_i$, independence gives

$$\mathbb{E}[\bar{Y}_i] = \mu, \qquad \operatorname{Var}(\bar{Y}_i) = G + \frac{V_E}{m}. \tag{6.3}$$

The factor $1/m$ follows from $\operatorname{Var}(m^{-1}\sum_j e_{ij}) = m^{-2}\sum_j V_E = V_E/m$. It is not obtained if the repeated environments share an unmodelled effect.

Proposition 6.1 (Balanced variance calibration)**.** *Under Equation (6.1), with $n \geq 2$ and $m \geq 2$, the estimators*

$$\begin{aligned} \widehat{V}_E &= \frac{\sum_{i=1}^{n}\sum_{j=1}^{m}(Y_{ij} - \bar{Y}_i)^2}{n(m-1)}, \\ \widehat{G} &= \frac{\sum_{i=1}^{n}(\bar{Y}_i - \bar{Y})^2}{n-1} - \frac{\widehat{V}_E}{m}, \qquad \widehat{\mu} = \bar{Y} \end{aligned} \tag{6.4}$$

are unbiased for V_E, G, μ, respectively.

Proof. Within a genotype, subtracting its mean removes both μ and X_i:

$$Y_{ij} - \bar{Y}_i = e_{ij} - \bar{e}_i.$$

Expanding the residual sum of squares gives

$$\begin{aligned} \sum_{j=1}^{m}(e_{ij} - \bar{e}_i)^2 &= \sum_j e_{ij}^2 - 2\bar{e}_i \sum_j e_{ij} + m\bar{e}_i^2 \\ &= \sum_j e_{ij}^2 - m\bar{e}_i^2. \end{aligned}$$

Its expectation is $mV_E - m(V_E/m) = (m-1)V_E$. Summing over n genotypes and dividing by $n(m-1)$ proves the first claim.

For the between-genotype term, put $T = G + V_E/m$. The identity

$$\sum_i (\bar{Y}_i - \bar{Y})^2 = \sum_i (\bar{Y}_i - \mu)^2 - n(\bar{Y} - \mu)^2$$

has expectation $nT - n(T/n) = (n-1)T$, because the genotype means are independent and $\operatorname{Var}(\bar{Y}) = T/n$. Consequently their sample variance estimates $G + V_E/m$ without bias. Subtracting $\widehat{V}_E/m$ gives

$$\mathbb{E}[\widehat{G}] = G + \frac{V_E}{m} - \frac{\mathbb{E}[\widehat{V}_E]}{m} = G.$$

This expectation argument does not require the two estimated terms to be independent. Finally, averaging the genotype means gives $\mathbb{E}[\widehat{\mu}] = \mu$. □

Unbiasedness does not ensure that every realised estimate of G is positive. When within-genotype noise is large relative to the between-genotype variation, the subtraction can be zero or negative. The supplied implementation rejects calibrations with $\widehat{G} \leq 10^{-8}$; it does not replace them with a small positive variance. This is a stated numerical admissibility rule, not a proof that smaller true variances cannot occur.

6.3 Derive the distribution of an uncertain breeding value

For the moment, regard μ, G, V_E as known, with $G, V_E > 0$. Write $d_i = \bar{y}_i - \mu$ for the centred mean of the observed measurements. The normal prior density for $X_i = x$ is proportional to $\exp(-x^2/(2G))$. The conditional measurement density is proportional to

$$\exp\left\{-\frac{1}{2V_E}\sum_j (y_{ij} - \mu - x)^2\right\}.$$

Separate the within-genotype variation from its mean:

$$\begin{aligned}\sum_j (y_{ij} - \mu - x)^2 &= \sum_j \{(y_{ij} - \bar{y}_i) + (d_i - x)\}^2 \\ &= \sum_j (y_{ij} - \bar{y}_i)^2 + m(d_i - x)^2. \end{aligned} \tag{6.5}$$

The cross term vanishes because $\sum_j (y_{ij} - \bar{y}_i) = 0$. The first term does not depend on x. Thus the genotype mean contains all the information about X_i supplied by these balanced measurements when the variance parameters are fixed.

Proposition 6.2 (Normal posterior and prediction reliability)**.** *Define*

$$R_{\text{pred}} = \frac{G}{G + V_E/m}, \qquad M_i = R_{\text{pred}}(\bar{y}_i - \mu), \qquad v = G(1 - R_{\text{pred}}).$$

Then

$$X_i \mid Y_{i1} = y_{i1}, \ldots, Y_{im} = y_{im} \sim N(M_i, v). \tag{6.6}$$

Across populations generated by the model,

$$\text{Var}(M) = R_{\text{pred}} G, \qquad \text{Cov}(X, M) = R_{\text{pred}} G, \qquad \text{Corr}(X, M)^2 = R_{\text{pred}}. \tag{6.7}$$

Proof. Multiplying prior and measurement densities gives a kernel with exponent

$$-\frac{1}{2}\left\{\frac{x^2}{G} + \frac{(x - d_i)^2}{V_E/m}\right\}.$$

Write $T_E = V_E/m$. Expanding and completing the square gives

$$\begin{aligned}\frac{x^2}{G} + \frac{(x - d_i)^2}{T_E} &= \left(\frac{1}{G} + \frac{1}{T_E}\right) x^2 - \frac{2d_i}{T_E} x + \frac{d_i^2}{T_E} \\ &= \frac{(x - M_i)^2}{v} + \frac{d_i^2}{T_E} - \frac{M_i^2}{v},\end{aligned}$$

where

$$v = \left(\frac{1}{G} + \frac{1}{T_E}\right)^{-1} = \frac{G T_E}{G + T_E}, \qquad M_i = v\frac{d_i}{T_E} = \frac{G}{G + T_E} d_i.$$

The terms not involving x are absorbed into the normalising constant. The remaining density is normal with the stated mean and variance.

For the prediction identities, $M = R_{\text{pred}}(X + \bar{e})$, with $\text{Var}(X + \bar{e}) = G + T_E$. Therefore

$$\text{Var}(M) = R_{\text{pred}}^2 (G + T_E) = R_{\text{pred}} G.$$

Independence of X and $\bar{e}$ gives

$$\text{Cov}(X, M) = R_{\text{pred}} \,\text{Cov}(X, X + \bar{e}) = R_{\text{pred}} G.$$

Squaring the correlation and substituting both variances gives

$$\mathrm{Corr}(X, M)^2 = \frac{(R_{\mathrm{pred}}G)^2}{G(R_{\mathrm{pred}}G)} = R_{\mathrm{pred}}.$$

These identities require positive variance for the correlation to be defined. □

The law of total variance makes the meaning of shrinkage explicit:

$$\begin{aligned} G = \mathrm{Var}(X) &= \mathrm{Var}\{\mathbb{E}[X \mid Y_1, \ldots, Y_m]\} + \mathbb{E}\{\mathrm{Var}(X \mid Y_1, \ldots, Y_m)\} \\ &= R_{\mathrm{pred}}G + v. \end{aligned} \tag{6.8}$$

The variance among predictions is smaller than the biological variance because the predictions omit unresolved variation. It is not evidence that the population itself has contracted. As V_E/m tends to zero, R_{pred} tends to one and v tends to zero. As measurement noise increases without bound, the predictions tend to zero while uncertainty tends to the prior variance G.

The implementation substitutes $\widehat{\mu}, \widehat{G}, \widehat{V}_E$ into these expressions. The resulting normal distributions are conditional on calibrated parameter values. They are not full Bayesian posteriors that also integrate over uncertainty in those values. Repeating calibration in the bootstrap addresses their sampling uncertainty under the specified model. Using predictions as error-free covariates discards the unresolved variance, a concern discussed by Hadfield et al. (2010).

6.4 Construct the likelihood for offspring counts

Write $\theta = (\alpha, b, H)^{\mathsf{T}}$ and

$$\eta_\theta(x) = \alpha + bx + \tfrac{1}{2}Hx^2, \qquad \lambda_\theta(x) = e^{\eta_\theta(x)}.$$

Given the calibrated measurements, the count probability for genotype i is obtained by averaging over its uncertain breeding value:

$$\begin{aligned} L_i(\theta) &= P(R_i = r_i \mid y_{i1}, \ldots, y_{im}; \theta) \\ &= \int_{-\infty}^{\infty} \frac{\lambda_\theta(x)^{r_i} e^{-\lambda_\theta(x)}}{r_i!} \phi(x; M_i, v)\, dx \\ &= \int \exp\{r_i \eta_\theta(x) - e^{\eta_\theta(x)} - \log(r_i!)\} \phi(x; M_i, v)\, dx. \end{aligned} \tag{6.9}$$

Here $\phi(x; M_i, v)$ is a normal density with variance v. The log likelihood is $\ell(\theta) = \sum_i \log L_i(\theta)$. The logarithms are of probabilities; no logarithm of r_i is taken.

For a zero count the Poisson factor is $e^{-\lambda_\theta(x)}$, since $0! = 1$. Zero counts therefore contribute ordinary, finite probabilities. For $v = 0$, the conditional distribution of X_i is concentrated at M_i, so

$$L_i(\theta) = \frac{\lambda_\theta(M_i)^{r_i} e^{-\lambda_\theta(M_i)}}{r_i!}.$$

That is the known-covariate limit. It is not the general case when $v > 0$. In particular, averaging a Poisson probability is not the same operation as evaluating it at the posterior mean or at the average rate.

This is a two-stage estimator. The measurement likelihood supplies the calibration; the count likelihood then estimates the fitness coefficients. Conditional on fixed calibration, genotypes contribute independent factors L_i. A joint analysis would also estimate the measurement parameters using all components of a joint likelihood. The example does not claim equivalent efficiency to such an analysis.

6.5 From a normal integral to an evaluated likelihood

6.5.1 The Hermite convention and normalisation

Substitute $x = M_i + \sqrt{v}\,t$ into Equation (6.9). Since

$$\phi(x; M_i, v)\,dx = \frac{1}{\sqrt{2\pi}} e^{-t^2/2}\,dt,$$

the integral is a standard normal expectation. Let t_j and u_j, $j = 1, \ldots, q$, be Gaussian quadrature nodes and weights for the weight function $e^{-t^2/2}$. Define $w_j = u_j/\sqrt{2\pi}$. Then

$$L_i(\theta) \;\simeq\; \sum_{j=1}^{q} w_j \exp\{r_i \eta_\theta(x_{ij}) - e^{\eta_\theta(x_{ij})} - \log(r_i!)\}, \quad x_{ij} = M_i + \sqrt{v}\,t_j. \tag{6.10}$$

The weights w_j are positive and sum to one. This is the convention used by `roots_hermitenorm`. It differs from the convention with weight e^{-t^2}, for which both the scaling of the nodes and the normalising constant differ.

Proposition 6.3 (Why the normal quadrature rule integrates polynomials)**.** *The q-node rule above integrates every polynomial of degree at most $2q - 1$ exactly under the standard normal distribution.*

Proof. Let He_q be the degree-q Hermite polynomial associated with the standard normal weight. Its defining derivative formula is

$$\mathrm{He}_q(t) = (-1)^q e^{t^2/2} \frac{d^q}{dt^q} e^{-t^2/2}.$$

If a is a polynomial of degree below q, integration by parts q times gives $\int a(t)\mathrm{He}_q(t)\phi(t; 0, 1)\,dt = 0$. The boundary terms vanish because a polynomial times a Gaussian density and its derivatives tends to zero at both endpoints; the final derivative of a is zero.

An orthogonal degree-q polynomial has q distinct real roots. To see why, multiply its factors corresponding to real sign changes. If there were fewer than q, this multiplier would have degree below q, while its product with He_q would have one nonzero sign on the real line except at isolated roots. Its Gaussian integral could not be zero, in contradiction to orthogonality. Thus the q nodes can be chosen as those distinct roots.

For a polynomial p of degree at most $2q - 1$, polynomial division gives $p = a\,\mathrm{He}_q + r$, where both a and r have degree at most $q - 1$. Orthogonality removes the first term from the integral. At each node, $\mathrm{He}_q(t_j) = 0$, so $p(t_j) = r(t_j)$. Interpolate r using the Lagrange polynomials l_j through the nodes:

$$r(t) = \sum_j r(t_j) l_j(t), \qquad w_j = \int l_j(t)\phi(t; 0, 1)\,dt.$$

Integrating gives $\int p\phi = \sum_j w_j p(t_j)$, proving exactness. The constant polynomial gives $\sum_j w_j = 1$. Applying exactness to l_j^2, of degree $2q - 2$, also gives

$$w_j = \sum_k w_k l_j(t_k)^2 = \int l_j(t)^2 \phi(t; 0, 1)\,dt > 0.$$

□

The general Gaussian quadrature construction is described in National Institute of Standards and Technology (2026). Exactness for polynomials explains checks such as $\sum_j w_j t_j^2 = 1$ and $\sum_j w_j t_j^4 = 3$. The Poisson-mixture integrand is not a polynomial, so this theorem does not make a finite-order likelihood exact. Its numerical accuracy must be assessed separately.

6.5.2 Evaluate sums without losing numerical range

For a particular genotype, set

$$a_{ij} = \log w_j + r_i \eta_\theta(x_{ij}) - e^{\eta_\theta(x_{ij})} - \log(r_i!).$$

Direct exponentiation can underflow when every probability is small. If $c_i = \max_j a_{ij}$, the identity

$$\log \sum_j e^{a_{ij}} = c_i + \log \sum_j e^{a_{ij} - c_i} \tag{6.11}$$

evaluates the same expression with all exponentials at most one. The implementation uses `logsumexp`, and evaluates $\log(r_i!)$ as $\log \Gamma(r_i + 1)$.

6.5.3 Derive the score used by the optimiser

Let

$$\boldsymbol{d}(x) = (1, x, x^2/2)^\mathsf{T}, \qquad \eta_\theta(x) = \boldsymbol{d}(x)^\mathsf{T}\theta.$$

The derivative of one log Poisson probability is

$$\begin{aligned} \nabla_\theta \{ r_i \eta_\theta(x) - e^{\eta_\theta(x)} - \log(r_i!) \} &= r_i \boldsymbol{d}(x) - e^{\eta_\theta(x)} \boldsymbol{d}(x) \\ &= (r_i - \lambda_\theta(x)) \boldsymbol{d}(x). \end{aligned}$$

For the finite quadrature sum, differentiating its logarithm therefore gives

$$\nabla_\theta \log L_i \simeq \sum_j \pi_{ij} (r_i - \lambda_\theta(x_{ij})) \boldsymbol{d}(x_{ij}), \qquad \pi_{ij} = \frac{e^{a_{ij}}}{\sum_k e^{a_{ik}}}. \tag{6.12}$$

The same identity holds for the exact integral with a conditional expectation in place of the sum when differentiation under the integral is justified. For the finite implemented sum the derivative is an ordinary derivative of a finite positive expression.

There are two different conditional distributions in this calculation. The normal $N(M_i, v)$ describes breeding value after observing the heights. The weights π_{ij} additionally use the count likelihood, and approximate the breeding-value distribution after observing both heights and reproduction. They sum to one, but are generally not the original normal quadrature weights. The implementation sums Equation (6.12) over genotypes and minimises the negative mean log likelihood; dividing by n does not change its minimiser.

6.5.4 Numerical checks are part of the estimator

The primary model uses 41 quadrature nodes initially and constrains $H \leq 0$. Its numerical coefficient bounds are

$$-7 \leq \alpha \leq 7, \qquad -5 \leq b \leq 5, \qquad -6 \leq H \leq 0.$$

These are bounds in the example's stated units. They are not universal biological limits. Multiple starting values reduce dependence on a particular initial guess, and a fit near a bound is flagged. Neither step proves global optimality of a marginal likelihood.

The integration check refits at increasing orders, including 81, 161 and 321. For consecutive orders, it requires both

$$\begin{aligned} \max_k \frac{|\theta_{\mathrm{new},k} - \theta_{\mathrm{old},k}|}{1 + |\theta_{\mathrm{new},k}|} &< 10^{-4}, \\ \frac{|\ell_{\mathrm{new}} - \ell_{\mathrm{old}}|}{n} &< 10^{-6}. \end{aligned} \tag{6.13}$$

When the comparison passes, the lower-order fit is returned with the higher check order recorded. An unresolved comparison raises an error. For the recorded default data, comparing 41 and 81 nodes changes $b, H, \mathcal{A}_g$ by less than 4×10^{-13}. That is a reproducible numerical result for this dataset, not an analytic error bound for every possible observation. An independent adaptive integral and a closed mean formula provide additional checks in the supplied tests.

6.6 From fitted coefficients to the variance share

The fitted log conditional mean is

$$\widehat{\ell}_A(\widehat{\mu} + x) = \widehat{\alpha} + \widehat{b}x + \tfrac{1}{2}\widehat{H}x^2.$$

Its derivatives are

$$\frac{d\widehat{\ell}_A}{dx} = \widehat{b} + \widehat{H}x, \qquad \frac{d^2\widehat{\ell}_A}{dx^2} = \widehat{H}.$$

Thus the fitted coefficients give slope and curvature at the estimated population mean, $x = 0$. If software instead fits cx^2, the second derivative is $2c$, so $H = 2c$. The design in the supplied code uses $x^2/2$ and requires no subsequent doubling.

Under the Gaussian quadratic model, the variance partition derived in Proposition 2.1 gives the plug-in quantities

$$\widehat{V_{\text{lin}}} = \widehat{b}^2\widehat{G}, \qquad \widehat{V_{\text{quad}}} = \tfrac{1}{2}\widehat{H}^2\widehat{G}^2, \qquad \widehat{\mathcal{A}_g} = \frac{\widehat{V_{\text{lin}}}}{\widehat{V_{\text{lin}}} + \widehat{V_{\text{quad}}}}. \tag{6.14}$$

They describe variation in log expected output across breeding values. Poisson variation in individual counts is not an extra term in their denominator. When both components vanish, the share is undefined.

For the generating parameters, the calculation is exact:

$$V_{\text{lin}} = 0.7^2(1) = 0.49, \qquad V_{\text{quad}} = \tfrac{1}{2}(-0.7)^2(1)^2 = 0.245, \qquad \mathcal{A}_g = \frac{0.49}{0.735} = \frac{2}{3}.$$

The fitted default value, approximately 0.610744, is a result of maximising the likelihood for the recorded simulated observations. There is no closed-form algebraic argument that a finite dataset must return that value or the generating value. Reporting both components alongside the ratio also shows whether a similar share is supported by a similar amount of variation.

6.7 Predict quantities on the observed-data scale

A graph of counts against observed heights asks for the count distribution given those measurements. It does not ask for the count distribution at an exact breeding value. Fix a genotype's measurement information and write $X \mid Y \sim N(M, v)$. In this section parameters are treated as fixed.

Proposition 6.4 (Predictive moments and zero counts)**.** *For a Poisson mixture with conditional rate $\lambda(X)$,*

$$\begin{aligned} \mu_R(Y) &= \mathbb{E}[R \mid Y] = \mathbb{E}[\lambda(X) \mid Y], \\ s_R^2(Y) &= \operatorname{Var}(R \mid Y) = \mathbb{E}[\lambda(X) \mid Y] + \operatorname{Var}(\lambda(X) \mid Y), \\ p_0(Y) &= P(R = 0 \mid Y) = \mathbb{E}[e^{-\lambda(X)} \mid Y]. \end{aligned} \tag{6.15}$$

The first two identities assume the indicated moments are finite.

Proof. Conditional on X, a Poisson variable has mean and variance $\lambda(X)$. Conditional independence of reproduction and the measurement environments therefore gives

$$\mathbb{E}[R \mid Y] = \mathbb{E}\{\mathbb{E}[R \mid X, Y] \mid Y\} = \mathbb{E}[\lambda(X) \mid Y].$$

The law of total variance, conditional on Y, gives

$$\begin{aligned}\operatorname{Var}(R \mid Y) &= \mathbb{E}\{\operatorname{Var}(R \mid X, Y) \mid Y\} + \operatorname{Var}\{\mathbb{E}[R \mid X, Y] \mid Y\} \\ &= \mathbb{E}[\lambda(X) \mid Y] + \operatorname{Var}(\lambda(X) \mid Y).\end{aligned}$$

Finally $P(R = 0 \mid X, Y) = e^{-\lambda(X)}$; averaging that probability over $X \mid Y$ proves the third identity. □

The supplied routine evaluates these as weighted normal expectations. If $\lambda_j = \lambda(M + \sqrt{v}\, t_j)$, it returns

$$\mu_R \simeq \sum_j w_j \lambda_j, \qquad s_R^2 \simeq \mu_R + \sum_j w_j \lambda_j^2 - \mu_R^2, \qquad p_0 \simeq \sum_j w_j e^{-\lambda_j}.$$

The second term in the variance accounts for uncertainty in the rate arising from uncertain breeding value. Replacing the mixture by a single Poisson distribution with rate μ_R would discard that contribution.

6.7.1 A closed-form prediction for the quadratic log mean

For the quadratic model, the first two rate moments can also be obtained analytically. This provides a check independent of the likelihood optimisation.

Proposition 6.5 (Normal average of an exponential quadratic)**.** *Let $X \sim N(M, v)$, with $v > 0$, and $\lambda(X) = \exp(\alpha + bX + HX^2/2)$. For $k > 0$, put $D_k = 1 - kHv$. If $D_k > 0$, then*

$$\mathbb{E}[\lambda(X)^k] = D_k^{-1/2} \exp\left\{ k\alpha + kbM + \frac{kHM^2}{2} + \frac{k^2 v (b + HM)^2}{2D_k} \right\}. \tag{6.16}$$

If $D_k \le 0$, that moment is not finite. For $v = 0$, it is simply $\lambda(M)^k$.

Proof. Write $X = M + \sqrt{v}\, T$, with $T \sim N(0, 1)$, and expand:

$$k\alpha + kbX + \frac{kH}{2} X^2 = k\alpha + kbM + \frac{kHM^2}{2} + k\sqrt{v}\,(b + HM)T + \frac{kHv}{2} T^2.$$

Multiplying by the standard normal density combines the last quadratic term with $-T^2/2$. Let $d = k\sqrt{v}\,(b + HM)$. The remaining integral is

$$\frac{1}{\sqrt{2\pi}} \int \exp\left(-\frac{D_k}{2} t^2 + dt \right) dt.$$

When $D_k > 0$, complete the square:

$$-\frac{D_k}{2} t^2 + dt = -\frac{D_k}{2} \left(t - \frac{d}{D_k} \right)^2 + \frac{d^2}{2D_k}.$$

The shifted normal integral contributes $D_k^{-1/2}$, and the constant contributes $\exp\{d^2/(2D_k)\}$. Restoring the terms taken outside the integral proves the formula. If $D_k < 0$, the exponent grows quadratically in both tails. If $D_k = 0$, a nonzero linear term grows in one tail, while a zero linear term leaves a constant integrand on the real line. Each case has an infinite integral. □

Taking $k = 1$ gives the fitted mean given measurement information; taking $k = 2$ supplies the second rate moment in Equation (6.15). The primary constraint $H \leq 0$ makes both conditions automatic. If a broader model allowed positive H, a finite probability integral would not by itself establish finite predictive moments. A Poisson probability is bounded by one, whereas $\lambda(X)$ and $\lambda(X)^2$ are not.

Substituting $M = 0, v = G$ in the first-moment formula also gives population mean output under the generating distribution:

$$\mathbb{E}[\lambda(X)] = \frac{e^{\alpha}}{\sqrt{1 - HG}} \exp\left\{ \frac{Gb^2}{2(1 - HG)} \right\}, \qquad 1 - HG > 0.$$

This shows explicitly why the setting $e^{\alpha} = 4$ describes output at the mean breeding value rather than an average over all genotypes.

6.7.2 Sampling bars for observed bins

The diagnostic plot sorts individuals by their observed mean measurements and forms ten groups. Conditional on all these measurements, group membership is fixed. For a group B containing n_B independent genotypes, define $\bar{R}_B = n_B^{-1} \sum_{i \in B} R_i$. Conditional independence gives

$$\begin{aligned} \mathbb{E}[\bar{R}_B \mid Y] &= \frac{1}{n_B} \sum_{i \in B} \mu_R(Y_i), \\ \mathrm{Var}(\bar{R}_B \mid Y) &= \frac{1}{n_B^2} \sum_{i \in B} s_R^2(Y_i). \end{aligned} \tag{6.17}$$

The second identity follows by expanding the variance of a sum; the cross-genotype covariance terms are zero under the model. The displayed half-width is

$$1.96 \, \frac{\sqrt{\sum_{i \in B} \widehat{s}_R^2(Y_i)}}{n_B}.$$

The multiplier uses a normal approximation for a bin average. The calculation conditions on fitted parameters and is displayed around the observed mean. It is not a confidence interval for the conditional fitness curve, a simultaneous band, or a calculation that includes fitting uncertainty. The predicted zero fraction in a group is likewise the average of the individual mixture probabilities, $n_B^{-1} \sum_{i \in B} \widehat{p}_0(Y_i)$.

Biological interpretation. An observed mean height does not identify an exact breeding value. Comparing its count directly with $W_A(M_i)$ would disregard the range of breeding values compatible with those measurements. Integrating over that range is required both when fitting counts and when checking predictions against observable quantities. Shared environments or related genotypes would also introduce covariance terms omitted by the bin formula.

6.8 A specific effect of substituting prediction variance

The reliability calculation distinguishes biological variance G from prediction variance. The following result isolates the consequence of substituting the latter while keeping slope and curvature fixed on the true breeding-value scale.

Proposition 6.6 (Fixed-coefficient shrinkage substitution)**.** *Let $L = V_{\mathrm{lin}} > 0$, $Q = V_{\mathrm{quad}} > 0$, and replace* $\mathbf{G}$ *by* $s\mathbf{G}$*, where $0 < s < 1$, without changing* $\boldsymbol{b}$ *or* $\mathbf{H}$*. Then*

$$L_{\mathrm{pred}} = sL, \qquad Q_{\mathrm{pred}} = s^2 Q, \qquad \mathcal{A}_{g,\mathrm{pred}} = \frac{L}{L + sQ} > \frac{L}{L + Q} = \mathcal{A}_g. \tag{6.18}$$

Proof. The component formulas give

$$\boldsymbol{b}^{\mathsf{T}}(s\mathbf{G})\boldsymbol{b} = sL, \qquad \tfrac{1}{2}\operatorname{tr}\{\mathbf{H}(s\mathbf{G})\mathbf{H}(s\mathbf{G})\} = s^2Q.$$

Cancel the positive s in their ratio. The difference between the new share and the target is

$$\frac{L}{L+sQ} - \frac{L}{L+Q} = \frac{LQ(1-s)}{(L+sQ)(L+Q)} > 0.$$

□

In the balanced scalar measurement model, $s = R_{\text{pred}}$. In the recurring guide example, $L = 4, Q = 0.125, s = 0.6$, so the substituted share is

$$\frac{4}{4 + 0.6(0.125)} = 0.981595\ldots,$$

compared with $4/(4 + 0.125) = 0.969697\ldots$. Both absolute components have decreased. If $L = 0 < Q$, the share stays zero; if $Q = 0 < L$, it stays one. If both are zero, it is undefined. At $s = 0$, both substituted components vanish and the ratio is undefined; the limit as positive s tends to zero must not be mistaken for its value at zero.

The deliberately naive fit in the worked example makes a different change. It sets the latent conditional variance to zero, refits α, b, H using the predictions as exact covariates, and substitutes their empirical variance for G. The fixed-coefficient proof therefore does not establish the direction of its overall error. For the recorded dataset, its share is approximately 0.673984, closer to 2/3 than the latent-value point estimate, while both component estimates are smaller than their generating values. Agreement of one ratio in one sample does not validate that observation model. Heterogeneous reliabilities across traits also need not produce a scalar covariance multiplier.

6.9 What the parametric bootstrap repeats

Write D for the complete observable dataset and $T(D)$ for the complete estimation procedure: calibrate the measurement model, fit the count model, and calculate the components and share. The target sampling distribution is the distribution of $T(D)$ under repeated data generated by the specified population model. Because its parameters are unknown, the parametric bootstrap substitutes their fitted values (Efron, 1979).

For each bootstrap replicate, the supplied algorithm generates

$$\begin{aligned} X_i^* &\sim N(0, \widehat{G}), \\ Y_{ij}^* &= \widehat{\mu} + X_i^* + e_{ij}^*, \qquad & e_{ij}^* &\sim N(0, \widehat{V}_E), \\ R_i^* \mid X_i^* &\sim \text{Poisson}\left[\exp\{\widehat{\alpha} + \widehat{b}X_i^* + \tfrac{1}{2}\widehat{H}(X_i^*)^2\}\right]. \end{aligned} \tag{6.19}$$

It then calculates $T(D^*)$, including new estimates of μ, G, V_E. Using fixed calibrated values during these refits would omit their sampling variation and their association with the fitted fitness coefficients. The simulator centres new breeding values at zero relative to $\widehat{\mu}$; recalibrating the new data includes the sampling variation in the estimated centre.

For N_{boot} completed replicates, let $T_1^*, \ldots, T_{N_{\text{boot}}}^*$ be the estimates of one quantity. Their empirical distribution is

$$\widehat{F}_{\text{boot}}(t) = \frac{1}{N_{\text{boot}}} \sum_{j=1}^{N_{\text{boot}}} \mathbf{1}_{\{T_j^* \le t\}}.$$

The percentile interval uses its 0.025 and 0.975 quantiles. This specifies how the interval is constructed. It does not prove exact 95% coverage. The procedure approximates a sampling

distribution under the fitted model; bias, boundaries, nonregular estimation or misspecification can affect coverage.

There are two separate sources of variation. With the original data held fixed, increasing N_{boot} makes the simulated empirical distribution less variable. It does not make the fitted biological or count model more appropriate. To assess repeated-sample coverage, one would simulate many independent original datasets, construct an interval for each, and calculate the fraction containing the relevant generating quantity.

The default reproduction uses 500 replicates with seed 20261006, after the observations generated with seed 20261005. All 500 recorded refits complete. That statement concerns computational success. The code records failures. If $N_{\mathrm{requested}}$ fits were requested, it returns intervals only when at least $\max\{2, \lfloor 0.9 N_{\mathrm{requested}} \rfloor\}$ have completed. The floor denotes rounding down to an integer. That numerical rule is not a statistical justification for ignoring failed samples: their causes and relation to the parameter estimates must still be examined. The successful default run does not establish interval coverage.

Biological interpretation. These intervals account for repeated sampling under the assumed additive trait, independent measurement environments and conditional Poisson reproduction. They do not express uncertainty that these assumptions are biologically correct. Adding bootstrap replicates cannot account for unmodelled relatedness, nonadditive variation, shared environments or a wrong fitness function.

6.10 A quartic alternative and the scope of the AIC comparison

The sensitivity fit extends the log mean to

$$\eta_4(x) = \alpha + bx + \tfrac{1}{2} H x^2 + c_3 x^3 + c_4 x^4. \tag{6.20}$$

It keeps the measurement calibration fixed between the two fitted families. The quadratic family has three fitness parameters; the quartic family has five. Applying the AIC expression $2k - 2\ell$ to their conditional likelihoods gives

$$\begin{aligned} \Delta\mathrm{AIC} &= \mathrm{AIC}_4 - \mathrm{AIC}_2 \\ &= (2 \cdot 5 - 2\ell_4) - (2 \cdot 3 - 2\ell_2) \\ &= 4 - 2(\ell_4 - \ell_2). \end{aligned} \tag{6.21}$$

A negative difference favours the more flexible fit under this criterion (Akaike, 1974). The reported default difference, approximately 3.410883, comes from the two numerical fitted log likelihoods.

This is an approximate conditional-likelihood comparison, not a formal test of the whole measurement-and-reproduction model. Both fits use the same estimated calibration. The primary family restricts $H \le 0$; the quartic family allows H up to three but constrains its highest coefficient to $-1 \le c_4 \le -10^{-8}$, with $-2 \le c_3 \le 2$. Negative c_4 makes the log mean decline in both tails and keeps population rate moments finite. The exactly quadratic case $c_4 = c_3 = 0$ lies on the limiting boundary of the quartic family, not in the interior of these numerical bounds. Ordinary unconstrained likelihood-ratio calibrations should not be inferred from the comparison.

6.10.1 Why the cubic term is needed when the centre moves

Suppose a generating curve is written in a coordinate x as

$$\ell(x) = \alpha + bx + \tfrac{1}{2} H x^2 - \kappa x^4.$$

Move the reference point by δ, so $x = z + \delta$. Expanding $(z+\delta)^2 = z^2 + 2\delta z + \delta^2$ and

$$(z+\delta)^4 = z^4 + 4\delta z^3 + 6\delta^2 z^2 + 4\delta^3 z + \delta^4$$

gives

$$\begin{aligned} \ell(z+\delta) &= \alpha_\delta + b_\delta z + \tfrac{1}{2} H_\delta z^2 - 4\kappa\delta z^3 - \kappa z^4, \\ \alpha_\delta &= \alpha + b\delta + \tfrac{1}{2} H\delta^2 - \kappa\delta^4, \\ b_\delta &= b + H\delta - 4\kappa\delta^3, \qquad H_\delta = H - 12\kappa\delta^2. \end{aligned} \tag{6.22}$$

A quartic curve with no cubic term at one centre generally has a cubic term at another. Including both terms allows the fitted family to represent that change of centre. In the design used here, c_3 multiplies z^3 and c_4 multiplies z^4 directly; the third and fourth derivatives at zero are $6c_3$ and $24c_4$. The coefficient H still multiplies $z^2/2$ and is the second derivative at zero.

6.11 Local quadratic variation and the complete quartic curve

Let $X \sim N(0, G)$, and consider the exact generating log mean

$$\ell(X) = \alpha + bX + \tfrac{1}{2} HX^2 - \kappa X^4, \qquad \kappa \geq 0. \tag{6.23}$$

The fourth-order term has first derivative $-4\kappa x^3$ and second derivative $-12\kappa x^2$, both zero at $x = 0$. Therefore its local quadratic approximation at the population mean still has slope b, curvature H, and the share derived from $b^2 G$ and $H^2 G^2/2$. Its complete population variance requires additional moments.

Proposition 6.7 (Variance and affine fit of the quartic curve)**.** *Under Equation (6.23),*

$$\begin{aligned} \operatorname{Var}\{\tfrac{1}{2} HX^2 - \kappa X^4\} &= \tfrac{1}{2} H^2 G^2 - 12 H\kappa G^3 + 96\kappa^2 G^4, \\ R^2_{\text{full}} &= \frac{b^2 G}{b^2 G + \tfrac{1}{2} H^2 G^2 - 12 H\kappa G^3 + 96\kappa^2 G^4}, \end{aligned} \tag{6.24}$$

provided the full-curve variance is positive. The second expression is the population affine-regression R^2 of the complete log mean on X.

Proof. The Gaussian moment results in Chapter 2 give

$$\mathbb{E}[X^2] = G, \qquad \mathbb{E}[X^4] = 3G^2, \qquad \mathbb{E}[X^6] = 15G^3, \qquad \mathbb{E}[X^8] = 105G^4.$$

Consequently,

$$\begin{aligned} \operatorname{Var}(X^2) &= \mathbb{E}[X^4] - \mathbb{E}[X^2]^2 = 3G^2 - G^2 = 2G^2, \\ \operatorname{Cov}(X^2, X^4) &= \mathbb{E}[X^6] - \mathbb{E}[X^2]\mathbb{E}[X^4] = 15G^3 - 3G^3 = 12G^3, \\ \operatorname{Var}(X^4) &= \mathbb{E}[X^8] - \mathbb{E}[X^4]^2 = 105G^4 - 9G^4 = 96G^4. \end{aligned}$$

Apply the variance-of-a-sum identity, retaining the cross term:

$$\begin{aligned} \operatorname{Var}\{\tfrac{1}{2} HX^2 - \kappa X^4\} &= \frac{H^2}{4} \operatorname{Var}(X^2) + \kappa^2 \operatorname{Var}(X^4) + 2\left(\frac{H}{2}\right)(-\kappa) \operatorname{Cov}(X^2, X^4) \\ &= \tfrac{1}{2} H^2 G^2 + 96\kappa^2 G^4 - 12 H\kappa G^3. \end{aligned}$$

All centred odd Gaussian moments vanish. Therefore

$$\operatorname{Cov}(X, X^2) = 0, \qquad \operatorname{Cov}(X, X^4) = 0,$$

and the linear term is uncorrelated with the even part. It follows that $\mathrm{Cov}\{X, \ell(X)\} = bG$. For $G > 0$, the population least-squares slope is $\mathrm{Cov}\{X, \ell(X)\}/G = b$. Its intercept is the population mean

$$\mathbb{E}[\ell(X)] = \alpha + \tfrac{1}{2}HG - 3\kappa G^2.$$

The fitted component therefore has variance b^2G, and the total variance is this component plus the even-term variance just derived. Dividing proves Equation (6.24). If $G = 0$, the target is constant and its regression R^2 is undefined. □

At $b = 0.7, H = -0.7, G = 1, \kappa = 0.08$, the local quadratic components remain 0.49 and 0.245, so their share is 2/3. For the complete curve, the two extra contributions to the even-part variance are

$$-12H\kappa G^3 = 0.672, \qquad 96\kappa^2 G^4 = 0.6144.$$

Thus

$$R^2_{\text{full}} = \frac{0.49}{0.49 + 0.245 + 0.672 + 0.6144} = 0.242406253\ldots.$$

The change is not a contradiction: the numerator describes the same linear term, while the full-curve denominator includes additional population variation absent from the local quadratic approximation.

Biological interpretation. Slope and curvature at the population mean do not determine how much higher orders contribute over all breeding values present. A fitted quadratic can estimate a local description accurately while describing the complete curve poorly. Its bootstrap interval cannot correct that difference in target. Even the local share extracted from a fitted quartic is not the R^2 of the complete fitted quartic unless the higher terms contribute no additional relevant variation.

6.12 Relating the derivations to the implementation

The functions in `examples/estimating_variance_share/model.py` correspond directly to the steps above: `calibrate` estimates the measurement model; `fit_surface` evaluates and optimises the mixture likelihood; `checked_surface` assesses integration stability; `conditional_prediction` produces observable-data means, variances and zero probabilities; `bootstrap` repeats the two-stage analysis; and `truth` calculates the generating local share and complete quartic R^2. The batch calculation and Shiny app call these same functions.

The accompanying `test_model.py` checks normal moments, separation of observations from simulation truth, the closed mean formula, an independent adaptive likelihood integral, the known-value limit, repeated-seed recovery, bootstrap recalibration, quartic variation, and explicit numerical failures. Those checks establish reproducibility and agreement for the tested cases. The proofs in this chapter identify the assumptions under which the formulas hold; they do not turn those finite checks into evidence that an observation model is suitable for every dataset.

Chapter 7

Converting phenotype-regression moments to breeding-value parameters

A published selection regression usually describes relative fitness as a function of measured phenotype. The variance share in this project instead uses log expected output conditional on breeding value. Their coefficients cannot generally be equated. Under stated conditions, the phenotype regression supplies selected moments, those moments can be transferred to breeding-value coordinates, and a log-quadratic model can be chosen to reproduce them. This chapter derives each operation and identifies when that model also recovers the actual conditional log-fitness derivatives.

The calculations use the tower property from Chapter 1, Gaussian fourth moments from Chapter 2, and the Gaussian selection formulas in Chapter 4. Matrix order matters throughout: covariance and curvature matrices need not commute.

7.1 Define the regression and its reference population

Let $\boldsymbol{Z}$ be the measured phenotype, let $\boldsymbol{x} = \boldsymbol{Z} - \mathbb{E}[\boldsymbol{Z}]$, and write $\mathbf{P} = \mathrm{Cov}(\boldsymbol{Z})$. Let $W \geq 0$ be output in the same population and selection episode, with $0 < \bar{W} = \mathbb{E}[W] < \infty$, and define relative fitness $w = W/\bar{W}$. Thus $\mathbb{E}[w] = 1$.

Consider the population least-squares regression

$$w = \alpha_w + \boldsymbol{\beta}^{\mathsf{T}}\boldsymbol{x} + \frac{1}{2}\boldsymbol{x}^{\mathsf{T}}\boldsymbol{\gamma}\boldsymbol{x} + \varepsilon, \qquad \boldsymbol{\gamma} = \boldsymbol{\gamma}^{\mathsf{T}}. \tag{7.1}$$

We use α_w to distinguish this relative-fitness intercept from the log-mean intercept in Chapter 6. This is the linear–quadratic phenotype regression of selection analysis (Lande and Arnold, 1983). Assume $\mathbb{E}[w^2] < \infty$, so that the population least-squares problem and the Gaussian-weighted moments used below are well defined. The normal equations for least squares give

$$\mathbb{E}[\varepsilon] = 0, \qquad \mathbb{E}[x_i\varepsilon] = 0, \qquad \mathbb{E}[x_i x_j \varepsilon] = 0 \quad \text{for every } i, j. \tag{7.2}$$

These equations require orthogonality to the fitted regressors. They do not require $\mathbb{E}[\varepsilon \mid \boldsymbol{x}] = 0$ at every phenotype. Consequently the moment calculations below can hold when the fitted polynomial is not the exact conditional mean.

The factor $1/2$ applies to the entire quadratic form. Expanding it gives

$$\frac{1}{2}\boldsymbol{x}^{\mathsf{T}}\boldsymbol{\gamma}\boldsymbol{x} = \frac{1}{2}\sum_j \gamma_{jj}x_j^2 + \sum_{j<k}\gamma_{jk}x_j x_k. \tag{7.3}$$

Thus a software coefficient c_{jj} multiplying x_j^2 corresponds to $\gamma_{jj} = 2c_{jj}$. A coefficient multiplying the single cross-product column $x_j x_k$ corresponds to γ_{jk} itself: the two symmetric matrix entries have already combined. This convention must be checked before using published coefficients (Stinchcombe et al., 2008). The coefficients must also use compatible trait units and the same population centring.

7.2 Recover selected phenotype moments

Assume now that $\boldsymbol{x} \sim N(\mathbf{0}, \mathbf{P})$. For this section, the formulas do not require inverses, although an identifiable full quadratic regression is most conveniently written on the positive-variance support.

Proposition 7.1 (Moments supplied by the phenotype regression)**.** *Under the regression orthogonality and Gaussian assumptions,*

$$\begin{aligned} \alpha_w &= 1 - \frac{1}{2}\operatorname{tr}(\boldsymbol{\gamma}\mathbf{P}), \\ \mathbb{E}[\boldsymbol{x}w] &= \mathbf{P}\boldsymbol{\beta}, \\ \mathbb{E}[\boldsymbol{x}\boldsymbol{x}^{\mathsf{T}}w] &= \mathbf{P} + \mathbf{P}\boldsymbol{\gamma}\mathbf{P}. \end{aligned} \tag{7.4}$$

The selected phenotype covariance is consequently

$$\mathbf{P}_{\mathrm{sel}} = \mathbf{P} + \mathbf{P}(\boldsymbol{\gamma} - \boldsymbol{\beta}\boldsymbol{\beta}^{\mathsf{T}})\mathbf{P}. \tag{7.5}$$

Proof. First take expectations in Equation (7.1). The linear term and residual have mean zero, while

$$\mathbb{E}[\boldsymbol{x}^{\mathsf{T}}\boldsymbol{\gamma}\boldsymbol{x}] = \sum_{j,k} \gamma_{jk}\mathbb{E}[x_j x_k] = \sum_{j,k} \gamma_{jk} P_{jk} = \operatorname{tr}(\boldsymbol{\gamma}\mathbf{P}).$$

Since $\mathbb{E}[w] = 1$, this proves the intercept formula. It is a normalisation of the fitted average over the population, not a statement that fitted relative fitness at the mean phenotype equals one.

Next multiply the regression by x_i and average:

$$\begin{aligned} \mathbb{E}[x_i w] &= \alpha_w \mathbb{E}[x_i] + \sum_j \beta_j \mathbb{E}[x_i x_j] + \frac{1}{2}\sum_{j,k} \gamma_{jk}\mathbb{E}[x_i x_j x_k] + \mathbb{E}[x_i \varepsilon] \\ &= \sum_j P_{ij}\beta_j. \end{aligned}$$

The mean and all third centred Gaussian moments vanish, and the residual term vanishes by the normal equations. In vector form this is $\mathbb{E}[\boldsymbol{x}w] = \mathbf{P}\boldsymbol{\beta}$.

For the second moment, multiply by $x_i x_l$. The linear and residual terms again vanish, leaving

$$\mathbb{E}[x_i x_l w] = \alpha_w P_{il} + \frac{1}{2}\sum_{j,k} \gamma_{jk}\mathbb{E}[x_i x_l x_j x_k].$$

The fourth-moment identity proved in Equation (2.11) is

$$\mathbb{E}[x_i x_l x_j x_k] = P_{il}P_{jk} + P_{ij}P_{lk} + P_{ik}P_{lj}.$$

Evaluate its three contributions separately:

$$\begin{aligned} \frac{1}{2}\sum_{j,k} \gamma_{jk} P_{il} P_{jk} &= \frac{1}{2} P_{il} \operatorname{tr}(\boldsymbol{\gamma}\mathbf{P}), \\ \frac{1}{2}\sum_{j,k} \gamma_{jk} P_{ij} P_{lk} &= \frac{1}{2}(\mathbf{P}\boldsymbol{\gamma}\mathbf{P})_{il}, \\ \frac{1}{2}\sum_{j,k} \gamma_{jk} P_{ik} P_{lj} &= \frac{1}{2}(\mathbf{P}\boldsymbol{\gamma}^{\mathsf{T}}\mathbf{P})_{il} = \frac{1}{2}(\mathbf{P}\boldsymbol{\gamma}\mathbf{P})_{il}. \end{aligned}$$

Symmetry of $\mathbf{P}$ and $\boldsymbol{\gamma}$ is used in the last two lines. Thus

$$\mathbb{E}[x_i x_l w] = \left\{\alpha_w + \frac{1}{2}\operatorname{tr}(\boldsymbol{\gamma}\mathbf{P})\right\} P_{il} + (\mathbf{P}\boldsymbol{\gamma}\mathbf{P})_{il} = P_{il} + (\mathbf{P}\boldsymbol{\gamma}\mathbf{P})_{il}.$$

Finally, relative-fitness weighting already has normalisation one. Its covariance is its second moment minus the outer product of its mean:

$$\begin{aligned}\mathbf{P}_{\mathrm{sel}} &= \mathbf{P} + \mathbf{P}\boldsymbol{\gamma}\mathbf{P} - (\mathbf{P}\boldsymbol{\beta})(\mathbf{P}\boldsymbol{\beta})^{\mathsf{T}} \\ &= \mathbf{P} + \mathbf{P}(\boldsymbol{\gamma} - \boldsymbol{\beta}\boldsymbol{\beta}^{\mathsf{T}})\mathbf{P}.\end{aligned}$$

□

These are moments under the nonnegative observed relative-fitness weights w. The fitted polynomial is used to calculate those moments through orthogonality. It need not be positive everywhere and should not be substituted as a probability weight over all phenotypes merely because its regression coefficients are available.

7.3 Condition breeding value on phenotype

Let $\boldsymbol{X} = \boldsymbol{A} - \mathbb{E}[\boldsymbol{A}]$ be centred breeding value, with covariance $\mathbf{G}$. Assume $(\boldsymbol{X}, \boldsymbol{x})$ is jointly Gaussian in compatible trait coordinates, and

$$\mathrm{Cov}(\boldsymbol{X}, \boldsymbol{x}) = \mathbf{G}. \tag{7.6}$$

For now take $\mathbf{P}$ positive definite. The cross-covariance assumption is additional information about the trait decomposition; it does not follow from fitting a phenotype regression.

Proposition 7.2 (Conditional breeding-value moments)**.** *Under these assumptions,*

$$\begin{aligned}\mathbb{E}[\boldsymbol{X} \mid \boldsymbol{x}] &= \mathbf{G}\mathbf{P}^{-1}\boldsymbol{x}, \\ \mathrm{Cov}(\boldsymbol{X} \mid \boldsymbol{x}) &= \mathbf{G} - \mathbf{G}\mathbf{P}^{-1}\mathbf{G}.\end{aligned} \tag{7.7}$$

Proof. Define the regression residual

$$\boldsymbol{U} = \boldsymbol{X} - \mathbf{G}\mathbf{P}^{-1}\boldsymbol{x}.$$

Its covariance with phenotype is

$$\mathrm{Cov}(\boldsymbol{U}, \boldsymbol{x}) = \mathbf{G} - \mathbf{G}\mathbf{P}^{-1}\mathbf{P} = \mathbf{0}.$$

Because $\boldsymbol{U}$ and $\boldsymbol{x}$ are linear transformations of a joint Gaussian vector, they are jointly Gaussian. Zero cross-covariance therefore makes them independent. This follows, for example, by factoring the joint Gaussian characteristic function with a block-diagonal covariance matrix. Consequently $\mathbb{E}[\boldsymbol{U} \mid \boldsymbol{x}] = \mathbf{0}$.

Expanding its covariance, with the matrix order retained, gives

$$\begin{aligned}\mathrm{Cov}(\boldsymbol{U}) &= \mathbf{G} - \mathbf{G}\mathbf{P}^{-1}\mathbf{G} - \mathbf{G}\mathbf{P}^{-1}\mathbf{G} + \mathbf{G}\mathbf{P}^{-1}\mathbf{P}\mathbf{P}^{-1}\mathbf{G} \\ &= \mathbf{G} - \mathbf{G}\mathbf{P}^{-1}\mathbf{G}.\end{aligned}$$

Since $\boldsymbol{X} = \mathbf{G}\mathbf{P}^{-1}\boldsymbol{x} + \boldsymbol{U}$, conditioning fixes the first term and leaves the independent residual covariance unchanged. This proves both statements. □

The matrix $\mathbf{G}\mathbf{P}^{-1}$ need not be symmetric. Its transpose is $\mathbf{P}^{-1}\mathbf{G}$, and these products must not be exchanged. For example, the conditional second moment is

$$\begin{aligned}\mathbb{E}[\boldsymbol{X}\boldsymbol{X}^{\mathsf{T}} \mid \boldsymbol{x}] &= \mathbf{S}_c + \mathbf{C}\boldsymbol{x}\boldsymbol{x}^{\mathsf{T}}\mathbf{C}^{\mathsf{T}}, \\ \mathbf{S}_c &= \mathbf{G} - \mathbf{G}\mathbf{P}^{-1}\mathbf{G}, \qquad \mathbf{C} = \mathbf{G}\mathbf{P}^{-1}.\end{aligned} \tag{7.8}$$

The two outer factors in the second term are $\mathbf{C}$ and $\mathbf{C}^{\mathsf{T}}$, not two identical copies of $\mathbf{C}$.

7.4 Transfer reproductive weights to breeding-value moments

We now need a condition that connects the phenotype regression to the reproduction of individuals with different breeding values. A sufficient condition is

$$\mathbb{E}[W \mid \boldsymbol{X}, \boldsymbol{x}] = \mathbb{E}[W \mid \boldsymbol{x}]. \tag{7.9}$$

It says that, once the measured phenotype is given, breeding value supplies no further information about expected reproductive output. Full conditional independence is stronger than needed. For the following calculation, it would also suffice to justify directly the corresponding conditional first- and second-moment factorizations.

Proposition 7.3 (Selected breeding-value moments)**.** *Under the regression, Gaussian, cross-covariance and selection-mediation conditions above, write* $\mathbf{K} = \boldsymbol{\gamma} - \boldsymbol{\beta}\boldsymbol{\beta}^{\mathsf{T}}$*. Then*

$$\boldsymbol{m} = \mathbb{E}_{\mathrm{sel}}[\boldsymbol{X}] = \mathbf{G}\boldsymbol{\beta}, \qquad \mathbf{G}_{\mathrm{sel}} = \mathbf{G} + \mathbf{G}\mathbf{K}\mathbf{G}. \tag{7.10}$$

Proof. The conditional mean condition implies

$$\begin{aligned} \mathbb{E}[\boldsymbol{X}w \mid \boldsymbol{x}] &= \mathbb{E}\{\boldsymbol{X}\,\mathbb{E}[w \mid \boldsymbol{X}, \boldsymbol{x}] \mid \boldsymbol{x}\} \\ &= \mathbb{E}[\boldsymbol{X} \mid \boldsymbol{x}]\,\mathbb{E}[w \mid \boldsymbol{x}]. \end{aligned}$$

Average across phenotypes and substitute Equation (7.7):

$$\begin{aligned} \mathbb{E}_{\mathrm{sel}}[\boldsymbol{X}] &= \mathbb{E}[\boldsymbol{X}w] \\ &= \mathbf{G}\mathbf{P}^{-1}\mathbb{E}\{\boldsymbol{x}\,\mathbb{E}[w \mid \boldsymbol{x}]\} \\ &= \mathbf{G}\mathbf{P}^{-1}\mathbb{E}[\boldsymbol{x}w] \\ &= \mathbf{G}\mathbf{P}^{-1}\mathbf{P}\boldsymbol{\beta} = \mathbf{G}\boldsymbol{\beta}. \end{aligned}$$

The same mediation condition applies to $\boldsymbol{X}\boldsymbol{X}^{\mathsf{T}}$. Using Equation (7.8), its selected second moment is

$$\begin{aligned} \mathbb{E}_{\mathrm{sel}}[\boldsymbol{X}\boldsymbol{X}^{\mathsf{T}}] &= \mathbf{S}_c\,\mathbb{E}[w] + \mathbf{C}\mathbb{E}[\boldsymbol{x}\boldsymbol{x}^{\mathsf{T}}w]\mathbf{C}^{\mathsf{T}} \\ &= \mathbf{G} - \mathbf{G}\mathbf{P}^{-1}\mathbf{G} \\ &\quad + \mathbf{G}\mathbf{P}^{-1}(\mathbf{P} + \mathbf{P}\boldsymbol{\gamma}\mathbf{P})\mathbf{P}^{-1}\mathbf{G} \\ &= \mathbf{G} - \mathbf{G}\mathbf{P}^{-1}\mathbf{G} + \mathbf{G}\mathbf{P}^{-1}\mathbf{G} + \mathbf{G}\boldsymbol{\gamma}\mathbf{G} \\ &= \mathbf{G} + \mathbf{G}\boldsymbol{\gamma}\mathbf{G}. \end{aligned}$$

Subtract the outer product of the selected mean:

$$\begin{aligned} \mathbf{G}_{\mathrm{sel}} &= \mathbf{G} + \mathbf{G}\boldsymbol{\gamma}\mathbf{G} - (\mathbf{G}\boldsymbol{\beta})(\mathbf{G}\boldsymbol{\beta})^{\mathsf{T}} \\ &= \mathbf{G} + \mathbf{G}(\boldsymbol{\gamma} - \boldsymbol{\beta}\boldsymbol{\beta}^{\mathsf{T}})\mathbf{G}. \end{aligned}$$

□

Biological interpretation. Individuals with the same phenotype can have different breeding values. Their conditional covariance remains part of the selected second moment. Discarding it would treat the prediction $\mathbf{G}\mathbf{P}^{-1}\boldsymbol{x}$ as an observed breeding value and underestimate the unresolved variation. Selection mediation is also substantive: if unmeasured traits, shared environments or genetic effects predict reproduction after the measured phenotype is given, the factorizations used above need not hold.

7.5 Invert the selected moments

Suppose $\mathbf{G}$ and the selected covariance $\mathbf{G}_{\mathrm{sel}}$ are positive definite. The exact Gaussian log-quadratic selection model in Proposition 4.2 has

$$\mathbf{G}_{\mathrm{sel}} = (\mathbf{G}^{-1} - \mathbf{H})^{-1}, \qquad \boldsymbol{m} = \mathbf{G}_{\mathrm{sel}}\boldsymbol{b}.$$

These equations can be solved without assuming that their matrices commute.

Proposition 7.4 (Moment-matched log-quadratic parameters)**.** *Under the selected-moment conditions above and positive definiteness,*

$$\boldsymbol{b} = \mathbf{G}_{\mathrm{sel}}^{-1}\mathbf{G}\boldsymbol{\beta}, \qquad \mathbf{H} = \mathbf{G}^{-1} - \mathbf{G}_{\mathrm{sel}}^{-1}, \qquad \mathbf{G}_{\mathrm{sel}} = \mathbf{G} + \mathbf{G}(\boldsymbol{\gamma} - \boldsymbol{\beta}\boldsymbol{\beta}^{\mathsf{T}})\mathbf{G}. \tag{7.11}$$

They define an integrable Gaussian log-quadratic weight with the specified selected mean and covariance.

Proof. Invert the selected-covariance identity:

$$\mathbf{G}_{\mathrm{sel}}^{-1} = \mathbf{G}^{-1} - \mathbf{H}.$$

Rearranging gives $\mathbf{H} = \mathbf{G}^{-1} - \mathbf{G}_{\mathrm{sel}}^{-1}$. Left-multiplying $\boldsymbol{m} = \mathbf{G}_{\mathrm{sel}}\boldsymbol{b}$ by $\mathbf{G}_{\mathrm{sel}}^{-1}$ gives $\boldsymbol{b} = \mathbf{G}_{\mathrm{sel}}^{-1}\boldsymbol{m} = \mathbf{G}_{\mathrm{sel}}^{-1}\mathbf{G}\boldsymbol{\beta}$. Both inverse covariances are symmetric, so $\mathbf{H}$ is symmetric.

To verify existence of the corresponding weight, multiply the original normal density by $\exp(\boldsymbol{b}^{\mathsf{T}}\boldsymbol{X} + \boldsymbol{X}^{\mathsf{T}}\mathbf{H}\boldsymbol{X}/2)$. Its exponent, apart from a constant, is

$$\begin{aligned} -\frac{1}{2}\boldsymbol{X}^{\mathsf{T}}(\mathbf{G}^{-1} - \mathbf{H})\boldsymbol{X} + \boldsymbol{b}^{\mathsf{T}}\boldsymbol{X} &= -\frac{1}{2}\boldsymbol{X}^{\mathsf{T}}\mathbf{G}_{\mathrm{sel}}^{-1}\boldsymbol{X} + \boldsymbol{m}^{\mathsf{T}}\mathbf{G}_{\mathrm{sel}}^{-1}\boldsymbol{X} \\ &= -\frac{1}{2}(\boldsymbol{X} - \boldsymbol{m})^{\mathsf{T}}\mathbf{G}_{\mathrm{sel}}^{-1}(\boldsymbol{X} - \boldsymbol{m}) + \frac{1}{2}\boldsymbol{m}^{\mathsf{T}}\mathbf{G}_{\mathrm{sel}}^{-1}\boldsymbol{m}. \end{aligned}$$

Positive definiteness of $\mathbf{G}_{\mathrm{sel}}$ makes this a finite Gaussian integral. After normalisation, the weighted distribution is $N(\boldsymbol{m}, \mathbf{G}_{\mathrm{sel}})$. □

The order in the slope formula is specifically $\mathbf{G}_{\mathrm{sel}}^{-1}\mathbf{G}\boldsymbol{\beta}$, not $\mathbf{G}\mathbf{G}_{\mathrm{sel}}^{-1}\boldsymbol{\beta}$. For another way to see the order, factor

$$\mathbf{G}_{\mathrm{sel}} = \mathbf{G}(\mathbf{I} + \mathbf{K}\mathbf{G}) = (\mathbf{I} + \mathbf{G}\mathbf{K})\mathbf{G}.$$

The first factorisation gives

$$\mathbf{G}_{\mathrm{sel}}^{-1}\mathbf{G} = (\mathbf{I} + \mathbf{K}\mathbf{G})^{-1}, \qquad \boldsymbol{b} = (\mathbf{I} + \mathbf{K}\mathbf{G})^{-1}\boldsymbol{\beta}.$$

Replacing $\mathbf{K}\mathbf{G}$ by $\mathbf{G}\mathbf{K}$ changes the expression unless the matrices happen to commute. Neither commutation nor simultaneous diagonalisation is assumed in the derivation.

Positive selected covariance is sufficient for an integrable matched model. It does not require stabilising curvature. If $\mathbf{G}_{\mathrm{sel}} \preceq \mathbf{G}$, inverse order gives $\mathbf{G}_{\mathrm{sel}}^{-1} \succeq \mathbf{G}^{-1}$, so $\mathbf{H} \preceq 0$; other selected covariance changes can give other curvature signs. Relative fitness cannot recover the absolute output multiplier: adding a constant to the matched log mean leaves the selected moments unchanged.

7.5.1 Exact derivatives versus matched moments

If the true W_A is exactly log quadratic in the breeding-value coordinates, then its Gaussian selection formulas must have the same mean and covariance, and the inversion recovers its $\boldsymbol{b}, \mathbf{H}$. If exact log-quadraticity fails while the moment conditions remain valid, the construction still gives a log-quadratic weight with matching first and second selected moments. Those integrated moments do not determine the local derivatives of an arbitrary function $\log \mathbb{E}[W \mid \boldsymbol{A}]$.

The distinction also applies to the variance share calculated from the converted parameters. Under exact closure it describes the stated log-quadratic relationship. Under moment matching it describes the chosen quadratic model, whose agreement in selected moments does not establish agreement in its complete distribution of log fitness.

7.6 A positive fitness model that satisfies the assumptions

The supplied checks include positive fitness functions, so the conversion can be assessed without treating a possibly negative regression polynomial as a fitness function. Let centred phenotype be $\boldsymbol{Z} = \boldsymbol{A} + \boldsymbol{E}$, where $\boldsymbol{A} \sim N(\mathbf{0}, \mathbf{G})$ and $\boldsymbol{E} \sim N(\mathbf{0}, \mathbf{V}_E)$ are independent. Take

$$f_Z(\boldsymbol{z}) = \exp\{c + \boldsymbol{q}^\mathsf{T}\boldsymbol{z} - \tfrac{1}{2}\boldsymbol{z}^\mathsf{T}\mathbf{C}\boldsymbol{z}\}, \qquad \mathbf{C} = \mathbf{C}^\mathsf{T} \succeq 0, \tag{7.12}$$

with $\mathbf{V}_E$ positive definite. This is a positive phenotype-dependent conditional mean; reproduction may have additional stochastic variation around it.

For a fixed breeding value $\boldsymbol{a}$, write $\mathbf{Q} = \mathbf{V}_E^{-1} + \mathbf{C}$. In the environmental integral, collect the terms involving $\boldsymbol{e}$:

$$\boldsymbol{q}^\mathsf{T}(\boldsymbol{a}+\boldsymbol{e}) - \frac{1}{2}(\boldsymbol{a}+\boldsymbol{e})^\mathsf{T}\mathbf{C}(\boldsymbol{a}+\boldsymbol{e}) - \frac{1}{2}\boldsymbol{e}^\mathsf{T}\mathbf{V}_E^{-1}\boldsymbol{e} = \boldsymbol{q}^\mathsf{T}\boldsymbol{a} - \frac{1}{2}\boldsymbol{a}^\mathsf{T}\mathbf{C}\boldsymbol{a} - \frac{1}{2}\boldsymbol{e}^\mathsf{T}\mathbf{Q}\boldsymbol{e} + (\boldsymbol{q} - \mathbf{C}\boldsymbol{a})^\mathsf{T}\boldsymbol{e}.$$

Completing the square in $\boldsymbol{e}$ and using the Gaussian normalising integral gives

$$\log W_A(\boldsymbol{a}) = c - \frac{1}{2}\log\det(\mathbf{V}_E) - \frac{1}{2}\log\det(\mathbf{Q}) + \boldsymbol{q}^\mathsf{T}\boldsymbol{a} - \frac{1}{2}\boldsymbol{a}^\mathsf{T}\mathbf{C}\boldsymbol{a} + \frac{1}{2}(\boldsymbol{q} - \mathbf{C}\boldsymbol{a})^\mathsf{T}\mathbf{Q}^{-1}(\boldsymbol{q} - \mathbf{C}\boldsymbol{a}). \tag{7.13}$$

The positive-definite $\mathbf{Q}$ ensures that the integral exists. Expand the last term before reading off the coefficients:

$$\frac{1}{2}\boldsymbol{q}^\mathsf{T}\mathbf{Q}^{-1}\boldsymbol{q} - \boldsymbol{a}^\mathsf{T}\mathbf{C}\mathbf{Q}^{-1}\boldsymbol{q} + \frac{1}{2}\boldsymbol{a}^\mathsf{T}\mathbf{C}\mathbf{Q}^{-1}\mathbf{C}\boldsymbol{a}.$$

Consequently,

$$\boldsymbol{b} = \boldsymbol{q} - \mathbf{C}\mathbf{Q}^{-1}\boldsymbol{q}, \qquad \mathbf{H} = -\mathbf{C} + \mathbf{C}\mathbf{Q}^{-1}\mathbf{C}. \tag{7.14}$$

These formulas provide an independent environmental-averaging route to the same target. The reproduction code fits the phenotype regression by quadrature and compares its moment conversion with this conditional expectation in one and two traits. The finite comparison checks the implementation. The agreement under the stated model follows from the derivations, including all Gaussian and mediation assumptions.

7.7 Centring, units and singular support

7.7.1 Centre at the population mean

The vanishing third moments used earlier refer to centred Gaussian phenotypes. If a raw polynomial is initially expressed about a different reference point, write its coordinate as $\boldsymbol{z} = \boldsymbol{x} + \boldsymbol{\delta}$. Then

$$\begin{aligned}\alpha_0 + \boldsymbol{\beta}_0^{\mathsf{T}}\boldsymbol{z} + \tfrac{1}{2}\boldsymbol{z}^{\mathsf{T}}\boldsymbol{\gamma}\boldsymbol{z} = \alpha_0 + \boldsymbol{\beta}_0^{\mathsf{T}}\boldsymbol{\delta} + \tfrac{1}{2}\boldsymbol{\delta}^{\mathsf{T}}\boldsymbol{\gamma}\boldsymbol{\delta} \\ + (\boldsymbol{\beta}_0 + \boldsymbol{\gamma}\boldsymbol{\delta})^{\mathsf{T}}\boldsymbol{x} + \tfrac{1}{2}\boldsymbol{x}^{\mathsf{T}}\boldsymbol{\gamma}\boldsymbol{x}.\end{aligned}$$

The centred linear coefficient is $\boldsymbol{\beta} = \boldsymbol{\beta}_0 + \boldsymbol{\gamma}\boldsymbol{\delta}$; it is not generally $\boldsymbol{\beta}_0$. The regression space is unchanged by the shift when all constant, linear and quadratic terms are retained, but its coefficients change.

7.7.2 Use compatible trait coordinates

For an invertible coordinate transformation $\boldsymbol{x}' = \mathbf{T}\boldsymbol{x}$, the same polynomial has

$$\boldsymbol{\beta}' = \mathbf{T}^{-\mathsf{T}}\boldsymbol{\beta}, \qquad \boldsymbol{\gamma}' = \mathbf{T}^{-\mathsf{T}}\boldsymbol{\gamma}\mathbf{T}^{-1}.$$

If breeding values are expressed in the same new coordinates,

$$\mathbf{G}' = \mathbf{T}\mathbf{G}\mathbf{T}^{\mathsf{T}}, \qquad \mathbf{P}' = \mathbf{T}\mathbf{P}\mathbf{T}^{\mathsf{T}}.$$

Substitution into the selected-moment formula gives $\boldsymbol{m}' = \mathbf{T}\boldsymbol{m}$ and $\mathbf{G}'_{\mathrm{sel}} = \mathbf{T}\mathbf{G}_{\mathrm{sel}}\mathbf{T}^{\mathsf{T}}$. Inverting the latter,

$$(\mathbf{G}'_{\mathrm{sel}})^{-1} = \mathbf{T}^{-\mathsf{T}}\mathbf{G}_{\mathrm{sel}}^{-1}\mathbf{T}^{-1},$$

so the converted parameters transform as

$$\boldsymbol{b}' = \mathbf{T}^{-\mathsf{T}}\boldsymbol{b}, \qquad \mathbf{H}' = \mathbf{T}^{-\mathsf{T}}\mathbf{H}\mathbf{T}^{-1}.$$

Thus the conversion is compatible with a coordinated change of trait units. Using standardised regression coefficients with an untransformed $\mathbf{G}$ is not such a coordinated change.

7.7.3 Restrict inverse operations to varying directions

The moment formulas involving only multiplication remain meaningful when covariances are singular, but inverses must be taken on the appropriate support. First remove zero-variance phenotype directions. If $\mathrm{Var}(\boldsymbol{v}^{\mathsf{T}}\boldsymbol{x}) = 0$, the centred phenotype component is zero almost surely. The cross-covariance condition then gives $\boldsymbol{v}^{\mathsf{T}}\mathbf{G}\boldsymbol{v} = 0$, so breeding value has no variation in that same direction. An orthonormal basis for the phenotype support therefore retains all relevant breeding-value variation and makes the reduced $\mathbf{P}$ invertible.

Breeding-value covariance can still have a smaller rank. Let $\mathbf{U}$ have orthonormal columns spanning the range of $\mathbf{G}$. Then $\boldsymbol{X} = \mathbf{U}\boldsymbol{X}_s$ almost surely, and $\mathbf{G}_s = \mathbf{U}^{\mathsf{T}}\mathbf{G}\mathbf{U}$ is positive definite. After calculating the selected moments, restrict them as

$$\boldsymbol{m}_s = \mathbf{U}^{\mathsf{T}}\boldsymbol{m}, \qquad \mathbf{S}_s = \mathbf{U}^{\mathsf{T}}\mathbf{G}_{\mathrm{sel}}\mathbf{U}.$$

If $\mathbf{S}_s$ is positive definite, the identifiable parameters are

$$\boldsymbol{b}_s = \mathbf{S}_s^{-1}\boldsymbol{m}_s, \qquad \mathbf{H}_s = \mathbf{G}_s^{-1} - \mathbf{S}_s^{-1}.$$

They describe the function on the breeding-value support. Derivatives in directions with no breeding-value variation are not determined by these moments. If selected covariance is degenerate within an initially varying direction, the finite positive-definite inverse construction fails; it should not be repaired by silently substituting an arbitrary inverse. If breeding value has no variation at all, the variance share is undefined.

7.8 Work through the scalar checkpoint

Take the guide's values

$$P = 1.5, \qquad \beta = 0.4, \qquad \gamma = -0.2, \qquad G = 0.5.$$

Normalisation first gives

$$\alpha_w = 1 - \frac{1}{2}\gamma P = 1 - \frac{1}{2}(-0.2)(1.5) = 1.15.$$

The selected phenotype mean is $P\beta = 0.6$, its second moment is

$$P + P^2\gamma = 1.5 + (1.5)^2(-0.2) = 1.05,$$

and its variance is $1.05 - 0.6^2 = 0.69$.

For breeding value, the conditional mean coefficient is $G/P = 1/3$ and the conditional variance is

$$G - \frac{G^2}{P} = \frac{1}{2} - \frac{1/4}{3/2} = \frac{1}{3}.$$

The selected mean and second moment are therefore

$$\begin{aligned} m &= \frac{G}{P}(P\beta) = G\beta = 0.2, \\ \mathbb{E}_{\mathrm{sel}}[X^2] &= G - \frac{G^2}{P} + \frac{G^2}{P^2}(P + P^2\gamma) \\ &= G + G^2\gamma = 0.5 + 0.25(-0.2) = 0.45. \end{aligned}$$

Subtracting the square of the mean gives

$$G_{\mathrm{sel}} = 0.45 - 0.2^2 = 0.41.$$

Only now invert the selected moments:

$$\begin{aligned} b &= \frac{0.2}{0.41} = \frac{20}{41} = 0.487804878\ldots, \\ H &= \frac{1}{0.5} - \frac{1}{0.41} = -\frac{18}{41} = -0.439024390\ldots. \end{aligned} \tag{7.15}$$

Substituting them back gives

$$(G^{-1} - H)^{-1} = \left(2 + \frac{18}{41}\right)^{-1} = 0.41, \qquad G_{\mathrm{sel}}b = 0.2.$$

This is an algebraic check of the inversion. The fitted polynomial $1.15 + 0.4x - 0.1x^2$ becomes negative sufficiently far into the tails. The checkpoint is therefore a calculation with compatible regression moments, not a proposal to use that polynomial as a nonnegative fitness function over an entire Gaussian population.

7.9 Why taking the logarithm of the regression is different

Let the fitted phenotype polynomial be

$$\widehat{w}(\boldsymbol{x}) = \alpha_w + \boldsymbol{\beta}^{\mathsf{T}}\boldsymbol{x} + \tfrac{1}{2}\boldsymbol{x}^{\mathsf{T}}\boldsymbol{\gamma}\boldsymbol{x}.$$

If $\alpha_w > 0$, continuity makes it positive in a neighbourhood of zero. Within that neighbourhood,

$$\nabla \log \widehat{w} = \frac{\nabla \widehat{w}}{\widehat{w}}.$$

Differentiate this product of $\widehat{w}^{-1}$ and its gradient:

$$\nabla^2 \log \widehat{w} = \frac{\nabla^2 \widehat{w}}{\widehat{w}} - \frac{(\nabla \widehat{w})(\nabla \widehat{w})^{\mathsf{T}}}{\widehat{w}^2}.$$

At zero, $\widehat{w} = \alpha_w$, $\nabla \widehat{w} = \boldsymbol{\beta}$, and $\nabla^2 \widehat{w} = \boldsymbol{\gamma}$. Thus

$$\nabla \log \widehat{w}(\mathbf{0}) = \frac{\boldsymbol{\beta}}{\alpha_w}, \qquad \nabla^2 \log \widehat{w}(\mathbf{0}) = \frac{\boldsymbol{\gamma}}{\alpha_w} - \frac{\boldsymbol{\beta}\boldsymbol{\beta}^{\mathsf{T}}}{\alpha_w^2}. \tag{7.16}$$

In the scalar checkpoint these give

$$b_{\mathrm{raw}} = \frac{0.4}{1.15} = \frac{8}{23} = 0.347826\ldots, \qquad H_{\mathrm{raw}} = -\frac{156}{529} = -0.294896\ldots.$$

They differ from Equation (7.15) because they differentiate the fitted phenotype-space projection. They do not perform environmental averaging or transfer selected moments to breeding value. Positivity near the mean permits these local derivatives; it does not make the polynomial positive over the whole phenotype distribution.

7.10 Conditions to retain when using the result

The conversion combines assumptions that belong to different steps. Gaussian phenotype moments and regression orthogonality supply selected phenotype moments. Joint Gaussian breeding values and phenotypes, the stated cross-covariance, and selection mediation transfer them to breeding value. Compatible centring, units and the half-quadratic coefficient convention make the matrix operations refer to the same quantities. Positive selected covariance permits inversion. Exact log-quadraticity is the additional condition that turns matched parameters into the true local derivatives.

Uncertainty in $\boldsymbol{\beta}, \boldsymbol{\gamma}, \mathbf{G}$, phenotype scaling and population centring must be propagated together. Applying the conversion to joint bootstrap or model draws preserves their dependencies; transforming separate interval endpoints generally does not. Near-singular selected covariance can make inverse-based parameters sensitive to small input changes. That statistical uncertainty is separate from the validity of the underlying moment and mediation assumptions.

The group `phenotype_regression_conversion` in `code/reproduce_result_checks.py` checks the scalar checkpoint and independently fitted positive-fitness examples. Its multivariate calculation retains off-diagonal coefficients and noncommuting matrix products. The calculation verifies those specified examples; the derivations here state the conditions required to use the same route with an existing selection regression.

Chapter 8

From the derivations to the reported results

The preceding chapters establish relationships among population quantities. A figure or fitted estimate additionally specifies numerical inputs and a calculation. This chapter makes that connection explicit. It develops the substitutions behind the main figures and visual examples, then explains what is computed from simulated observations. The source paths in Appendix A locate the complete calculations, including their random seeds and saved results.

8.1 What can be derived and what must be computed?

For a stated Gaussian population and an exact quadratic target, the component formulas are identities. Their numerical evaluation is a substitution into a proved expression. The population parameters can instead be used to generate a finite random sample. A discrepancy measured in that sample is a random numerical result; its particular value depends on the sample, although its definition follows from the theory. Finally, an estimate obtained by maximising a likelihood and its bootstrap interval depend on the observed data, optimisation and refitting procedure. A proof of the likelihood formula does not specify the maximiser without performing that calculation.

These distinctions determine what reproduction means. An analytical result is reproduced by following its proof under its assumptions. A generated example is reproduced by using its recorded parameters and random stream. A fitted result is reproduced by fitting the same observation model to the same observations and repeating its stated uncertainty calculation. The repository supplies all three forms of material.

8.2 Constructing the conditional fitness curve

Main Figure 1 uses $Z = A + E$, independent $E \sim N(0, 0.49)$, optimum $\theta = 1$ and phenotype-fitness width $\omega = 0.6$. The population has mean breeding value zero and breeding-value variance $G = 0.49$. The environmental calculation in Chapter 1 gives

$$W_A(a) = \sqrt{\frac{0.36}{0.36 + 0.49}} \exp\left[-\frac{(a-1)^2}{2(0.36 + 0.49)}\right], \tag{8.1}$$

$$\ell_A(a) = \frac{1}{2} \log \frac{0.36}{0.85} - \frac{(a-1)^2}{1.70}. \tag{8.2}$$

The overall fitness multiplier is one in this example. Differentiating before substituting the reference mean gives

$$\ell'_A(a) = \frac{1-a}{0.85}, \qquad \ell''_A(a) = -\frac{1}{0.85}.$$

Consequently $b = 20/17 = 1.1764705882$ and $H = -20/17$. The breeding-value variance determines how often different parts of this curve occur; it does not enter this environmental averaging integral. In contrast, the environmental variance changes the curve itself.

The points in this construction are phenotype-dependent fitness values assigned after drawing environmental deviations. They are not Poisson offspring-count observations. Repeating the environmental average numerically checks the integration, with a recorded maximum absolute discrepancy of about 0.001817 over the tested values. That discrepancy is a Monte Carlo result, not an additional term in W_A.

Biological interpretation. Individuals sharing a breeding value can experience different environments and receive different reproductive weights. The conditional mean averages those differences. The curve relevant to breeding values therefore has a different curvature from the original phenotype-fitness function.

8.3 Checking a population identity with finite samples

Main Figure 2 generates 360 combinations of $\boldsymbol{b}$, $\mathbf{H}$ and $\mathbf{G}$ in dimensions 2, 4 and 8. For each combination it draws 20,000 Gaussian breeding-value vectors. The exact population value is

$$A_{\text{theory}} = \frac{\boldsymbol{b}^{\mathsf{T}}\mathbf{G}\boldsymbol{b}}{\boldsymbol{b}^{\mathsf{T}}\mathbf{G}\boldsymbol{b} + \frac{1}{2}\operatorname{tr}(\mathbf{HGHG})}.$$

For a particular generated sample, form $L_i = \boldsymbol{b}^{\mathsf{T}}\boldsymbol{X}_i$ and $Y_i = L_i + \{\boldsymbol{X}_i^{\mathsf{T}}\mathbf{H}\boldsymbol{X}_i - \operatorname{tr}(\mathbf{HG})\}/2$. The sample comparison is

$$r^2 = \frac{\{\sum_i (L_i - \bar{L})(Y_i - \bar{Y})\}^2}{\sum_i (L_i - \bar{L})^2 \sum_i (Y_i - \bar{Y})^2}. \tag{8.3}$$

Sample centring remains necessary: a finite sample from a centred population need not have sample mean zero. The constant used to centre the quadratic term does not change a correlation.

For each parameter combination the script records $|A_{\text{theory}} - r^2|$. Across the 360 cases, the saved median is 0.0023672601, the 95th percentile is 0.0103264192, and the maximum is 0.0252707809. Chapter 2 proves the population identity; these discrepancies measure finite-sample agreement with it. The seed 20260905 and the original sequence of random draws determine the recorded values.

8.4 Contracting covariance without changing the surface

The fixed-surface comparison uses

$$\mathbf{G}_0 = \begin{pmatrix} 1 & 0.3 \\ 0.3 & 0.6 \end{pmatrix}, \qquad \boldsymbol{b} = \begin{pmatrix} 0.75 \\ 0.30 \end{pmatrix}, \qquad \mathbf{H} = -0.8\mathbf{I}.$$

First perform the matrix–vector multiplication:

$$\mathbf{G}_0\boldsymbol{b} = \begin{pmatrix} 0.75 + 0.09 \\ 0.225 + 0.18 \end{pmatrix} = \begin{pmatrix} 0.84 \\ 0.405 \end{pmatrix}.$$

The linear variance is therefore $0.75(0.84) + 0.30(0.405) = 0.7515$. Next,

$$\mathbf{G}_0^2 = \begin{pmatrix} 1.09 & 0.48 \\ 0.48 & 0.45 \end{pmatrix}, \qquad V_{\text{quad}} = \frac{0.8^2}{2}(1.09 + 0.45) = 0.4928.$$

The share is $0.7515/(0.7515 + 0.4928) = 0.6039540304$.

For the narrower population $\mathbf{G}_1 = 0.11\mathbf{G}_0$, Chapter 3 gives

$$V_{\text{lin}}(\mathbf{G}_1) = 0.11(0.7515) = 0.082665, \tag{8.4}$$

$$V_{\text{quad}}(\mathbf{G}_1) = 0.11^2(0.4928) = 0.00596288, \tag{8.5}$$

$$\mathcal{A}_g(\mathbf{G}_1) = \frac{0.082665}{0.082665 + 0.00596288} = 0.9327200425. \tag{8.6}$$

Both absolute variances decrease. Their different scaling rates increase the fraction contributed by the linear term. This is the calculation behind Main Figure 3, not a change in the curvature or the population mean.

8.5 Following the specified covariance trajectory

Main Figure 4 takes four traits with $\mathbf{G}_t = g_t\mathbf{I}$, $\mathbf{B} = \mathbf{I}$, $\mathbf{\Gamma} = 0.3\mathbf{I}$ and $\rho = 0.9$. The scalar recursion becomes

$$g_{t+1} = 0.1 + 0.9\frac{g_t}{1 + 0.3g_t}, \qquad g_0 = 4.$$

Its positive fixed point solves $0.3(g^*)^2 + 0.07g^* - 0.1 = 0$, giving

$$g^* = \frac{-0.07 + \sqrt{0.07^2 + 0.12}}{0.6} = 0.4723532349.$$

The contraction result in Chapter 5 explains why iterating from $g_0 = 4$ approaches this value. It does not specify the directional gradient. The figure separately prescribes its norm:

$$\|\boldsymbol{b}_t\| = \begin{cases} 1, & 0 \le t \le 40, \\ \exp\{-0.06(t-40)\}, & 40 < t \le 100. \end{cases}$$

For each generation,

$$V_{\text{lin}}(t) = g_t \|\boldsymbol{b}_t\|^2, \qquad V_{\text{quad}}(t) = \frac{4}{2}(0.3)^2 g_t^2 = 0.18g_t^2. \tag{8.7}$$

Thus the initial components are 4 and 2.88 and the initial share is $4/6.88 = 0.5813953488$. At generation 40 the iterated variance is 0.4723535595, giving components 0.4723535595 and 0.0401612193 and share 0.9216389049. At generation 100 the prescribed gradient norm is $e^{-3.6}$, and substitution into Equation (8.7) gives share 0.0087044928.

Biological interpretation. The first phase illustrates how reproduction and stabilising selection jointly change covariance under the stated closure while the gradient is maintained. The later decrease in the share follows from the imposed decline in the gradient. It is not a prediction of a moving population mean, which this covariance recursion does not calculate.

8.6 One trait: broad, narrow and centred at the peak

Appendix Figure A1 and the one-trait tutorial use one curve, $\ell_A(a) = a - a^2/2$. Let $A \sim N(m, v)$ and write $x = a - m$. Expanding the actual polynomial, without a remainder, yields

$$\ell_A(m + x) = m - \frac{m^2}{2} + (1 - m)x - \frac{x^2}{2}.$$

Its local slope is $b = 1 - m$ and its curvature is $H = -1$. The population-fitted line includes the expected quadratic contribution:

$$\widehat{\ell}_1(m + x) = m - \frac{m^2}{2} - \frac{v}{2} + (1 - m)x.$$

The tangent omits $-v/2$ and therefore differs in intercept. The component calculation gives

$$V_{\text{lin}} = (1 - m)^2 v, \qquad V_{\text{quad}} = \frac{v^2}{2}.$$

For $v > 0$, their ratio is $\mathcal{A}_g = (1 - m)^2/\{(1 - m)^2 + v/2\}$.

Population	m	v	V_{lin}	V_{quad}	$\mathcal{A}_g$
Broad	0	1	1	0.5	2/3
Narrow, same mean	0	0.1	0.1	0.005	20/21
Narrow, at the peak	1	0.1	0	0.005	0

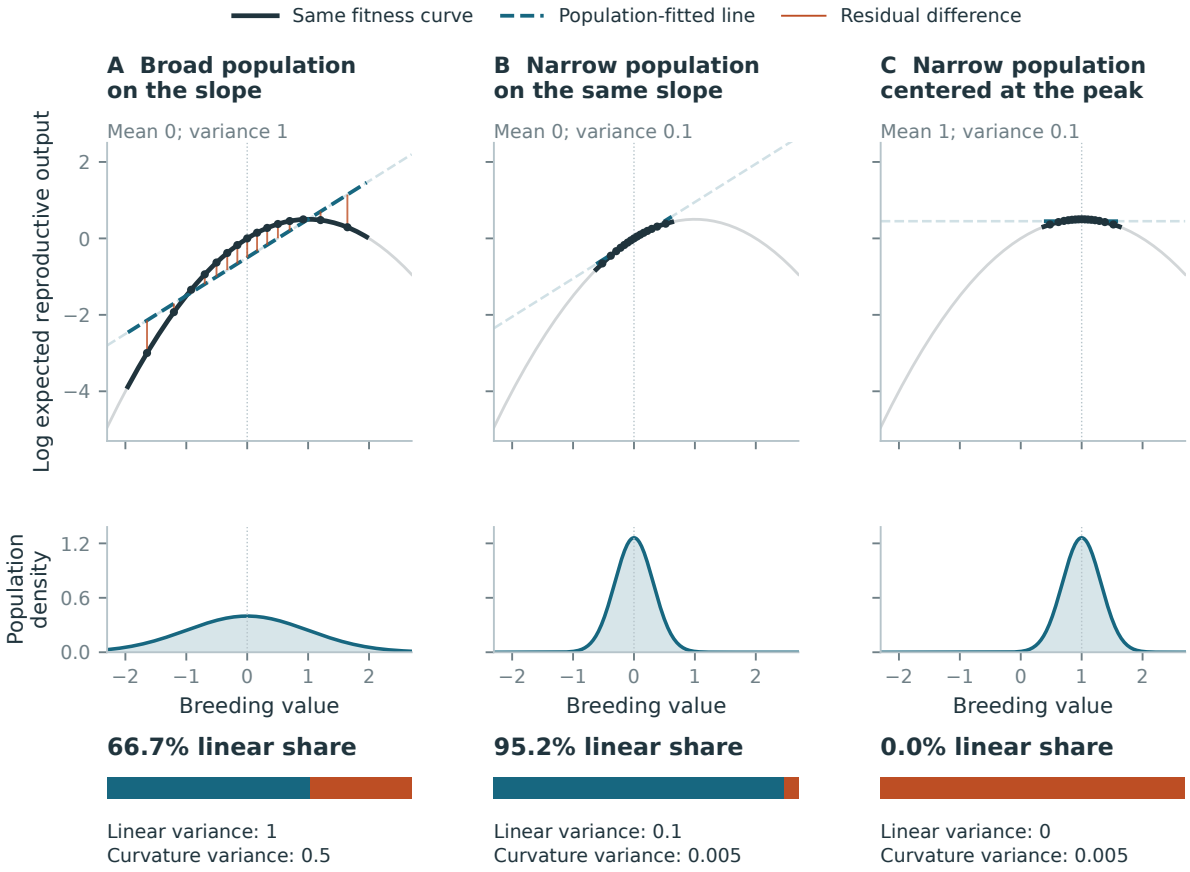


Figure 8.1: The curve is the same in all three populations. Population variance changes the weights assigned to its values; moving the mean to the peak also changes the local slope. The displayed densities are separate from the fitness scale. All values in the table are population calculations, not estimates from the plotted points.

Biological interpretation. A narrow population has a large linear share in the second case because its mean is on the slope. Narrowness by itself does not give a large share: at the peak the first-order component vanishes while curvature still produces variation.

8.7 Rotating the population on a fixed two-trait surface

The geometry example takes $\mathbf{\Gamma} = 0.8\mathbf{I}$ and an optimum whose original coordinates are $(0.9375, 0.375)^{\mathsf{T}}$. An orthonormal coordinate change places the direction towards the optimum along the first axis. This preliminary relabelling does not change any variance component. The mean remains zero and the gradient is $(g, 0)^{\mathsf{T}}$, where

$$g^2 = (0.8)^2(0.9375^2 + 0.375^2) = 0.6525.$$

Now physically rotate the covariance, keeping the fitness surface and trait axes fixed:

$$\mathbf{G}_\varphi = \mathbf{R}_\varphi \begin{pmatrix} 0.20 & 0 \\ 0 & 0.005 \end{pmatrix} \mathbf{R}_\varphi^{\mathsf{T}}, \qquad \mathbf{R}_\varphi = \begin{pmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{pmatrix}.$$

Matrix multiplication gives $G_{\varphi,11} = 0.20\cos^2\varphi + 0.005\sin^2\varphi$. Since only the first gradient coordinate is nonzero,

$$V_{\mathrm{lin}}(\varphi) = 0.6525\{0.20\cos^2\varphi + 0.005\sin^2\varphi\}, \tag{8.8}$$

$$V_{\mathrm{quad}}(\varphi) = \frac{0.8^2}{2}\,\mathrm{tr}(\mathbf{G}_\varphi^2) = \frac{0.8^2}{2}(0.20^2 + 0.005^2) = 0.012808. \tag{8.9}$$

The second equality follows because rotation preserves the covariance eigenvalues, and the curvature is isotropic in this example. It need not hold for arbitrary curvature.

Rotation	V_{lin}	V_{quad}	$\mathcal{A}_g$
0°	0.1305	0.012808	0.910626
45°	0.06688125	0.012808	0.839276
90°	0.0032625	0.012808	0.203012

At 90°, the long axis is perpendicular to the gradient, but variance 0.005 remains in the gradient direction. A section along the long axis therefore has zero first-order slope even though the full population has a positive linear component. A one-dimensional slice cannot replace the full two-trait calculation.

There is another angle in the teaching figures, defined between centred fitness functions under the inner product $\langle f, h\rangle = \mathbb{E}[fh]$. If both the linear component and total variance are positive, let ψ be the angle between L and $L + Q_c$. Orthogonality gives

$$\cos\psi = \frac{\mathbb{E}[L(L+Q_c)]}{\sqrt{\mathbb{E}[L^2]}\sqrt{\mathbb{E}[(L+Q_c)^2]}} = \sqrt{\frac{V_{\mathrm{lin}}}{V_{\mathrm{lin}} + V_{\mathrm{quad}}}}, \qquad \mathcal{A}_g = \cos^2\psi.$$

This angle is between functions, with the population supplying their inner product. It is not the physical rotation φ of the covariance ellipse. When L is the zero function, its ordinary angle is undefined; a drawn 90° triangle then represents a limit.

8.7.1 The same share from different population changes

The four geometry comparison cases alter covariance scale s and the distance to the optimum by a factor d. With isotropic fixed curvature and the original direction to the optimum retained, the new gradient is $d\boldsymbol{b}_0$ and the new covariance is $s\mathbf{G}_0$. Hence

$$V_{\mathrm{lin}}(s,d) = sd^2L_0, \qquad V_{\mathrm{quad}}(s,d) = s^2Q_0, \qquad \mathcal{A}_g(s,d) = \frac{d^2L_0}{d^2L_0 + sQ_0}.$$

Case A has $(s,d) = (1,1)$ and gives 0.603954. Case B has $(0.11,1)$ and gives 0.932720. Case C has $(0.11,0)$ and gives zero. Case D has $(1,1/\sqrt{0.11})$, so

$$\mathcal{A}_g(1,1/\sqrt{0.11}) = \frac{L_0/0.11}{L_0/0.11 + Q_0} = \frac{L_0}{L_0 + 0.11Q_0} = \mathcal{A}_g(0.11,1).$$

The identical shares in B and D describe different absolute components. B narrows the population. D retains its covariance and places its mean farther from the peak. The share alone does not identify which biological difference occurred.

8.8 Reproductive contributions of a fixed sample of parents

For a displayed sample of n parents with known log conditional means ℓ_i, the equal-frequency and expected-output weights are

$$q_i^{(0)} = \frac{1}{n}, \qquad q_i^{(W)} = \frac{\exp(\ell_i)}{\sum_j \exp(\ell_j)}.$$

Fit an affine predictor $\widehat{\ell}_i$ using an intercept and all relevant breeding-value coordinates. The line-based reproductive weights are $q_i^{(1)} = \exp(\widehat{\ell}_i)/\sum_j \exp(\widehat{\ell}_j)$. For any specified group S, its population fraction is $\sum_{i\in S} q_i^{(0)}$ and its expected reproductive fraction is $\sum_{i\in S} q_i^{(W)}$. The difference is entirely a difference of weights on the same parents.

The two-trait companion recovers case 25 from the Main Figure 2 random stream and retains all 20,000 parents for calculation. The group is defined by a first-trait breeding value more than one sample standard deviation below its mean. It contains 3,180 parents, so its frequency is $3180/20000 = 0.159$. Its expected reproductive fraction under the curved relationship is 0.2846675; the fitted plane assigns it 0.3412681.

The displayed shifts standardise each coordinate by that trait's starting sample standard deviation. In this coordinate system the full shift has length 0.6380495, whereas the plane-based shift has length 0.8299486. The reported length overestimate is

$$\frac{0.8299486}{0.6380495} - 1 = 0.3007591.$$

This is a comparison of two shift lengths in the displayed coordinates. It is not the norm of their vector difference divided by the full shift, and the coordinatewise standardisation is not generally the $\mathbf{G}^{-1}$ norm used in Chapter 4. The example illustrates changed reproductive contributions without changing the definition of that theorem's error bound.

For the 800-genotype estimation example, realised counts r_i give an additional set of weights, $q_i^{(\mathrm{obs})} = r_i/\sum_j r_j$, provided total reproduction is positive. The sample has 2,862 offspring and 88 zero counts. For this teaching comparison, the group is defined from the separately saved true simulated breeding values: it includes values more than one sample standard deviation above their mean. The group contains 122 genotypes, with frequency $122/800 = 0.1525$, and contributes 585 observed offspring, a fraction $585/2862 = 0.2044025$. Its conditional expected fraction under the

generating curve is approximately 0.212940. The difference between those last two fractions is sampling variation in reproduction. The true breeding values and generating curve are used only to check this teaching comparison; they are not inputs to the estimator. None of these weighted parental distributions specifies offspring breeding values without a transmission model.

8.9 The mathematics guide's recurring example

The guide's five figures and eleven checkpoints use the preceding derivations with a small set of parameters. In its phenotype construction, $\theta = 4$, $\omega^2 = 1$ and $V_E = 1$. Conditional averaging gives $\ell_A(a) = c - (a-4)^2/4$. At the reference mean zero this is $\ell_0 + 2a - a^2/4$, with $b = 2$ and $H = -0.5$. The following calculations make the numerical checks reproducible without reading values from a graph.

8.9.1 Conditional averages and finite differences

For outputs $(0, 2)$ the conditional sample mean is 1 and its logarithm is zero. For $(1, 3)$ the mean is 2 and its logarithm is $\log 2$. The first group's mean of logged counts is not finite. These are checks on the order of operations, rather than estimates from the full teaching dataset.

For a quadratic polynomial $\ell_0 + bx + Hx^2/2$,

$$\frac{\ell(\varepsilon) - \ell(-\varepsilon)}{2\varepsilon} = \frac{2b\varepsilon}{2\varepsilon} = b, \tag{8.10}$$

$$\frac{\ell(\varepsilon) - 2\ell(0) + \ell(-\varepsilon)}{\varepsilon^2} = \frac{H\varepsilon^2}{\varepsilon^2} = H. \tag{8.11}$$

With $\varepsilon = 0.2$, the changes from ℓ_0 are 0.39 and -0.41. The two finite differences are $0.80/0.40 = 2$ and $-0.02/0.04 = -0.5$. They are exact here because the function is quadratic; general smooth functions also have finite-difference approximation error.

8.9.2 Components, units and selected-parent moments

With $G = 1$, the components are $b^2G = 4$ and $H^2G^2/2 = 0.125$, so $\mathcal{A}_g = 4/4.125 = 32/33$. The centred quadratic term is $Q_c = -0.25(x^2 - 1)$. At $x = 2$ the product LQ_c is $4(-0.75) = -3$; at $x = -2$ it is $(-4)(-0.75) = 3$. Equal Gaussian weights make these paired contributions cancel. Chapter 2 establishes this for the whole distribution.

Changing trait units to $x' = 10x$ gives $G' = 100$, $b' = 0.2$ and $H' = -0.005$. Substitution gives $(0.2)^2(100) = 4$ and $(-0.005)^2(100)^2/2 = 0.125$, so neither component changes.

With $\Gamma = 0.5$, selected variance is $G_{\text{sel}} = (1 + 0.5)^{-1} = 2/3$ and selected-parent displacement is $(2/3)2 = 4/3$. Ignoring curvature predicts displacement 2. Its relative error is $(2-4/3)/(4/3) = 1/2$. Replacing $b = 2$ by $b = 10$ changes the share to $100/100.125 = 0.998752$, but the full and linear displacements become $20/3$ and 10, retaining the same relative error $1/2$.

8.9.3 Inheritance and phenotype-regression checkpoints

The guide takes $B = 1$, $\gamma = 0.5$, $\rho = 0.5$ and $g_0 = 1$. Iteration gives

$$g_1 = \frac{5}{6}, \qquad g_2 = \frac{27}{34}, \qquad g_3 = \frac{149}{190}, \qquad g_4 = \frac{827}{1058}.$$

The decimal values are 0.833333, 0.794118, 0.784211 and 0.781664. Solving the scalar fixed-point equation gives $g^* = (\sqrt{17} - 1)/4$. With $\rho = 1$ instead, $g_4^{-1} = g_0^{-1} + 4\gamma = 3$, so $g_4 = 1/3$.

The final checkpoint takes $P = 1.5$, $\beta = 0.4$, $\gamma = -0.2$ and $G = 0.5$ in the raw relative-fitness regression convention. Here γ denotes the regression coefficient, not the stabilising parameter in the preceding paragraph. Chapter 7 gives

$$\alpha = 1 - \tfrac{1}{2}(-0.2)(1.5) = 1.15, \tag{8.12}$$

$$G_{\mathrm{sel}} = 0.5 + (0.5)^2\{-0.2 - (0.4)^2\} = 0.41, \tag{8.13}$$

$$b = \frac{0.5(0.4)}{0.41} = 0.4878048780, \tag{8.14}$$

$$H = \frac{1}{0.5} - \frac{1}{0.41} = -0.4390243902. \tag{8.15}$$

The guide's tenth checkpoint concerns logical scope: a current-population approximation check is not a test of an inheritance recursion. The simulation calculation below explains why.

8.10 The fitted estimation example

Chapter 6 derives the calibration, conditional likelihood and uncertainty procedure. The default example generates 800 genotypes with three independent noisy height observations per genotype and one reproductive count. Its generating parameters are $b = 0.7$, $H = -0.7$ and $G = 1$, giving

$$V_{\mathrm{lin}} = 0.7^2 = 0.49, \qquad V_{\mathrm{quad}} = \frac{(-0.7)^2}{2} = 0.245, \qquad \mathcal{A}_g = \frac{0.49}{0.735} = \frac{2}{3}.$$

The observation seed is 20261005; the bootstrap seed is 20261006. The recorded results are:

Quantity	Generating value	Estimate	Conditional 95% interval
b	0.7	0.682560	[0.600334, 0.778885]
H	−0.7	−0.769799	[−0.916719, −0.630133]
G	1	1.002149	[0.876642, 1.136257]
V_{lin}	0.49	0.466889	[0.354749, 0.599602]
V_{quad}	0.245	0.297570	[0.195636, 0.421178]
$\mathcal{A}_g$	2/3	0.610744	[0.533024, 0.685713]

The fitted measurement variance is 0.979881 and the corresponding reliability is 0.754189. Recalibrating and refitting all 500 bootstrap samples gives the stated intervals; no reference bootstrap fit failed. The naive analysis, which treats predictions as known breeding values and refits the fitness relationship, gives 0.673984. It changes fitted slope and curvature as well as variance, so it is not the same experiment as substituting a smaller G while holding b, H fixed.

The three estimation figures display the observed counts and conditional mean, component estimates with bootstrap intervals, and observable-data diagnostics. For the default data, the quartic-minus-quadratic conditional AIC difference is 3.410883. Raising quadrature order from 41 to 81 changes each of $b, H, \mathcal{A}_g$ by less than 4×10^{-13} for this fit. These checks support this numerical calculation under its observation model; they do not prove that the model is correct.

The separate quartic generating example adds $-0.08a^4$. Its second derivative at the mean leaves the local quadratic share at 2/3, while its full-curve affine R^2 is 0.242406. The eighth-moment derivation in Chapter 6 accounts for that difference. Changing functional form and increasing numerical integration accuracy address different sources of error.

8.11 What the multilocus simulation assesses

The simulation uses four additive traits and Gaussian stabilising selection, $\ell(\boldsymbol{a}) = -\frac{1}{2}(\boldsymbol{a}-\boldsymbol{\theta})^{\mathsf{T}}\boldsymbol{\Gamma}(\boldsymbol{a}-\boldsymbol{\theta})$, with $\boldsymbol{\Gamma} = \mathrm{diag}(0.10, 0.05, 0.04, 0.03)$. At each generation, set $\boldsymbol{x}_i = \boldsymbol{a}_i - \bar{\boldsymbol{a}}$ and expand:

$$\ell(\bar{\boldsymbol{a}} + \boldsymbol{x}_i) = \ell(\bar{\boldsymbol{a}}) + \{\boldsymbol{\Gamma}(\boldsymbol{\theta} - \bar{\boldsymbol{a}})\}^{\mathsf{T}}\boldsymbol{x}_i - \frac{1}{2}\boldsymbol{x}_i^{\mathsf{T}}\boldsymbol{\Gamma}\boldsymbol{x}_i.$$

Thus $\boldsymbol{b} = \boldsymbol{\Gamma}(\boldsymbol{\theta} - \bar{\boldsymbol{a}})$ and $\mathbf{H} = -\boldsymbol{\Gamma}$ are known exactly for that generation's mean. The analysis estimates covariance using $\widehat{\mathbf{G}} = (N-1)^{-1}\sum_i \boldsymbol{x}_i\boldsymbol{x}_i^{\mathsf{T}}$, inserts it into the Gaussian component formulas, and compares the resulting share with Equation (8.3) for the observed finite population.

Each of 20 replicates has $N = 2000$, 200 loci (50 per trait) and 200 recorded generations. Allelic-effect standard deviations are $(0.346, 0.283, 0.200, 0.141)$. The first optimum coordinate increases by 0.3 per generation for 40 generations to 12, then remains fixed; the other coordinates remain zero. No environmental residual is added. Seeds 42–61 define the replicate runs.

For replicate r and generation t, let $d_{rt} = |\widehat{\mathcal{A}}_{g_{rt}} - r_{rt}^2|$. The reported statistic is the median of the $20 \times 200 = 4000$ values, namely 0.0156678857. A table of those diagnostic values is sufficient to reproduce that summary. Recreating every individual population instead requires the SLiM model, version and seed sequence; these are provided separately in the repository.

The polynomial expansion remains exact for these individuals, but their finite breeding-value distribution need not be Gaussian. Their third and fourth moments, and finite-sample estimation, can therefore produce a discrepancy between the Gaussian component formula and the sample squared correlation. This correlation uses the theoretical linear term; it is not necessarily the best unrestricted multivariate affine fit to a non-Gaussian sample. Chapter 2 gives the general moment terms that the Gaussian simplification removes.

Biological interpretation. The comparison assesses a current-population approximation in a multilocus design. Allele frequencies and associations change during that simulation. Its agreement with the variance formula consequently cannot validate the separate recursion that holds $\mathbf{B}$ fixed. It also does not test the selected-parent displacement bound unless that additional comparison is explicitly computed.

8.12 Reproducing the numerical work

From the repository root, the principal commands are

```
bash run_all.sh --quick
bash run_all.sh
bash run_all.sh --visuals
bash run_all.sh --simulate
bash run_shiny.sh
```

The first regenerates the main figures, mathematical checks and archived simulation summary. The second adds the fitted estimation example and all 500 bootstrap refits. The third rebuilds the visual examples and offline explorers. The fourth runs fresh simulations and requires SLiM 5.2. The final command starts the Shiny app, which uses the same model and analysis code as the batch estimation example.

The result catalogue and reproduction guide identify exact inputs and outputs. The two offline explorers calculate the same Gaussian geometry or sample reproductive weights derived above. Changing an app control specifies a new example; it does not introduce a new theorem. The figures, observation model and proof assumptions should be interpreted together.

Appendix A

Result-by-result derivation map

This map covers all 53 records in the existing result catalogue. Some describe the same mathematics at different locations, and some describe figures, estimation procedures or teaching resources. They are not 53 distinct theorems. Each link goes to the appropriate proof or calculation.

A.1 Analytical results, examples and resources

Result or resource	Derivation or calculation
Main Figure 1: Construct conditional expected fitness before taking logs *Worked or numerical calculation*	p. 63; p. 5
Main Figure 2: Gaussian population projection identity *Worked or numerical calculation*	p. 13; p. 64
Main Figure 3: Proportional covariance contraction on a fixed surface *Worked or numerical calculation*	p. 17; p. 64
Main Figure 4: Scalar covariance trajectory with a separately changing gradient *Worked or numerical calculation*	p. 35; p. 65
Conditional mean, logarithm and reproductive weighting *Analytical derivation*	p. 1; p. 1; p. 3
Gaussian environmental averaging changes breeding-value curvature *Analytical derivation*	p. 5; p. 63
Local quadratic approximation and the full nonquadratic curve *Analytical derivation*	p. 9; p. 51
Linear and centred-quadratic Gaussian moments *Analytical derivation*	p. 9; p. 10
Affine regression intercept, slope and R^2 *Analytical derivation*	p. 12
Noise reduces both R^2 values but their ratio recovers the share *Analytical derivation*	p. 4; p. 13
Proportional covariance scaling and the omitted fraction *Analytical derivation*	p. 17; p. 64
A smaller covariance can reduce the linear share *Explicit counterexample*	p. 18
Changing trait units leaves the component variances unchanged *Analytical derivation*	p. 18
Whitening, curvature eigenvalues and the separate Rayleigh quotient *Analytical derivation*	p. 20; p. 20; p. 67
Zero terms, undefined correlation and singular covariance support *Analytical derivation*	p. 12; p. 14
Asymmetry and non-Gaussian fourth moments can change the partition *Analytical derivation*	p. 15
Exact selected-parent mean and covariance *Analytical derivation*	p. 23; p. 24; p. 24; p. 24

Result or resource	Derivation or calculation
A large share does not ensure an accurate parental shift or perfect rankings *Explicit counterexample*	p. 26; p. 26; p. 69
Absolute covariance-weighted curvature bounds relative displacement error *Analytical derivation*	p. 27; p. 28
Fixed allelic effects do not imply fixed genic covariance *Analytical derivation*	p. 30
Fixed-baseline covariance interval and matrix fixed-point bounds *Analytical derivation*	p. 32; p. 33
Scalar fixed point, contraction and a transient increase *Analytical derivation*	p. 35; p. 37
No-restoration boundary and retained null-space variation *Analytical derivation*	p. 37
Prediction shrinkage can mimic proportional biological contraction *Analytical derivation*	p. 43; p. 49
Convert compatible phenotype regression moments to breeding-value parameters *Analytical derivation*	p. 54; p. 55; p. 57; p. 58
Appendix S1: Gaussian moments and population projections *Analytical derivation*	Chapter 2; Chapter 3
Appendix S2: Gaussian selection, environmental averaging and displacement bound *Analytical derivation*	p. 5; p. 23; p. 24; p. 27
Appendix S3: Fixed-baseline inheritance and its limits *Analytical derivation*	Chapter 5
Appendix S4: Forward-time finite-population approximation *Worked or numerical calculation*	p. 70
Appendix S5: Moment conversion from phenotype regressions *Analytical derivation*	Chapter 7
Appendix S6: Interpretation, assumptions and boundaries *Interpretation*	p. 80; Chapter B
Appendix S7: Visual one-trait and rotation examples *Worked or numerical calculation*	p. 65; p. 67
SLiM approximation check over 4,000 generation records *Worked or numerical calculation*	p. 70
One fixed curve with broad, narrow and peak-centred populations *Worked or numerical calculation*	p. 65
Rotate the same covariance eigenvalues on a fixed surface *Worked or numerical calculation*	p. 67
Average phenotype-dependent expected output at each breeding value *Worked or numerical calculation*	p. 5; p. 69
Compare the linear and centred quadratic contributions *Worked or numerical calculation*	p. 10; p. 69
Reproductive weighting gives a selected-parent distribution *Worked or numerical calculation*	p. 26; p. 26; p. 69
Iterate the scalar covariance closure *Worked or numerical calculation*	p. 37; p. 69
From phenotype moments to log conditional-fitness parameters *Worked or numerical calculation*	p. 61; p. 69
Every numerical checkpoint and the multivariate extension *Worked or numerical calculation*	p. 69; Chapter 2; Chapter 4; Chapter 5; Chapter 7
Two-stage estimation from repeated measurements and offspring counts *Statistical procedure*	p. 42; p. 44; p. 47; p. 70

Result or resource	Derivation or calculation
Joint refitting and model-conditional uncertainty *Statistical procedure*	p. 50; p. 70
Latent integration, recovery and explicit numerical failures *Statistical procedure*	p. 45; p. 46; p. 46
Model shape, diagnostics and the local-versus-full distinction *Analytical derivation*	p. 51; p. 51; p. 52; p. 70
Observed counts and latent conditional relationship *Statistical procedure*	p. 47; p. 47; p. 70
Component estimates, naive comparison and conditional interval *Statistical procedure*	p. 50; p. 70
Observable-bin predictions and zero-count checks *Statistical procedure*	p. 47; p. 49; p. 52
Runnable scientific source, reference data and result checks *Reproduction resource*	p. 71; p. 80
Observation means, population-fitted lines and parental shift *Worked or numerical calculation*	Chapter 1; p. 65; p. 69
Rotate, narrow or move a Gaussian population on one fixed surface *Worked or numerical calculation*	Chapter 3; p. 67; p. 68
Frequency, expected-output and recorded-count weights *Worked or numerical calculation*	p. 1; p. 68
Change measurement uncertainty, sample size and model shape *Reproduction resource*	p. 44; p. 47; p. 50; p. 71

A.2 Equation-by-equation map

The following 106 source-qualified equation identifiers connect the main account and supplementary derivations to this companion. These are stable source labels, not printed equation numbers. The same label can occur in different sources. Definitions and model assumptions are identified separately from proofs and worked calculations. The source versions inspected on 2026-11-13 are identified by hashes in the equation map. The map contains identifiers and concepts; the source documents are not needed to build this companion.

A.2.1 Main text and mathematical appendices

Source label	Concept and treatment	Derivation
`eq:target`	Conditional expected reproduction and its logarithm *definition*	p. 1; p. 2
`eq:selection_weight`	Normalised reproductive weighting of parental frequencies *proof + definition*	p. 1; p. 1
`eq:log_ratio`	Log differences as proportional changes in conditional mean output *proof*	p. 2
`eq:jensen`	Noncommutation of averaging and taking logarithms *proof*	p. 3; p. 3
`eq:taylor`	Local quadratic expansion of log conditional expected output *definition + proof*	p. 8; p. 9

Source label	Concept and treatment	Derivation
`eq:centered_terms`	Mean-centred quadratic component *definition + proof*	p. 10; p. 10
`eq:vlin`	Variance of the linear component *proof*	p. 11
`eq:vquad`	Gaussian variance of the quadratic component *proof*	p. 11; p. 11
`eq:Ag`	First-order share of quadratic log-target variance *definition*	p. 12; p. 11
`eq:affine_projection`	Population affine projection and curvature-adjusted intercept *proof*	p. 13; p. 13
`eq:r2_identity`	Regression and squared-correlation interpretations of the share *proof*	p. 13; p. 13
`eq:nested_r2`	Ratio of nested fit statistics and the complementary fraction *proof*	p. 14; p. 14
`eq:scaling`	Linear and quadratic components under proportional covariance scaling *proof*	p. 17; p. 17
`eq:whitening`	Gradient and Hessian in population-standardised coordinates *definition + proof*	p. 20; p. 20
`eq:log_quadratic_weight`	Exact quadratic reproductive-weight model *definition*	p. 23; p. 23
`eq:selected_mean`	Exact selected-parent mean displacement *proof*	p. 24; p. 24
`eq:selected_covariance`	Exact covariance under parental reproductive weights *proof*	p. 24; p. 24
`eq:counterexample`	Large variance share with a one-half displacement overestimate *calculation*	p. 26; p. 69
`eq:T`	Population-scaled stabilising curvature *definition*	p. 25; p. 25
`eq:response_bound`	Covariance-scaled relative displacement-error bound *proof*	p. 28; p. 28
`eq:fixedB_recursion`	Covariance update under fixed genic baseline and association retention *definition + proof*	p. 31; p. 31
`eq:genic_baseline`	Genic covariance from allele frequencies and fixed allelic effects *proof*	p. 30; p. 30
`eq:loewner_interval`	Invariant order interval for the covariance map *definition + proof*	p. 34; p. 34
`eq:fixedpoint_bounds`	Bounds on every positive covariance fixed point *proof*	p. 34; p. 34
`eq:scalar_recursion`	One-trait or isotropic covariance update *definition + proof*	p. 35; p. 35
`eq:scalar_root`	Unique positive scalar covariance fixed point *proof*	p. 35; p. 35
`eq:rho_one`	Precision accumulation with complete association retention *proof*	p. 38; p. 37

Source label	Concept and treatment	Derivation
`eq:log_error_model`	Observation model with a constant conditional log gap *definition + proof*	p. 4; p. 13
`eq:ebv_inflation`	Share after substituting shrunken prediction covariance *proof*	p. 49; p. 49
`eq:LA_regression`	Raw relative-fitness phenotype regression and quadratic convention *definition*	p. 54; p. 54
`eq:LA_conversion`	Conversion of selected moments into log-quadratic parameters *proof*	p. 57; p. 58
`eq:fitness_model`	Phenotype-fitness model with independent Gaussian environment *definition*	p. 5; p. 5
`eq:environmental_average`	Gaussian environmental convolution of the fitness curve *proof*	p. 5; p. 5
`eq:precision_order`	Inverse-order covariance contraction under stabilising curvature *proof*	p. 33; p. 33
`eq:LA_moments`	Phenotype-regression normalisation and selected moments *proof*	p. 55; p. 55
`eq:LA_selected_mean`	Selected breeding-value mean from phenotype regression *proof*	p. 57; p. 57
`eq:LA_selected_cov`	Selected breeding-value covariance from phenotype regression *proof*	p. 57; p. 57

A.2.2 Appendix S7: visual walkthrough

Source label	Concept and treatment	Derivation
`eq:visual_model`	Fixed one-trait curve and Gaussian populations *definition*	p. 65
`eq:visual_variances`	Variance components for the three one-trait populations *calculation + proof*	p. 65; p. 11
`eq:visual_line`	Population-fitted line for the one-trait walkthrough *calculation + proof*	p. 65; p. 13
`eq:visual_rotation_surface`	Two-trait surface expressed along and across the gradient *definition + calculation*	p. 67; p. 21
`eq:visual_rotation_variances`	Full-population components under physical covariance rotation *calculation + proof*	p. 67; p. 21

A.2.3 Supplementary File 1: mathematics guide

Source label	Concept and treatment	Derivation
eq:guide-target	Conditional expected reproductive output *definition*	p. 1
eq:guide-log-target	Logarithm of conditional expected output *definition*	p. 2
eq:guide-jensen	Averaging versus logging individual outputs *proof*	p. 3; p. 3
eq:guide-selection-weight	Normalised reproductive weighting of breeding values *proof + definition*	p. 1; p. 1
eq:guide-phenotype-fitness	The guide's Gaussian phenotype-fitness example *definition + calculation*	p. 5; p. 69
eq:guide-environmental-average	Conditional mean for the guide's environmental parameters *proof + calculation*	p. 5; p. 69
eq:guide-quadratic-example	Expansion of the guide's exact quadratic log mean *calculation*	p. 69; p. 5
eq:guide-finite-slope	Symmetric finite-difference estimate of slope *calculation + proof*	p. 69; p. 9
eq:guide-finite-curvature	Symmetric second difference for curvature *calculation + proof*	p. 69; p. 9
eq:guide-local-description	One-trait quadratic approximation *definition + proof*	p. 8; p. 9
eq:guide-environmental-integral	Integral over environmental deviations in the guide example *definition + proof*	p. 5; p. 5
eq:guide-general-completing-square	Completing the environmental square *proof*	p. 5
eq:guide-environmental-average-general	General Gaussian environmental convolution *proof*	p. 5; p. 5
eq:guide-normal-moments	Centred Gaussian moments through order four *proof*	p. 10; p. 10
eq:guide-centered-Q	Centred scalar quadratic component *definition + proof*	p. 10; p. 10
eq:guide-vlin-one	Scalar linear variance *proof*	p. 11
eq:guide-vquad-one	Scalar quadratic variance *proof*	p. 11; p. 10
eq:guide-correlation	Squared correlation between the linear term and full quadratic target *proof*	p. 13
eq:Ag	First-order variance share *definition*	p. 12; p. 14
eq:guide-projection	Population affine-regression R-squared equals the share *proof*	p. 13; p. 13
eq:guide-one-trait-slope	Scalar least-squares slope equals the local slope under Gaussian quadratic conditions *proof*	p. 12; p. 13
eq:guide-nested-r2	Ratio of noisy affine and quadratic fit statistics *proof*	p. 14; p. 14
eq:guide-scaling	Proportional covariance scaling of the two components *proof*	p. 17; p. 17

Source label	Concept and treatment	Derivation
eq: guide-selected-moments	Scalar selected-parent variance and mean shift *proof*	p. 26; p. 24
eq: guide-response-bound-one	One-dimensional curvature bound on relative displacement error *proof + calculation*	p. 26; p. 28
eq:guide-recursion	Scalar fixed-baseline inheritance closure *definition + proof*	p. 35; p. 31
eq:guide-scalar-bounds	Numerical endpoints of the scalar invariant interval *calculation + proof*	p. 34; p. 37
eq:guide-general-fixedpoint-quadratic	Polynomial satisfied by the scalar equilibrium *proof*	p. 35
eq:guide-scalar-root	Positive scalar equilibrium root *proof*	p. 35; p. 35
eq:guide-contraction	Exact distance formula and scalar contraction inequality *proof*	p. 36; p. 36
eq:guide-rho-one	Scalar precision after repeated updates with full retention *proof*	p. 38; p. 37
eq:guide-LA	Scalar phenotype regression of relative fitness *definition*	p. 54; p. 54
eq:guide-intercept	Relative-fitness intercept and its numerical value *proof + calculation*	p. 55; p. 61
eq:guide-first-moment	First selected phenotype moment *proof*	p. 55; p. 55
eq:guide-second-moment	Second selected phenotype moment *proof*	p. 55; p. 55
eq: guide-conditional-A-one	Breeding-value mean and variance conditional on phenotype *proof*	p. 56; p. 56
eq:guide-BV-moments	Selected breeding-value moments in one trait *proof*	p. 57; p. 61
eq:guide-conversion-one	Scalar inversion from selected moments to log-quadratic parameters *proof + calculation*	p. 58; p. 61
eq:guide-ebv	Share under fixed-coefficient prediction-variance substitution *proof + calculation*	p. 49; p. 49
eq:guide-trace-reduction	Four-index reduction of quadratic variance to a trace *proof*	p. 11; p. 11
eq:guide-multi-variances	Multivariate linear variance, quadratic variance and orthogonality *proof*	p. 11; p. 11
eq:guide-multi-selected	Multivariate selected-parent mean and covariance *proof*	p. 24; p. 24
eq:guide-whitened-full	Exact selected displacement in covariance-scaled coordinates *proof*	p. 27; p. 25
eq:guide-whitened-error	Displacement error in covariance-scaled coordinates *proof*	p. 27
eq:guide-multi-bound	Relative displacement bound and Frobenius upper bound *proof*	p. 28; p. 28

Source label	Concept and treatment	Derivation
eq:guide-genic-baseline	Multitrait genic covariance from loci *proof*	p. 30; p. 30
eq:guide-matrix-interval-map	Invariance of the covariance order interval *proof*	p. 34; p. 34
eq:guide-multi-conditioning	Multivariate breeding-value conditioning on phenotype *proof*	p. 56; p. 56
eq:guide-multi-raw-derivatives	Derivatives of the logged phenotype-regression polynomial *proof*	p. 62; p. 61

A.2.4 Supplementary File 3: worked estimation

Source label	Concept and treatment	Derivation
eq:A	Gaussian latent breeding-value population in the estimation example *definition*	p. 41; p. 41
eq:measurement	Repeated noisy trait measurements *definition*	p. 41; p. 41
eq:count	Conditional Poisson reproductive-count model *definition*	p. 41; p. 41
eq:surface	Quadratic log conditional expected count *definition*	p. 41
eq:linear	Generating linear variance in the worked example *calculation + proof*	p. 47; p. 11
eq:quadratic	Generating quadratic variance in the worked example *calculation + proof*	p. 47; p. 11
eq:share	Generating variance share in the worked example *calculation*	p. 47; p. 70
eq:Ehat	Unbiased within-genotype measurement-variance calibration *proof*	p. 42; p. 42
eq:Ghat	Unbiased between-genotype variance calibration *proof*	p. 42; p. 42
eq:shrink	Calibrated prediction reliability and centred conditional mean *proof + calculation*	p. 43; p. 43
eq:posterior	Normal latent distribution conditional on repeated measurements *proof*	p. 43; p. 43
eq:likelihood	Poisson likelihood integrated over uncertain breeding value *proof*	p. 44; p. 44
eq:derivatives	Slope and curvature of the fitted quadratic log mean *proof + calculation*	p. 47
eq:quarticvar	Complete even-part variance with a quartic term *proof*	p. 52; p. 52
eq:quarticR2	Full-curve affine R-squared for the quartic generating function *proof + calculation*	p. 52; p. 52

A.2.5 Unlabelled intermediate displays

A further 87 unlabelled displays were inspected. They supply definitions, intermediate algebra and numerical substitutions within the derivations above. The machine-readable equation map groups them by concept and identifies their companion targets; no additional result was left without a derivation or calculation.

A.3 The eleven interpretation questions

The questions in Appendix S6 can be followed through the derivations as follows.

Q1: what the share measures
A fraction of variance in a quadratic log conditional-mean description under the stated population. See p. 12.

Q2: established tools and the question addressed
The prerequisite sources and their roles are separated from the comparisons made here. See Chapter B.

Q3: why average before taking logs
Log conditional mean and conditional mean log are different targets. See p. 3.

Q4: gene action and a cutoff
A population fitness description alone identifies neither gene action nor a universal threshold. See p. 12.

Q5: which Gaussian assumptions matter
Third and fourth moments show which identities fail outside the Gaussian model. See p. 15.

Q6: estimation
Observation and latent-value uncertainty enter the likelihood and refitting procedure. See Chapter 6.

Q7: zero gradient and zero variance
Zero shares, undefined ratios and reduced-rank supports are treated separately. See p. 14.

Q8: rankings and selection accuracy
Exact weighting and counterexamples distinguish a close fit from an accurate parental prediction. See Chapter 4.

Q9: covariance, dimensions and orientation
Whitening, Rayleigh quotients and physical rotation answer different comparison questions. See Chapter 3.

Q10: inheritance and equilibrium
The fixed baseline and retention rule are explicit modelling assumptions. See Chapter 5.

Q11: use of the descriptive share
Report the share with both absolute components to distinguish different population changes. See p. 68.

A.4 Commands associated with the results

Commands start at the repository root. Inputs, outputs and assumptions remain in the machine-readable catalogue. Executing these implementations does not replace the analytical arguments.

main figures
Generate the four main figures from recorded parameters and seeds.

```
python code/figures.py
```

core mathematics
Check 30 matrix cases, coordinate changes, proportional scaling, bounds and scalar trajectories.

```
python code/verify_mathematics.py
```

result checks
Independent small numerical examples of the identities, scope conditions and counterexamples; no random simulation or document rebuild.

```
python code/reproduce_result_checks.py
```

simulation archive
Validate and summarise the 20 archived diagnostic tables; does not run SLiM.

```
python code/summarize_simulation.py
```

fresh simulation
Run seeds 42–61 with SLiM 5.2, reconstruct diagnostics and compare every diagnostic table with the archive.

```
bash run_all.sh -simulate
```

guide figures
Regenerate all five mathematics-guide figures and check their stated values.

```
python code/guide_figures.py
```

intro checks
Print quadrature checks of one-trait population regressions and selected-parent weighting.

```
python tutorials/introduction/check_examples.py
```

intro figures
Rebuild three introductory teaching figures from their specified examples.

```
python tutorials/introduction/make_figures.py
```

one trait
Calculate and quadrature-check the three one-trait populations, then draw Figure A1.

```
python tutorials/one_trait_visual/make_figure.py
```

geometry values
Quadrature-check all rotation and comparison cases without rebuilding visual documents.

```
python tutorials/geometry/code/geometry.py
```

geometry visuals
Rebuild the two-page geometry figure and offline explorer; Figure A2 is page one.

```
bash tutorials/geometry/run.sh
```

parental weighting
Recover the fixed parental cloud, compare frequency and reproductive weights, and rebuild its figures and offline explorer.

```
bash tutorials/parental_weighting/run.sh
```

estimation
Recreate the default observations, latent-value fit, 500 bootstrap refits and three estimation figures. The seed is 20261005; the bootstrap seed is 20261006.

```
bash run_estimation.sh
```

estimation checks

Run eight scientific checks: moments, observation/truth separation, independent integration, recovery, bootstrap calibration, quartic scope, integration failure and failed calibration.

```
python -m unittest discover -s examples/estimating_variance_share -p
test_model.py
```

quartic case

Generate the optional misspecified quadratic analysis of data from a quartic log mean. These fitted results are an exploratory case; the cited 0.2424 is the exact generating full-curve R^2.

```
bash run_estimation.sh -quartic 0.08 -output outputs/estimation_quartic
```

shiny

Launch the local Shiny for Python teaching app. Its download records the displayed data, settings, estimates and any completed bootstrap.

```
bash run_shiny.sh
```

reference comparison

After a complete default run, compare figure parameters/results, archived-simulation summary, observations and every bootstrap draw with the reference records.

```
python code/check_results.py default
```

Appendix B

Mathematical concepts, biological questions and sources

This index connects the mathematical tools to the questions they answer in the companion. The references provide the established background. The derivations in the chapters show how those tools apply to the specified population and observation models.

B.1 Probability and population description

Conditional expectation and the tower property. Conditional expectation gives mean reproductive output at each breeding value. The tower property then explains why averaging individual output within breeding-value classes preserves the expected-output weighting of parental classes. Chapter 1 proves the relevant identities. The covariance form connects this calculation to the selection component of the Price identity (Price, 1970); it does not by itself supply a transmission term.

Gaussian moments and the moment-generating function. Gaussian symmetry removes odd moments, while the fourth moment is a sum over covariance pairings (Isserlis, 1918). Chapter 2 derives the required identities and carries their indices through the quadratic-form variance. The same identities enter phenotype-regression conversion in Chapter 7. Their use is a statement about the breeding-value or phenotype distribution, not a statement about whether gene action is additive.

Projection and explained variance. Population least squares is an orthogonal projection under the inner product defined by population frequencies. Its normal equations identify the fitted intercept and slope. Chapter 2 derives those equations directly, so the reader can see why a tangent and a fitted line generally have different intercepts. Chapter 3 then connects squared correlation to an angle between functions. This projection angle is distinct from the physical orientation of genetic covariance.

B.2 Linear algebra, selection and inheritance

Congruence, similarity and matrix norms. Congruence preserves positive semidefiniteness; similarity preserves eigenvalues and trace. These are different operations, both needed when transforming covariance and curvature. Chapters 3–5 explain them at the point of use. The linear algebra and convex-set background is treated in Boyd and Vandenberghe (2004). In biological terms, these tools separate a change of measurement units from a change in the population, and distinguish absolute curvature from its fraction of fitness variation.

Directional evolvability. The Rayleigh quotient $\boldsymbol{u}^{\mathsf{T}}\mathbf{G}\boldsymbol{u}$ measures variance along a unit direction. The connection to directional evolvability is developed by Hansen and Houle (2008). Chapter 3 shows exactly which norm is held fixed in that comparison. This is essential because fixing the gradient's ordinary norm and fixing its whitened norm define different comparisons.

Completing the square and change of weights. Multiplying a Gaussian population density by an exponential quadratic weight changes its precision and mean. Chapter 4 performs the factorisation and normalisation explicitly, including the condition that the integral be finite. Conditional averaging over environmental effects and Gaussian selection calculations have a quantitative-genetic precedent in Lande (1979). Their exactness in this companion depends on the stated distributions and functional form.

Genic covariance and associations between loci. The distinction between a genic baseline and association covariance is central to selection-induced changes in genetic variance (Bulmer, 1971, 1974; Barton and Turelli, 1991). Chapter 5 shows how the particular scalar retention assumption generates the model's recursion. These sources support the biological background; they do not make that one-parameter closure universal. Broader quantitative-genetic context is available in Walsh and Lynch (2018).

Existence, uniqueness and convergence. Brouwer's finite-dimensional fixed-point theorem establishes existence for a continuous map from a nonempty compact convex set into itself. Chapter 5 verifies every condition for the covariance order interval. A standard account is Granas and Dugundji (2003). The scalar contraction argument is stronger: it also yields uniqueness and convergence, following the principle associated with Banach (1922). Existence alone does not imply either conclusion for the general matrix recursion.

B.3 Estimation and interpretation

Prediction uncertainty and Gaussian conditioning. Repeated measurements lead to a conditional distribution of breeding value, not a perfectly known value. Chapter 6 derives its mean, variance and reliability, then integrates the reproductive likelihood over that distribution. Hadfield et al. (2010) explains the inferential problems that can arise when predicted random effects are treated as observed quantities in a second analysis. The companion's fixed-coefficient shrinkage calculation identifies one specific effect; it is not a universal formula for every such analysis.

Latent integration and Gaussian quadrature. The likelihood integrates a count probability over the conditional breeding-value distribution. Its Gauss–Hermite approximation uses an explicitly stated normalisation. The general quadrature framework is documented by National Institute of Standards and Technology (2026), and the probabilists' Hermite convention used by the code is documented by The SciPy Community (2026). Polynomial exactness is a property of the quadrature rule; a Poisson–Gaussian mixture integrand is not generally a polynomial of the required degree.

Bootstrap uncertainty and model comparison. Bootstrap calculations approximate the sampling distribution of an estimator by repeating its analysis under a specified generating distribution (Efron, 1979). Chapter 6 identifies that distribution and shows why the calibration must be repeated as well as the fitness fit. The AIC comparison uses the likelihood and parameter-count penalty of Akaike (1974). In this example both calculations remain conditional on the chosen observation model.

Phenotype regressions and breeding-value parameters. Lande and Arnold (1983) provides the population-regression framework for selection on correlated phenotypes. The quadratic coefficient convention matters (Stinchcombe et al., 2008). Chapter 7 derives the moment conversion

under explicit joint-distribution and selection-mediation assumptions. It also explains why differentiating the logarithm of a fitted raw-scale polynomial is a different operation. Log-link fitness models (Morrissey and Goudie, 2022) and the distinction between latent and observed scales (de Villemereuil et al., 2016) provide related methodological context.

B.4 How to assess a calculation

A reader checking a result can ask four questions in sequence. First, which probability distribution supplies the expectation or covariance? Second, is the fitness relationship a local polynomial approximation or the exact model used for weighting? Third, does the calculation describe parents or require an additional transmission model? Fourth, are its inputs population quantities, estimates or realised observations? The answer to each question fixes the assumptions needed for the next step.

This sequence prevents several common substitutions: a sample of predictions for the true breeding-value distribution; a high explained fraction for a small absolute error; a phenotype regression for the conditional fitness surface without a conversion argument; or a covariance fixed point for a full evolutionary trajectory. The result map in Appendix A points to the relevant derivation whenever one of these distinctions matters.

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